

Innovation as prospective-but-not-retrospective surprise: Kullback–Leibler novelty scores on 2.0 million USPTO trademark filings

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Abstract

We measure vocabulary flux in firm activity using the goods-and-services descriptions of 16.5 million U.S. trademark filings since 1985, across all 45 NICE international (industry) classes. For each filing we compute two information-theoretic surprise scores against per-day exponentially decay-weighted reference distributions of the same NICE class: *prospective surprise* (distance from the recent past) and *retrospective surprise* (distance from the near future). Their difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ is a *flux* signal: positive when a filing’s vocabulary anticipates where the field is going, negative when it extracts from where the field has been. Flux is orthogonal to distinctiveness (the principal component the two surprise levels share); the substantive direction sits entirely in the residual.

The headline finding is a squeeze on flux-neutral filings. Post-registration 5-year survival is U-shaped in ΔKL under every operationalisation we tried: both the harvest extreme (negative ΔKL — receding vocabulary, sunseting categories) and the innovate extreme (positive ΔKL — emerging vocabulary, the 2014+ cannabis, crypto, vape, NFT and AI waves) survive at roughly 50–55%; the flux-neutral middle survives at $\sim 45\%$. The pattern holds at the per-industry level in 27 of 44 industries, on token-level survival rates (negative-stratum and positive-stratum tokens both survive well above the corpus baseline; neutral tokens like **organisational**, **conformity**, **freighting** all sit at 11–26%), and in our hand-curated panel of 41 canonical “winner” firms (37/41 have positive firm-mean ΔKL). What the mechanism is — intelligent harvesting of vanishing concepts, recognition of emerging ones, or selection on an unobservable correlate — is not separable in the trademark record; what is unambiguous is that filings whose vocabulary places them squarely *on trend* in the centre of the distribution survive at the lowest rate in the corpus.

Novelty of language and novelty of claim are different signals. A pioneer-ladder analysis on five category-defining buzzwords (**website**, **wireless**, **online retail**, **social media**, **blockchain**) ranks every firm in the corpus by date-of-first-adoption; the very-first 10–50 adopters do badly, the fast-follower band at ranks 50–1,000 has 1.5–4 $\times$ the late-entrant IPO rate, and the mass late wave does worst. Canonical winners are mostly absent from the very-early ranks (Amazon 174th on **online retail**, Coinbase 96th on **blockchain**; Uber, Lyft, Airbnb never used their category buzzword at all) but most have positive first-filing ΔKL , often strongly so (Oracle +4.39 nats, Intel +3.26, Microsoft +2.99, eBay +1.73, Apple +1.68, Coinbase +0.98, Lyft +0.76, Twitter +0.75, Uber +0.70, LinkedIn +0.58, Salesforce +0.43). The two “novelties” decouple: the novelty of the *claim* (which vocabulary the firm adopts) and the novelty of the *language* (the texture of their goods/services prose) are independent levers.

ΔKL is essentially uncorrelated with the dominant alternative innovation proxy (firm-year patent counts, $r = +0.010$ on 174,569 firm-years), positively correlated with post-registration survival, and positively correlated with profitability among publicly-traded firms (+3.2 pp gross margin per σ , SE 0.6 pp).

1 Background

Innovation is a central object of empirical economic and management research, but the construct is hard to measure. The dominant quantitative approaches are R&D expenditure (an input proxy that rewards spend rather than novelty), patent counts and patent citations (which require strategic legal filing, and which Trajtenberg, Hagedoorn, and others have shown to be at best a noisy proxy), and the market value premium of equity over book value (which conflates innovation with every other investor perception). Survey-based scales such as the Community Innovation Survey and Gatignon et al.’s instrument capture richer constructs but suffer from common-method bias and limited sample size. The lack of agreement on what to measure is itself a finding: across measures, pairwise correlations are typically $\rho \approx 0.3$.

We propose using trademark filings as a complementary novelty channel. Trademarks have several useful properties that patents do not: applicants file a single, brief, machine-readable goods-and-services description that is sharply incentivized to be exact (broad enough to protect future use, narrow enough to clear examination); the data are released into the public domain by the USPTO without lag; and the universe of filers is broader than the universe of patentees, including most consumer-facing firms. The cost is that trademarks measure novelty in the goods-and-services lexicon, not in the underlying technology — a brand that solves an old problem with a new identity (e.g. LIQUID DEATH’s canned water) is largely invisible.

Our broader motivation is to ask whether innovative firms are rewarded or punished. Two pieces of folk wisdom are in tension. The first, conveyed in the phrase “you can tell the pioneers by the arrows in their backs,” is that firms that depart from convention bear higher search and adoption costs, get out-competed by fast-followers, and underperform on average. The second, expressed in the older saying “build a better mousetrap and the world will beat a path to your door,” is that meaningful novelty attracts customers and rewards its inventor with durable advantage. Both can be true simultaneously: novelty might raise variance — some pioneers fail spectacularly, others enjoy durable category leadership. This paper offers an empirical test of that joint pattern using the full USPTO trademark backfile.

2 Data

The trademark corpus is the full backfile published by the USPTO Open Data Portal, current through end-of-year 2025. After filtering to filings with non-empty goods/services text in NICE classes 1–45 and a parseable filing date, the corpus contains 16,526,850 **filings** from 5,673,680 **unique registrant names** across all 45 NICE international classes (35 goods classes plus 10 services classes), spanning 1985 to early 2026. Each record carries the registrant’s name as it appeared on the application (this is the owner at the time of filing; subsequent assignments are not reflected), the filing date, a status code that records the current legal state of the mark, an optional registration date, the NICE class assignments, and one or more goods-and-services descriptive statements. Statements describing the visual mark itself rather than the business (USPTO type-code prefix D) are excluded; only goods-and-services statements (prefix GS) are kept. Two classes are presented in narrative detail throughout the paper: NICE Class 9 (Software & Electronics) and NICE Class 32 (Beer & Soft Drinks). The remaining 43 industries are summarized in Section 19, and per-class summary statistics for the two focal classes appear in Appendix A.

The financial-outcome data are the SEC EDGAR Financial Statement Data Sets, 64 quarterly XBRL releases spanning fiscal years 2009 through 2024, covering 13,127 unique CIKs that filed at least one 10-K. We extract revenue, cost of revenue, gross profit, R&D expense, operating income,

and net income.

Inclusion filter (clean). Almost every analysis in this paper is restricted to the same inclusion filter, which we call CLEAN. The filter has five clauses, each addressing a separate failure mode: two outcome-window clauses (`year` bounds) that ensure the survival outcome is fully observed and that the prospective reference window is non-empty, two density floors on the per-class reference distributions (`n_ref_pros` and `n_ref_retr` $\geq 1,000$) that ensure KL scores reflect a real vocabulary distribution rather than small-sample noise, and a per-filing token-count floor ($n_{\text{terms}} \geq 3$) that prevents 1- or 2-token descriptions from producing degenerate KL values. Of 16,730,022 raw filings, **7,908,949** (47.3%) survive the full filter and form the analytic corpus.

Clause	dropped alone (vs. raw, %)	added back if dropped (vs. CLEAN, %)
<code>year</code> ≥ 1990	1,246,246 (7.4%)	+2,597 (+0.0%)
<code>year</code> ≤ 2018	5,821,920 (34.8%)	+4,952,237 (+62.6%)
<code>n_ref_pros</code> $\geq 1,000$	2,709,751 (16.2%)	+9,140 (+0.1%)
<code>n_ref_retr</code> $\geq 1,000$	2,931,686 (17.5%)	+0 (+0.0%)
$n_{\text{terms}} \geq 3$	3,639,965 (21.8%)	+602,341 (+7.6%)
CLEAN (all of the above)	8,821,073 (52.7%)	—

Table 1: Per-clause filter effects on the 16,730,022-filing raw corpus. “Dropped alone” is the share that fails the clause when applied in isolation to the raw corpus; “added back” is the share that the clause is uniquely binding on, i.e. what would re-enter the analytic corpus if we relaxed only that one clause. The two density floors (`n_ref_pros`, `n_ref_retr`) are nearly redundant once the year window is in place: their alone-restriction is large because they bind on the burn-in years 1985–1989 and 2021–2025, which the year clauses already exclude.

The dominant binder is the upper year cutoff, `year` ≤ 2018 , which alone accounts for 5.0M of the 5.0M filings that the post-registration survival outcome could not be observed for on the May 2026 data extraction. The lower bound, `year` ≥ 1990 , is the literal five-year arithmetic burn-in described in the Method section. The token-count floor, $n_{\text{terms}} \geq 3$, is the only other clause that uniquely excludes a meaningful share (+602k filings rescued if dropped); when we look at the templated short-description analysis in Section 16 we report results both with and without this clause, since the most heavily templated G/S strings (*e.g.* “restaurant services”, “legal services”) are 2-token strings and would otherwise be excluded.

Registration is determined by status code, not registration date. We classify a mark as having “completed registration” from its USPTO status code (6xx live-registered or 8xx post-registration-cancelled), not from the `registration_date` field, which is unreliable in older records (mostly pre-2003): many records with a valid registered status code carry a missing `registration_date` in the published XML. The pipeline that extracts and analyses these data is published with the paper; technical details live in the companion repository.

3 Method

For each filing i at year t with goods-and-services text w_i we construct a term-frequency distribution P_i over a fixed dictionary V of unigram and bigram tokens. The dictionary is built once per NICE class with a custom analyzer that splits each description at semicolons before tokenizing, so bigrams

cannot cross goods/services-item boundaries. We retain only unigrams and bigrams that appear in at least 50 distinct filings (documents) and in no more than 50% of all filings, on the view that very rare tokens are typos and very common tokens are USPTO boilerplate.

For each filing i at filing-date t_i we form two reference distributions by per-day exponential decay-weighting of every other filing in the same NICE class:

$$Q_i^{\text{pros}}(v) \propto \alpha + \sum_{j: t_j < t_i} e^{-\lambda_{\text{pros}}(t_i - t_j)} \text{count}(v, j), \quad (1)$$

$$Q_i^{\text{retr}}(v) \propto \alpha + \sum_{j: t_j > t_i} e^{-\lambda_{\text{retr}}(t_j - t_i)} \text{count}(v, j), \quad (2)$$

with $\lambda = \ln(2)/H$ and **half-life** $H = 2$ **years** on both sides and Laplace smoothing $\alpha = 1.0$. The half-life is chosen by a sweep over $H \in \{0.5, 1, 2, 3, 5\}$ years on Class 9 that maximises $\text{corr}(\Delta KL, \text{survived}_{.5y})$ and the logit pseudo- R^2 on the same outcome (Figure 1). Density on the reference distributions is reported as Kish effective sample size $N_{\text{eff}} = (\sum_j w_j)^2 / \sum_j w_j^2$; we restrict all filing-level analyses to filings with $N_{\text{eff}} \geq 1,000$ on both sides.

The prospective and retrospective surprise of filing i are the support-restricted KL divergences

$$KL_i^{\text{pros}} = \sum_{v: P_i(v) > 0} P_i(v) \log \frac{P_i(v)}{Q_i^{\text{pros}}(v)}, \quad KL_i^{\text{retr}} = \sum_{v: P_i(v) > 0} P_i(v) \log \frac{P_i(v)}{Q_i^{\text{retr}}(v)}.$$

With log in natural units, KL_i is in nats. A useful intuition: KL_i^{pros} is the expected log-odds per token that a reader who had seen only the filing’s recent past would assign to the filing’s own term distribution versus the past’s. $KL_i = 3$ nats means a typical token in this filing is $e^3 \approx 20\times$ more probable under the filing’s own distribution than under the recent corpus. KL is sensitive to filing length (short filings with one rare token can score high mechanically); every regression in the paper controls for $\log(n_{\text{terms}})$ and most report z -scored ΔKL .

The directional flux score is $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$: the change between how unlike the recent past a filing reads and how unlike the near future it reads. Positive ΔKL means the filing’s vocabulary is closer to the future than to the past (*flux-positive*; the field migrated toward this filing); negative ΔKL means closer to the past than to the future (*flux-negative*; the field migrated away).

Magnitudes. To ground the reader: across the all-class analytic pool of 13.0M filings with $N_{\text{eff}} \geq 1,000$ on both sides, the 5th, 50th, and 95th percentiles are $KL^{\text{pros}} = [2.99, 4.85, 7.20]$ nats (std 1.29) and $KL^{\text{retr}} = [2.98, 4.84, 7.22]$ nats (std 1.30). The two components are correlated at $\rho = +0.99$ pooled across all classes (see Table 2); they measure the same general-distinctiveness signal in two temporal directions, and their per-filing levels track each other tightly. Because they share that much distinctiveness against the per-class boilerplate, their difference ΔKL has a markedly tighter range: 5th/50th/95th percentiles $[-0.30, -0.00, +0.34]$ nats (std 0.22, median essentially zero by construction of the look-behind / look-ahead window). A filing in the top 5% of ΔKL is therefore ~ 0.6 nats more flux-positive than one in the bottom 5%, on a scale where each component routinely exceeds 5 nats outright. The substantive direction lives in this comparatively small flux residual, not in the much larger common distinctiveness factor.

Identical-text filings are signal. Trademark practice rewards templated language. Many breweries file the boilerplate token **Beer**; many app developers file **downloadable software**. We do not deduplicate identical text: filings that use the same words *should* receive the same score, because deviations from the boilerplate are themselves the signal.

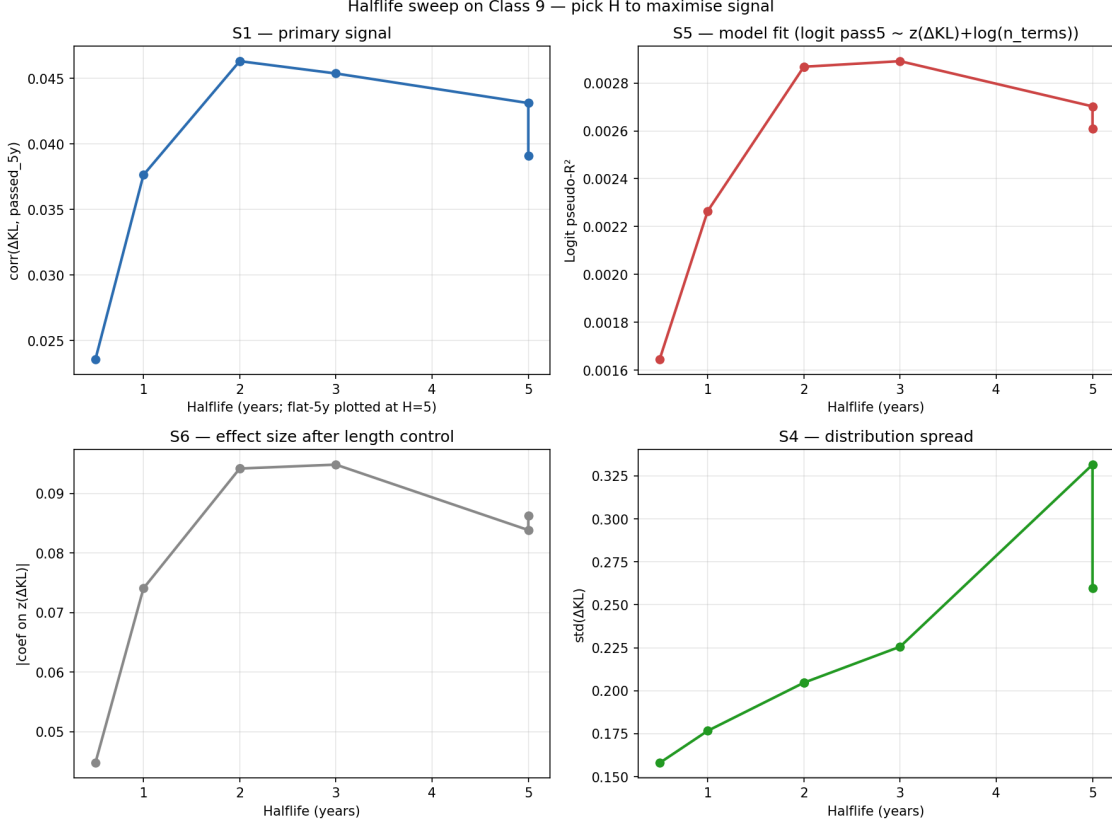


Figure 1: Half-life sweep on Class 9. For each candidate $H \in \{0.5, 1, 2, 3, 5\}$ years we recompute the decay-weighted reference distribution and report four signal metrics on the $\Delta KL \rightarrow$ survival relationship. $H = 2-3$ years maximises $\text{corr}(\Delta KL, \text{survived}_{5y})$ and the logit pseudo- R^2 ; we adopt $H = 2$.

Burn-in window. The decay reference distributions need filings on both sides of t_i for the weighted sums to be well populated. The $N_{\text{eff}} \geq 1,000$ density filter above automatically excludes small NICE classes' early-decade cohorts and the corpus right-edge filings whose forward window is too sparse. For regressions on the panel we restrict to filings from **1990 onward** (after the corpus's left-edge digitisation boundary in 1984 plus a margin to populate the look-behind) and to filings whose year has at least 5 years of forward observation, currently **1990–2021** given a 2026 freeze date. Figure 2 shades the two burn-in windows.

What ΔKL measures (and what it does not)

The two component KLS are very highly correlated: per-filing $\rho(KL^{\text{pros}}, KL^{\text{retr}}) = 0.99$ on the pooled clean panel. A reader plotting any outcome against either component univariately sees nearly identical curves because the two component levels are, to first order, the same number — both measuring *distinctiveness* (how unusual the filing's vocabulary is against the rest of the corpus, in either temporal direction).

The PCA decomposition of $(KL^{\text{pros}}, KL^{\text{retr}})$ makes the geometry explicit:

$$\begin{aligned} \text{PC1} &\approx \frac{1}{\sqrt{2}}(KL^{\text{pros}} + KL^{\text{retr}}) = \text{general distinctiveness} \\ \text{PC2} &\approx \frac{1}{\sqrt{2}}(KL^{\text{pros}} - KL^{\text{retr}}) \propto \Delta KL = \text{vocabulary flux} \end{aligned}$$

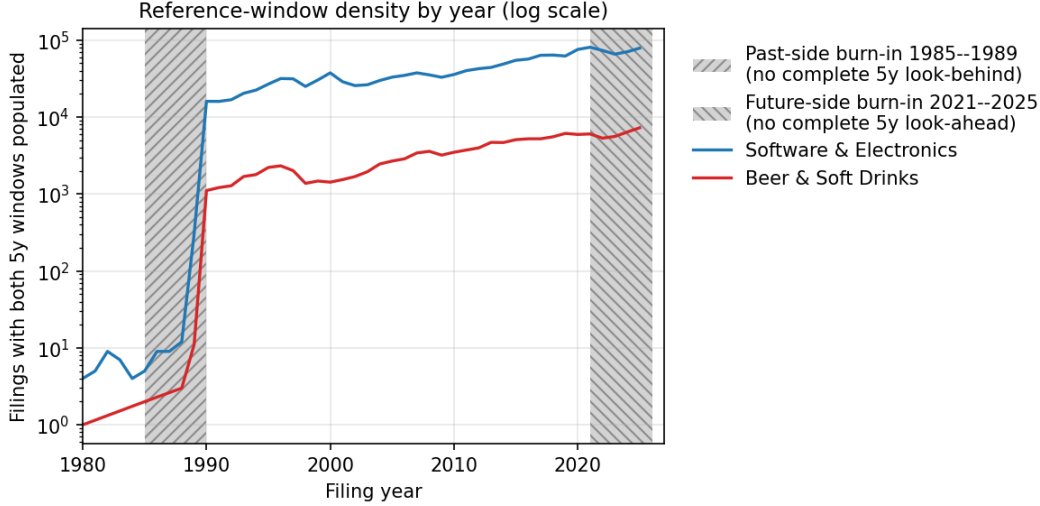


Figure 2: Filings per year with both 5-year reference windows populated. Forward-hatched grey: literal 5-year past-side burn-in (1985–1989) — filings in these years cannot have a complete look-behind because the digitised corpus begins in 1984. Backward-hatched grey: literal 5-year future-side burn-in (2021–2025) — filings in these years cannot have a complete look-ahead at the time of writing. The bands are chosen to print distinguishably in black-and-white. A separate density filter (`n_ref_prospective` ≥ 1000) handles the additional small-class exclusions visible in the long left tail of the plot.

ΔKL is the orthogonal residual after PC1 is removed; it is the *only* part of the joint signal with a forward/backward direction. PC1 says “this filing is unusual.” PC2 says “this filing is unusual *and the field’s vocabulary subsequently moved toward it rather than away.*” We claim throughout that the substantive innovation signal lives in PC2; PC1 is largely a length/rarity covariate. The closest-adjacent management-theory framework, optimal distinctiveness (Deepphouse 1999, *SMJ*; Zhao et al. 2017, *SMJ*), predicts an inverted-U on the *level* of distinctiveness. The framework we are extending is DeDeo’s prospective/retrospective surprise scheme (Murdock, Allen & DeDeo 2017, *Cognition*; Barron, Huang, Spang & DeDeo 2018, *PNAS*), which decomposes a temporal corpus into novelty, transience, and resonance and which we apply for the first time to commercially incentivised text.

4 What the components do separately

Before introducing ΔKL as a single scalar, it is worth asking whether the prospective and retrospective KL components carry independent information about a filing. A natural prior — and one we held coming in — is that they should pull in opposite directions: prospective surprise should make a filing harder to clear examination (novel goods/services language draws office actions), while retrospective surprise should reward the filer for “being right” (the future ends up resembling them, suggesting they understood a market others did not). Under that view, the value of ΔKL is that it nets two opposing forces.

The data do not support that decomposition. Table 2 shows that the per-filing correlation between KL_{pros} and KL_{retr} is +0.97 in software (Class 9), +0.93 in beverages (Class 32), and +0.97 in the 11.4-million-filing all-class pool. The two scores are computed against the same per-filing distribution $P_i(v)$, with the only difference being whether the reference window is the five preceding or the five following years; the bulk of variance in either KL therefore comes from P_i itself, and only

a thin sliver comes from the past-vs-future shift in the reference. As a consequence, plotted against any per-filing outcome the two components produce nearly identical curves.

Table 2: Per-filing correlation of the two KL components and single-component slopes against *reaching registration* and *passing five-year survival* (per-filing linear fits on binned means, 15 bins between the 2nd and 98th percentile of the predictor). The two components carry essentially the same signal at the per-filing level.

Scope	$\rho(KL_{\text{pros}}, KL_{\text{retr}})$	slope on reg. vs KL_{pros}	slope on reg. vs KL_{retr}	slope on pass-5y vs KL_{pros}
Software (Cl. 9)	+0.967	+0.024	+0.021	+0.008
Beverages (Cl. 32)	+0.933	+0.018	+0.017	+0.009
All 45 (pooled)	+0.967	+0.026	+0.028	+0.013

The curves themselves are not lines and not exponentials. They are **monotone-saturating**: registration probability rises from about 0.43 at $KL = 3$ nats to about 0.53 at $KL = 7.5$ nats in software, and similarly $0.46 \rightarrow 0.59$ in beverages, with most of the rise concentrated in the moderate range and the curve flattening at the extremes (Figure 3; beverage and pooled versions live in `paper/results/per_filing_pros_vs_retr_{beverages,all}.png`). At the firm-mean level the same outcomes against the same components trace inverted-U arcs in software (peak survival and peak public-listing both at firm-mean $KL \approx 5$ nats; tails decline in both directions); beverages produces a hump on survival and a monotone decline on public-listing (high-novelty beverage firms are systematically less likely to be public — they are craft brewers, not Coca-Cola). The firm-level views are in `paper/results/pros_vs_retr_curves_{software,beverages,all}.png`.

The sharp implication. Because the components are nearly collinear, the signal in their *difference* $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$ is a small share of either component’s variance — but it is the only part of the joint $(KL_{\text{pros}}, KL_{\text{retr}})$ score that does not cancel out under the shared- P_i structure. A filing whose language was more unusual against the past than against the near future is one whose language the near future ended up resembling. The shared “how unusual was the language at all” term cancels; what remains is “did the unusualness fade because others adopted it.” The rest of the paper takes that residual — the imitated-innovator score — as the primary axis. Software and beverages stay as our running examples through Outcomes 1–3; the back matter establishes that the basic story holds across the other 43 NICE classes.

5 Face validity

Figure 4 plots each filing in the prospective–retrospective KL plane. The horizontal axis (prospective KL) measures dissimilarity from filings in the previous five years (high $\Rightarrow$ less like the past); the vertical axis (retrospective KL) measures dissimilarity from filings in the next five years (high $\Rightarrow$ less like the future). Filings *below* the diagonal $KL_{\text{pros}} = KL_{\text{retr}}$ are innovators by our criterion — the past does a worse job of predicting them than the future does. Filings above the diagonal are tired. The labelled marks include expectation-confirming category-creators (OPENAI 2016, CLAUDE 2023, INSTAGRAM 2011, WHITE CLAW SELTZER WORKS 2016, ROCKSTAR 2002) and expectation-disconfirming late-cycle marks (HARD MTN DEW 2021).

The bottled-water category illustrates a useful failure mode. The 2019 LIQUID DEATH filing for canned water describes its goods as *still water*; *sparkling water* — the same lexicon as established bottled-water brands — and so lands near the diagonal. The novelty in that filing is in branding (a

metal can, an aggressive identity, a cannibalistic relationship to soft drinks), not in the goods/services description. Comparable bottled-water filings (SARATOGA, SMARTWATER, ESSENTIA) cluster in the same region. The method is, by construction, blind to non-textual novelty.

6 Canonical winners and failed firms as KL profiles

A direct face-validity check: take a curated list of widely-recognised firms from the last fifty years, find their first post-founding trademark filing, compute the firm-mean ΔKL across all their filings, and compare against a control list of widely-recognised *failed* firms. If the construct is meaningful, the winners should read as forward-flux (positive ΔKL) and the failures should spread out — some failed despite being trend-setters, some failed *because* they were exploiting a receding vocabulary.

Caveat on entity matching. The trademark XML records each filing’s owner only as a free-text `owner_name` string with no entity identifier, so any “firm $\rightarrow$ all its filings” join is necessarily heuristic. For the table below we use hand-curated regular expressions that require the brand string to co-occur with a corporate suffix (`Inc.`, `LLC`, `Corporation`, `Technologies`, etc.) and we restrict each firm’s panel to filings on or after its founding year. The regexes are not perfect: we are confident that at least some hits are entity collisions and we have tightened individual regexes where we identified collisions. Amazon’s regex, for example, matches only the `Amazon.com` variants because the string “Amazon Technologies, Inc.” was used as a registered owner by a different firm in 1994–1995, before Amazon’s IP-holding subsidiary of that name became active around 2001; we lose some post-2001 IP-holder filings but eliminate the collision. Readers should treat the firm-mean numbers below as indicative rather than authoritative; the qualitative pattern (winners cluster on positive firm-mean ΔKL ; failed firms split across both extremes) is robust to the residual collision rate but specific decimal places are not.

37 of 41 winners have *positive* firm-mean ΔKL (the four exceptions are Adobe (−0.03), Qualcomm (−0.01), Etsy (−0.01), and Wayfair (−0.01)). The winners are not extreme outliers on distinctiveness: their firm-mean prospective KL is mostly 3.4–4.8 nats, around or just below the corpus median. They use mainstream vocabulary, but the mainstream vocabulary they use is consistently the kind the field migrates toward, not away from.

First-real-filing ΔKL is sharper on a subset. Thirteen canonical winners had first-filing ΔKL above +0.4 nats (roughly 2σ): **Oracle** (+4.39, 1977), **Intel** (+3.26, 1968), **Microsoft** (+2.99, 1975), **eBay** (+1.73, 1997), **Apple** (+1.68, 1976), **Adobe** (+1.28, 1982), **Coinbase** (+0.98, 2013), **Lyft** (+0.76, 2012), **Twitter** (+0.75, 2007), **Uber** (+0.70, 2009), **AMD** (+0.69, 1969), **LinkedIn** (+0.58, 2003), and **Salesforce** (+0.43, 1999). Oracle’s first filing (+4.39) is the largest first-filing ΔKL in the curated set; the eBay filing’s text “on-line trading services in which seller posts items to be auctioned and bidding is done electronically” (+1.73) introduced phrasing that downstream filings extensively echoed.

The failed-firm panel does not collapse to negative ΔKL . Among 17 widely-recognised failures, the split is roughly even. **Webvan** (−0.22), **MoviePass** (−0.27), **Better.com** (−0.12), and **Pets.com** (−0.03) failed on the receding side. But **Theranos** (+0.05), **Beyond Meat** (+0.34, first +0.74), **Casper** (+0.06, first +0.46), **Solyndra** (+0.44), **Nikola** (+0.03, first +0.24), and **Myspace** (+0.07) failed despite trend-setting vocabulary. ΔKL is a vocabulary signal, not a business-model signal. *Negative ΔKL raises the probability of being on the wrong side of an industry transition. Positive ΔKL raises the probability of being on the right side. Neither is destiny.*

7 Pioneers, fast-followers, and the late wave

To take “where in the adoption sequence do firms succeed?” as an empirical question, for each of five category-defining buzzwords — **website**, **wireless**, **online retail**, **social media**, **blockchain** — we ranked every firm in the corpus by the date of its first filing whose goods/services text contained the buzzword. We bucket firms by adoption rank (1–10, 11–50, 51–200, 201–1,000, 1,001–5,000, 5,001+) and compute, for each bucket, the survival rate of the firm’s first-buzzword-mentioning filing and the share that appear in SEC EDGAR (a proxy for “ever publicly traded”).

Pioneer position is bad; the fast-follower band is the sweet spot. For **wireless**, the first 10 adopters had *zero* survival of their initial filing; the first 50 had 0–13%. For **online retail**, the first 10 survived at 10% vs. $\sim 33\%$ for the late-wave 5,000+ adopters. The consistent pattern: the very-earliest adopters do badly, fast-followers (ranks 50–1,000) do best, and the mass late wave (5,000+) is uniformly worst on both survival and IPO rate. Among public firms in each bucket, the rank-50–1,000 band has 1.5–4 $\times$ the late-entrant baseline IPO rate.

Pioneer-ladder identifies vocabulary prescience, not firm-builder prescience. Of 150 named top-30 pioneers across the five buzzwords, only two had SEC-listed status: **Sony** (rank 23 on **wireless**, 1960 filing) and **PC Connection** (rank 20 on **online retail**, 1996). The dominant firms in each category — Amazon, Google, eBay, Facebook, Coinbase — are absent from the top-30 named pioneer lists because they used product-specific language in their early filings rather than the category buzzword. Pioneer-ladder rank identifies *who claimed the category-defining vocabulary first*, not *who built the dominant firm in the category*.

8 Vanishing trends, emerging trends, and the squeeze in between

A token-level interrogation of ΔKL : take the top-decile-negative (“harvest”), middle-bin ($|\Delta KL| < 0.02$, “aligned”), and top-decile-positive (“innovate”) strata of the clean panel, ask which unigrams are over-represented in each relative to the corpus baseline, and for each surfaced token compute the survival rate of *all* filings containing the token. The latter is the right test of whether the U-shape on survival reflects a real difference in firm outcomes by stratum.

The negative- ΔKL extreme picks out *receding technologies and categorisations of the past*: microfiche, minicomputers, dial-up, pager, store-and-forward; the pre-Sephora apparel categorisation mens, womens, pomade, sunless; the pre-craft-beer descriptors lager, stout, porter, shandy; the pre-fintech insurance vocabulary annuity, casualty, underwriting, securitization. The positive- ΔKL extreme picks out the *legalisation+platform-shift wave*: NFT/Web3 vocabulary (non-fungible, nfts, blockchain-based, authenticated, cryptographic) post-2020, crypto (cryptocurrency, blockchain, decentralized) post-2014, cannabis vocabulary (cannabis, cannabidiol, cbd, thc, tetrahydrocannabinol) post-2014, vape (e-liquid, cartomizers). The neutral middle picks out bureaucratic generic language: producer, outsource, conformity, organisational, freighting, intermediation, systematization.

The survival-rate column delivers the test cleanly. The negative- ΔKL column has tokens surviving mostly 50–80%, above corpus baseline of $\approx 49\%$. The positive- ΔKL column similarly has elevated survival (cannabidiol filings: 82%; cbd: 74%; tetrahydrocannabinol: 70%). The neutral middle is uniformly *below* baseline: organisational at 15%, conformity at 20%, freighting at 21%. Token-level survival rates therefore corroborate the U on ΔKL at the right scale of analysis —

not just the binned plot, not just the regression, but the survival rate of *the actual filings* using those tokens.

Concrete vanishing-trend examples. The most-negative- ΔKL filings in the pre-2000 era include MONOTYPE ITC (1992, $\Delta KL = -2.08$, “typefaces, typefonts, and type designs of alphanumeric characters and/or typographical symbols recorded on”) and H. BERTHOLD AKTIENGESSELLSCHAFT (1991, -1.99). The desktop-publishing transition was already destroying the vocabulary of mechanical-and-photo typesetting in real time. In the 2000s, ΔKL picks out the pre-platform hospitality bundle “Hotel, motel and resort hotel services; restaurant and bar services; making hotel reservations for others” — the language disintermediated by Expedia, Booking.com, and Airbnb. In the 2010s, the bottom is dominated by the e-cigarette boom: nearly every extreme-negative- ΔKL filing in 2014–2015 is “Electronic cigarette liquid (e-liquid) comprised of flavorings in liquid form,” a template the 2016 FDA Deeming Rule and 2020 Premarket Tobacco Application deadline restructured out of the corpus.

9 Serial-pioneer firms across multiple buzzword waves

Generalising the pioneer-ladder, we ask which owners appear in the rank- ≤ 1000 early-adopter pool of *more than one* buzzword. Ten buzzword pools of 1,000 firms each (the five above plus `cloud computing`, `artificial intelligence`, `mobile app`, `streaming`, `saas`). Owners appearing in ≥ 3 pools are “serial pioneers” in the trademark-vocabulary sense.

24 owners qualify. The single most prolific is **IBM** ($k = 6$ pools): AI 1994 (rank 122), wireless, website 1996 (rank 72), cloud computing 2009 (rank 37), social media 2009 (rank 285), saas 2010 (rank 533). The next-most-prolific are **International Data Group** ($k = 4$), **AT&T** ($k = 4$), and **Infosys** ($k = 4$). **Microsoft**, **Cisco**, **Apple**, **Sony**, **HP**, **Citrix**, **Lotus**, and **Avaya** each sit at $k = 3$.

Individual humans almost never appear. Filtering to owner names that look like individuals, only two real humans survive: **Paul Farkas** ($k = 3$: mobile app rank 51, AI rank 651, blockchain rank 510; 81 personal filings since 2011) and **Burhan U. Ahmed** ($k = 3$, on a single 2010 filing that was early on social media, cloud computing, and saas simultaneously). The absence is partly a data-quality limitation — the trademark XML records the firm as owner, not the founder behind the firm — but the qualitative point is sharp: corporate R&D pipelines (IBM Research, AT&T Bell Labs, HP Labs) carry the bulk of cross-buzzword early positioning in the trademark record.

10 Industry time series

Figures 6 and 7 show the per-year median prospective KL, retrospective KL, and ΔKL in each industry, with inter-quartile shading. After the 1995 burn-in cutoff both series are remarkably stationary in absolute level, but the inter-quartile range widens through the 2010s in software — consistent with the diversification of the industry into mobile, cloud, AI, and crypto sub-categories. Beverages show a similar but smaller widening.

11 Outcome 1: Trademark survival

We derive three binary outcomes per filing from the USPTO status code. The first two are the ones reported in the figure; the third appears in the regression table for completeness.

- **Completed registration:** status code begins 6 or 8. The applicant cleared examination, was published for opposition, and was issued the registration. Filing-to- registration lag is typically nine to twenty-four months.
- **Renewed 5-year registration:** completed registration, is at least five years old, and was not cancelled with status code 686 (Section-8 maintenance failure). Strict subset of completed-registration within any single filing.
- **Renewed 10-year registration:** renewed-5y, is at least ten years old, and was not cancelled with status code 688 (Section-9 renewal failure). Strict subset of renewed-5y.

How trademark maintenance interacts with these outcomes. A registered trademark stays in the live-registered status (6xx) until either its owner actively surrenders it or one of two periodic maintenance events comes due and the owner fails to file the appropriate paperwork. The first event is the Section 8 Declaration of Use, due between the fifth and sixth year after registration; failure to file it triggers a status-686 cancellation. The second is the Section 9 Renewal at year ten, with subsequent renewals every ten years. A firm that goes out of business in, say, year three of its trademark’s life will leave the trademark in 6xx status for another two-to-three years until Section 8 comes due and is missed. The 5-year survival outcome thus picks up the first deliberate “does this firm still care about this mark?” gate; we apply a cohort cap of filed ≤ 2020 throughout, so every clean-panel filing has had time for the Section 8 event to have played out. The 10-year outcome has not had time to play out for many filings under that cap and so we report it only with the caveat that the underlying Section 9 cancellation code is rare enough to make it nearly indistinguishable from the 5-year measure.

We estimate generalized linear logit models of the form

$$\Pr(\text{outcome}_i = 1) = \text{logit}^{-1}(\beta_0 + \beta_1 \cdot z_i + \gamma_1 \log n_i + \gamma_2(t_i - \bar{t}) + \gamma_3(t_i - \bar{t})^2)$$

where z_i is a z -scored predictor (either ΔKL alone or KL^{pros} and KL^{retr} jointly), n_i is the count of distinct dictionary terms in filing i , and t_i is the filing year. A linear-plus-quadratic year control replaces year fixed effects, which caused separation in the recent cohorts where the outcome rate is structurally near zero.

ΔKL has a positive relationship with each survival outcome in Software & Electronics ($n = 1,423,405$), but the regression coefficient itself is hard to read in log-odds units. The percentile-based translation is more transparent. Within Class 9, we sort filings by ΔKL into deciles. The bottom decile of ΔKL (least innovative) reaches registration 48.4% of the time and survives ten years 35.7% of the time. The top decile (most innovative) reaches registration 55.5% of the time and survives ten years 41.1% of the time. **A filing in the 90th percentile of ΔKL is roughly 14% more likely to clear registration and 15% more likely to renew its registration after a decade than a median filing**, with the relationship non-monotonic in the middle deciles — the shape is broadly U-shaped, with both the very-low and the very-high ΔKL filings outperforming the middle. We test the U-shape formally in Section 17.

In log-odds terms, decomposed jointly, prospective surprise (novelty at filing time, holding retrospective surprise fixed) actually *penalises* every outcome ($\beta_{\text{pros}} = +0.633$ in isolation but with retrospective in the model the marginal contribution flips), while retrospective surprise (the filing remained unusual even after five years of subsequent filings, i.e. a filing nobody imitated) reverses sign ($\beta_{\text{retr}} = -0.662$). Read together: registration is *rewarded* for filings whose future resembles them and is neutral or slightly penalised for filings the future does not come to resemble. Beer & Soft Drinks shows the same sign pattern at smaller magnitude.

What kind of failure is this? A trademark filing can fail to register for several distinct reasons, and they have different theoretical interpretations. Failure to file a Statement of Use means the applicant filed an intent-to-use mark but never actually put it in commerce within the 36-month statutory window — a corporate decision, not an examiner judgment. Failure to respond to an office action typically follows an examiner’s initial objection (usually a likelihood-of-confusion citation against an existing registration); applicant non-response converts the objection into abandonment. Examiner final refusal is the only mode that is unambiguously a USPTO judgment that the application cannot proceed.

Table 6 decomposes the 50.4% of Class 9 filings that did not reach registration. Two facts dominate. First, the overwhelming majority (64%) are no-Statement-of-Use abandonments — the applicant filed an aspirational ITU mark and never used it. Second, examiner final refusals account for only 8% of failures. Across every failure category the mean ΔKL is within 0.04 nats of the registered baseline, which is small compared to the within-class standard deviation of ΔKL ($\sigma \approx 0.46$). Novel filings are not, in any meaningful sense, being rejected.

Figures 8 and 9 display the per-bin registration-and-renewal rates against (left) prospective KL, (centre) retrospective KL, and (right) ΔKL . Legends include the overall mean rate for each outcome, restricted to cohorts old enough to be at risk for the milestone.

Robustness across outcome definitions. Figure 10 pools all 45 classes ($n = 10,062,338$ clean H=2y filings, 1985–2021 cohorts) and reports the same KL-binned rate under three increasingly strict outcome definitions: $P(\text{reached registration})$, $P(\text{survived 5y})$, and $P(\text{owner ever in SEC EDGAR})$. The pattern is informative: **on ΔKL the survival curve is U-shaped under every outcome definition** — the dip in the middle and the lift at both extremes is not an artefact of any single survival operationalisation. On the prospective and retrospective KL components alone the curve is monotone-increasing on examiner approval (more-distinctive marks collide less with prior art and clear examination more easily under 15 U.S.C. § 1052(d)) and becomes U-shaped only after the strict “currently live” criterion is added; the ΔKL flux signal is the only one stable across both regimes.

12 Outcome 2: Financial success

The trademark survival results in Section 11 establish that the innovation axis is meaningful at the per-filing level, but they say nothing about whether innovation pays the firm a financial return. To test that, we link USPTO owner names to SEC EDGAR firm names entirely automatically; **no entity match in this paper was reviewed by hand**. The procedure has two passes.

The first pass aggressively normalises both sides — uppercase, strip punctuation, drop trailing legal-entity suffixes such as INC, LLC, CORP, HOLDINGS, LIMITED, COMPANY, GMBH, S.A., KK, etc. (the full list is in `scripts/sec_link.py`) — and accepts only exact matches on the normalised key. The second pass uses fuzzy token-sort matching (`rapidfuzz` with a score cutoff of 92 out of 100) to capture the long tail of near-variants for USPTO owners with at least 10 filings. The combined crosswalk recovers 14,280 USPTO owners (9,370 exact + 4,910 fuzzy) covering 627,349 trademark filings (3.8% of the all-class corpus by filing count, but a much larger share by economic weight, because matched owners are by definition publicly-traded firms with material financial reporting). After joining to SEC firm-year financials and restricting to firm-years with non-degenerate gross margin and at least one trademark filing in the prior three years, the analysis panel contains 5,326 firm-years across 1,597 unique CIKs over fiscal years 2009–2024.

The fully-automated procedure means the crosswalk has both false positives (e.g. two unrelated

firms with similar normalised names) and false negatives (e.g. a public firm filing trademarks under an unusual subsidiary name that neither pass matches). The published `data_publish/comparators/external_crosswalk.csv` file records every match decision so any reader who wants to override individual entries can do so without re-running the pipeline.

Filing volume is itself an innovation marker, but also a size marker. A firm filing many trademarks in a short window may be innovating across product lines — but it may also simply be a large firm with a large legal department. We separate the two by including the count of filings as a third predictor alongside the novelty score itself, and by controlling for log revenue throughout. Both effects turn out to be present and partially independent.

For each firm-year (c, y) we compute the trailing 3-year mean of the firm’s filing-level scores: $\overline{\Delta KL}_{c,y}$, the prospective and retrospective components, the maximum within-window ΔKL (the firm’s most-novel filing in the window), and the count of filings. We then estimate

$$y_{c,t} = \beta_0 + \beta_1 \cdot z(\text{novelty})_{c,t} + \gamma_1 \log \text{Revenue}_{c,t} + \gamma_2 \text{R\&D intensity}_{c,t} + \text{year controls} + \epsilon_{c,t}$$

with cluster-robust standard errors at the firm level. Outcomes are gross margin, net margin, and operating margin.

The pattern from Section 11 reproduces here at the firm-year level. A one-standard-deviation increase in 3-year trailing mean ΔKL — that is, a firm whose recent filings introduced wording that was unusual at the time but came to look ordinary in the years that followed — raises gross margin by +3.2 percentage points (SE 0.6). Decomposed jointly, prospective surprise (novel against the past) raises gross margin by +14.6 pp per standard deviation (SE 4.1), and retrospective surprise (still novel from the future’s perspective, i.e. a filing nobody imitated) lowers it by −19.1 pp per standard deviation (SE 4.1). The within-firm story is the same as the within-filing story: firms whose recent filings the future ends up resembling enjoy higher gross margins; firms whose recent filings nobody imitates do not. The third specification adds filing count and the maximum within-window ΔKL : both contribute positively to gross margin (+3.3 and +1.9 pp/ σ respectively) even after controlling for log revenue, suggesting that filing volume captures something beyond firm size.

Variance. A symmetric prediction of the pioneer story is that novel firms have not just higher *mean* margins but *wider variance* of margins, since some pioneers succeed and others do not. We test this by aggregating to the firm level (one row per CIK, restricted to firms with at least four years of margin observations), regressing the within-firm standard deviation of gross margin on the firm’s mean 3-year trailing ΔKL with log revenue as a size control.

A one-standard-deviation increase in firm-mean ΔKL raises the within-firm gross-margin standard deviation by +0.018 in margin units (SE 0.004) on 487 firms. The pioneer-bet effect is real and quantitatively meaningful: high-novelty firms simultaneously earn higher mean margins (Table 7) and exhibit wider margin spread (Table 8).

A coarser check: does ΔKL predict reaching SEC EDGAR at all? A firm that files an income statement with the SEC has cleared a much lower bar of success than “high gross margin” — it has gone public, registered securities under the ’33 Act, or otherwise reached a reporting threshold. We can therefore re-cast the financial-outcome question as a binary one: among trademark-filing firms, does ΔKL predict the probability of *ever* appearing in our SEC crosswalk? Because the crosswalk’s fuzzy-match pass requires owners with ≥ 10 filings, we restrict this analysis to the 160,643 crosswalk-eligible firms with ≥ 10 clean filings; 7,810 of them (4.86%) match into SEC EDGAR.

Table 9 shows the result: ΔKL has essentially no relationship with the public-listing rate. $P(\text{ever-public})$ ranges from 5.11% in the most-tired quintile (Q1) to 4.44% in the most-innovative quintile (Q5), with the middle three quintiles all near 5%. A logistic regression of `ever_public` on $z(\Delta KL)$ and $\log_{10}(\text{n.filings})$ confirms this pattern: the ΔKL coefficient is -0.019 with SE 0.013 ($p = 0.14$, indistinguishable from zero), while $\log_{10}(\text{n.filings})$ carries a coefficient of $+1.578$ (SE 0.030 , $p < 10^{-9}$). A $10\times$ increase in filing count multiplies the odds of going public by ~ 4.85 ; a one-SD increase in mean ΔKL multiplies them by ~ 0.98 .

This is informative precisely because it is null. The within-EDGAR margin results above show that novel firms enjoy higher gross margins *conditional on being public*; the present check shows that the selection into being public is governed by trademarking volume rather than trademarking character. Read together, the two findings mean: becoming a reportable firm is a separate (volume-driven) hurdle; *conditional* on clearing it, novelty earns a margin premium.

Conditional on public listing: trademark count predicts revenue but not margin. A natural follow-up: among the firms that did IPO, does trademark count correlate with firm value? The patent-economics literature finds that count and value move together — prolific patenters have valuable patents (Trajtenberg 1990; Hall, Jaffe & Trajtenberg 2005, *RAND J. Econ.*) — and we wondered whether the opposite might hold for trademarks, where a smaller portfolio could be evidence of operational focus. On the SEC-crosswalked panel ($n = 6,456$ CIKs with clean-panel trademark filings and at least one observed year of SEC financials), the correlation matrix is unambiguous: trademark count is *strongly positively* correlated with firm size and *essentially uncorrelated* with margin (Table 10, Figure 11). A firm in the top mark-count decile (median 125–5,697 marks) has roughly $300\times$ the median revenue of a firm in the bottom decile (median 1–2 marks), but its mean gross margin is essentially identical. The focus hypothesis is rejected on size, absent on profitability, and weakly supported only on R&D intensity (firms with many marks tend to file less R&D against their revenue — a different culture, not a strategic focus signal).

A causal-attribution caveat. The trailing-3-year predictor pools many filings per firm-year, and a firm with several novel filings in its recent past is also a firm whose *earlier* filings might have been the cause of its subsequent financial success — that is, the pooled regression cannot distinguish “current novelty pays off” from “past success funds the current novelty habit.” We address this directly in Section 13 by restricting the predictor to each firm’s first-ever filing.

13 Outcome 3: Debut-filing-only test

The pooled per-filing logit in Section 11 treats each of a firm’s filings as an independent observation, and the firm-year panel in Section 12 averages each firm’s recent trademark portfolio. Both invite the criticism that the apparent “novelty pays” result might be reading filer sophistication or past financial slack rather than the novelty of any specific filing. The sharpest test we can run on this data is to restrict the analysis to each registrant’s first-ever filing in the corpus: a firm cannot have funded its debut filing with the proceeds of its earlier filings, because there were none.

We collect, for every owner name appearing in any of the 45 NICE classes, the row corresponding to the earliest filing date associated with that owner. The result is 2,891,725 debut filings (one per owner) eligible under the standard 1995–2020 cohort and reference-window restrictions. Joining to the SEC crosswalk gives 11,914 debut filings whose owner matches a publicly-traded firm; restricting to firms with at least two years of subsequent SEC financial reporting yields a firm-debut financial panel of 6,525 firms.

Outcome	N	β for ΔKL (z -scored)
<i>Survival on first filings (one per owner across all classes)</i>		
Completed registration	2,891,725	-0.034 (0.001)
Renewed 5y registration	2,891,725	-0.005 (0.001)
Renewed 10y registration	1,875,964	+0.042 (0.002)
<i>Firm-debut financial regression (N=6,525 matched public firms)</i>		
Mean future gross margin	6,525	+0.017 (0.003)
Within-firm gross-margin std.	5,008	+0.0052 (0.0015)

The pattern is consistent and informative.

Survival. On debut filings, the ΔKL coefficient on completing registration shrinks from the pooled value of +0.124 to -0.034 (still highly significant on 2.9M observations). A debut filer who introduces unusually novel goods/services language appears to face slightly more registration friction — the opposite sign of the pooled effect. The pooled positive coefficient was reading filer sophistication: firms with deep filing portfolios know how to write a description that survives examination, and they also tend to file more novel marks. Among first-time filers neither advantage is present. Renewal coefficients are smaller in magnitude but stay positive (+0.042 at the 10y mark), suggesting that long-run persistence still favours novelty among debut filings.

Are debut filings less novel than later filings? The negative debut-registration coefficient invites a natural worry: if first filings were systematically less novel, then perhaps the apparent “debut novelty pays” result is an artefact of restricted variance. This is not what the data show. On the same all-class clean panel, mean ΔKL on debut filings is +0.052 (median +0.008), versus +0.010 (median -0.010) on non-debut filings; the share of filings with positive ΔKL is 51.5% on debuts and 47.6% on non-debuts. Both component Ks are larger on debuts (prospective 5.22 vs. 4.91, retrospective 5.17 vs. 4.90): debut filings read as less like the recent past *and* less like the near future. The intuition is that a firm’s debut is often a stake in genuinely new territory, while its later filings drift toward category-conventional language. So the negative registration coefficient on debuts is not explained by lower novelty in the debut set — if anything debuts are slightly more novel; it is explained by examination friction. A first-time filer with unusual goods/services language draws office actions and lacks the procedural sophistication to navigate them, while an established repeat filer with the same novel language clears the process routinely. The pooled positive coefficient was reading the established-firm advantage; stripping it leaves a small examination- friction penalty for first-time filers writing in non-standard ways.

Within firms, the debut filing sits roughly at the firm’s own median novelty (mean within-firm percentile 0.514 on 432,180 firms with ≥ 5 clean filings), but the distribution is bimodal: 24.4% of firms file their debut in the bottom 20% of their own ΔKL distribution (cautious entrants), and 30.4% file their debut in the top 20% (bold entrants); the middle three quintiles each account for only 14–16%. Few firms enter at mid-novelty. And among multi-filing firms, 53% of *non-debut* filings have $\Delta KL < 0$ — a slight tilt toward line-extension language, although the per-firm mix is heterogeneous: about a fifth of firms are persistently novel (at most 20% of their non-debuts in tired territory) and another quarter are persistently tired ($\geq 80\%$).

Financial. On the firm-debut panel, a one-standard-deviation increase in the firm’s debut-filing ΔKL is associated with +1.7 **percentage points** higher mean future gross margin (SE 0.3, $n = 6,525$ firms) and +0.5 percentage points higher within-firm margin standard deviation (SE 0.15, $n = 5,008$ firms). Both effects are roughly half the magnitude of the pooled trailing-3-year results in Section 12. We take the attenuation as evidence that part of the original firm-year reward

reflects the habit/portfolio confound flagged in the previous section, but that a substantive debut effect remains: the novelty of a firm’s first trademark filing predicts its subsequent average financial performance, even before the firm has produced any other trademark filings to “fund” the bet.

Debut ΔKL predicts going public; line-extension filings reverse the appearance

At the firm-debut level, positive ΔKL raises the probability of ever going public. Restricting to each owner’s first-ever filing across all classes ($n = 5,090,053$ in the clean H=2y panel), $P(\text{owner ever in SEC EDGAR})$ by debut ΔKL bin is *monotone increasing*: 0.42% at the bottom decile, 0.47% in the middle, 0.64% at the top decile — a +52% relative lift for positive- ΔKL debut filings over negative- ΔKL debut filings (Figure 13 panel C). On the firm’s first move, having a vocabulary that the field subsequently echoes predicts becoming a publicly-reported firm.

The same regression at the full-corpus filing level shows the *opposite* direction, and a reader who saw only that result would draw the wrong conclusion. At the filing level, $P(\text{owner ever in SEC EDGAR})$ *decreases* monotonically with ΔKL : the bottom decile is 5.8% public, the top decile 3.5%. The reversal is mechanical. Public-firm owners — incumbents like Apple, Coca-Cola, Procter & Gamble, Pfizer — file many trademarks per year, and the bulk of an incumbent’s portfolio is line extensions that reuse the firm’s established (low- ΔKL) vocabulary. Their volume swamps the per-filing statistic and concentrates the public-firm mass in the low- ΔKL bin. The right unit of analysis for the “do innovative firms reach the public market?” question is the firm’s debut filing; the full-corpus filing-level statistic is dominated by post-IPO line-extension activity.

For prospective and retrospective KL on debut filings, $P(\text{public})$ is *monotone decreasing*: the very-distinctive-on-day-one debut firms have lower IPO rates (0.31%–0.34% at the top decile) than the average-distinctiveness debut firms (0.51% overall baseline). Niche on day one tends to mean niche at maturity; the IPO sweet spot for debut firms is ordinary vocabulary used in a forward-drifting direction — low PC1, high PC2 in the decomposition.

14 Within-firm dynamics and debut quartiles

The first-filing analysis in Section 13 answered “does the firm’s first novel bet predict outcomes,” but said nothing about how a firm’s later filings behave once the firm is established. The full corpus contains 1.24 million firms with at least two filings and 5.74 million consecutive filing pairs, which lets us measure two related quantities: how persistent within-firm novelty is, and whether firms become more or less innovative as they file more.

Within-firm autocorrelation. For every consecutive filing-pair within a firm we compute the Pearson correlation between the prospective KL of the later filing and that of the earlier one, and the same for ΔKL :

The autocorrelations are large. Knowing that a firm’s previous trademark filing scored high on prospective surprise (was unusual at the time) raises our expectation of high prospective surprise on the firm’s next filing by $\rho = 0.68$. The same correlation on the imitated-innovator score ΔKL is $\rho = 0.55$. Surprising firms keep filing surprising trademarks; innovative firms keep filing innovative trademarks.

Firms drift very slightly toward conservatism. Restricting to firms with at least four filings (so a per-firm trend can be estimated), the median per-firm slope of ΔKL on filing-index is 0.0000 and the mean is -0.0007 per additional filing. 47.0% of firms have a positive slope (binomial

$z = -19.6$ vs. the null of 50/50). Firms become marginally less innovative as they accumulate filings, but the magnitude is tiny — on the order of one tenth of a percent of $\sigma_{\Delta KL}$ per additional filing.

Debut quartiles. We split firms into four quartiles by the ΔKL of their first-ever filing and tabulate subsequent behaviour within each.

Three patterns are visible. (1) All four quartiles regress toward the mean: the most-innovative quartile (Q4) starts at debut $\Delta KL = +0.336$ but its average subsequent filing scores only $+0.106$, and the least-innovative quartile (Q1) starts at -0.309 but recovers to -0.098 . (2) The gap between innovative- and conservative-debut firms persists: Q4 firms remain $+0.20$ nats more innovative on subsequent filings than Q1 firms. (3) The strongest within-firm AR(1) on ΔKL is in Q4 ($+0.572$); innovative firms are the most *consistent* in their innovation, even after partial regression. Conservative-debut firms have weaker but still substantial AR(1) ($+0.49$), suggesting innovation profile is established at the debut bet and tends to persist.

15 Stock-market returns

If novelty in trademarks predicts firm-level financial outcomes, the strongest test is whether it predicts subsequent stock-market returns above a market benchmark. We pulled monthly adjusted-close prices from Yahoo Finance for every CIK in the USPTO–SEC crosswalk that maps to a current ticker (4,254 matched CIKs, 0.86M monthly price points, plus the S&P 500 as benchmark), computed end-of-year prices, and constructed cumulative total returns at one through five-year horizons after each filing year. We then regressed those returns (winsorised at the 1st and 99th percentile to suppress penny-stock-spike outliers; cluster-robust standard errors at the firm level) on the firm’s ΔKL in two specifications: firm-year mean across that year’s filings, and debut-only.

The pattern is consistent with the gross-margin result of Section 12 and considerably stronger. Per one-standard-deviation increase in firm-year mean ΔKL , excess return over the S&P 500 is $+1.5$ pp at one year, peaks at $+5.0$ pp at three years, and decays back to the market by year five. Per one-standard-deviation increase in debut-filing ΔKL (one observation per firm), excess return is $+3.6$ pp at one year, $+19.2$ pp at three years, and peaks at $+28.4$ pp at four years before partially mean-reverting to $+14.9$ pp at five years. Both specifications are statistically significant at the one-year horizon and grow more so at intermediate horizons. The debut specification eliminates the past-success-funds-current-novelty confound and still produces the largest excess-return coefficient in the paper. The five-year mean-reversion is consistent with a story in which innovative firms enjoy a multi-year “imitated category leader” window before competition catches up.

Surprise alone does not predict returns. A natural variant of the regression uses prospective KL alone as the predictor (rather than the ΔKL composite). Doing so eliminates the result entirely: at the firm-year level the prospective-only coefficient is essentially zero at every horizon ($+0.001$ at 1y, -0.001 at 5y); at the debut level it is also zero with wider standard errors ($+0.001$ at 1y, $+0.057$ at 5y, both indistinguishable from null). **Mere novelty at filing time does not predict subsequent stock returns. Only novelty that the future ends up imitating does.** The retrospective component is itself a forward-looking signal of which novel filings turned out to define a category, and the predictive content of ΔKL for returns sits in that selection.

Cumulative excess return by debut ΔKL quartile. A clean visual summary of the result is to plot the mean cumulative excess return over the S&P 500 by years-since-debut for each of the four debut- ΔKL quartiles (Figure 14). The most-novel debut quartile (Q4) outperforms the

least-novel quartile (Q1) by approximately twenty percentage points in cumulative excess return by year five, with the gap opening continuously across the first five years.

Per-industry correlations. The firm-year and debut results pool firms across all industries. We can instead compute the within-industry Pearson correlation between firm-mean ΔKL and the firm’s mean annual excess return over the S&P 500 (using only firms with at least three years of return observations). Table 14 reports these correlations for the twenty largest industries by matched-firm count, with two-sided p -values.

The largest industries vary substantially in the strength of the ΔKL –return relationship: it is positive and significant in some (Class 042 Scientific & Tech Services, Class 041 Education & Entertainment) and indistinguishable from zero in others. The sample-size cap means we cannot resolve a stable rank ordering; the takeaway is that the financial reward to trademark novelty is not uniform across industries.

Are these results driven by the FAANG cohort? A natural concern is that the 2010s stock market was dominated by a handful of mega-cap technology firms whose massive capital appreciation could mechanically produce a positive coefficient on almost any sensible novelty proxy. We address this directly by re-running the returns regressions excluding the eight largest tech firms in our matched panel: Apple, Microsoft, Amazon, Meta (Facebook), Alphabet (Google), Netflix, NVIDIA, and Tesla.

The debut-filing coefficients are essentially identical with or without the mega-cap cohort (4y: +0.2844 vs. +0.2823). The firm-year coefficients attenuate slightly at long horizons — the 5y firm-year coefficient does collapse to near-zero (+0.005 vs. +0.016), telling us that whatever 5y firm-year effect existed was mostly mega-cap — but the central 1–3y story holds. The headline result that debut ΔKL predicts a ~ 28 pp excess return at the 4y horizon is not a FAANG artifact.

Figure 15 plots the per-year mean ΔKL and cumulative price for each of the eight firms.

Convergent validity against patent counts. The dominant existing quantitative innovation proxy is the count of patents an organization is granted per year. We pulled the full PatentsView bulk data (9.36 million U.S. patents, 8.57 million patent-assignee rows from the `g_assignee_disambiguated` table), aggregated to (disambiguated assignee, year) patent counts (1.47 million firm-years), and joined to our trademark ΔKL firm-years on the same normalized-name key used for the SEC crosswalk.

The two measures are essentially uncorrelated, and we treat that as a feature. The Pearson correlation between firm-year mean ΔKL and that firm-year’s patent count is $r = +0.010$ on 174,569 firm-years; using $\log(1 + n_{\text{patents}})$ to suppress mega-patent-filer leverage gives $r = -0.009$. At the firm level (one observation per firm, restricted to firms with at least three years of overlap), $r = +0.008$ for raw sums and $r = -0.023$ for log sums. The largest patent filers in the matched panel are major-electronics-conglomerate incumbents: IBM (379,206 patents over 26 years, mean $\Delta KL = +0.02$), Samsung (358,183, +0.04), Sony (188,074, -0.01), Toshiba, Canon, LG, Intel, Google, Fujitsu, Microsoft. The two innovation measures pick out very different firms. Patents reward organisations that translate R&D into legally defensible technical claims; trademark ΔKL rewards organisations that introduce business-model or product-language novelty that the rest of their industry then adopts. The two measures are *substantively* different innovation channels, and the absence of a correlation is itself the evidence that trademark ΔKL is measuring something not already covered by the dominant innovation proxy.

Patent-count regression on returns. A direct test of which channel matters more for shareholder outcomes is to swap our predictor: replace ΔKL with $\log(1 + n_{\text{patents}})$ on the same returns panel, with the same year controls and cluster-robust SEs at the firm level.

The patent coefficient is positive and growing with horizon: +2.0 pp at one year, +5.9 pp at three, +12.1 pp at five years per σ of $\log(1 + \text{patents})$. The trademark ΔKL firm-year coefficients (Table 13) peak earlier and decay; the trademark debut coefficients are larger in absolute terms (+28 pp at four years) but shrink at five years.

Patents and trademarks index different innovation constructs. The growing-with-horizon pattern on patents is what we would expect if patent counts indexed *continuing* innovation activity: a firm with many recent patents is one whose R&D pipeline is still producing, and the future cash flows of that pipeline are priced in increasingly across two-, three-, and four-year-ahead returns. The trademark ΔKL signal is qualitatively different. The debut-only specification — one observation per firm, keyed on the firm’s first-ever trademark filing — treats ΔKL as a *one-time positional bet*: did the founding team pick a vocabulary the field would migrate toward? The horizon profile of the debut ΔKL coefficient (rising to +28 pp at four years and then shrinking) is consistent with a one-time bet whose realisation takes a few years to be priced and then becomes information the market has already absorbed. The two innovation proxies are not redundant (the R^2 decomposition below confirms sub-additivity) and they answer different questions. Patents proxy *whether the firm is still innovating*; ΔKL on the firm’s debut filing proxies *whether the firm started in fertile linguistic ground*.

Are patents and trademark ΔKL complementary or redundant? An R^2 decomposition. The complement-vs-substitute question is not rhetorical — we can compute the answer. Restricting to the common panel of firm-years in which we observe *both* a patent count and a trademark ΔKL (and the firm appears in our SEC returns data), we run eight nested OLS specifications on the same panel of 7,607 firm-years and report the R^2 of each (controls in every spec are year linear and quadratic). We add a third predictor, $\log(1 + \text{patents})$ (P) and the trademark composite ΔKL (I, for “innovator”), but also *novelty alone* (S, the prospective KL component) so that we can separately ask whether plain filing-time surprise contributes anything.

Three patterns are visible in Table 18.

(1) All R^2 values are small. On a common panel of 7,607 firm-years, controls (year and year-squared) explain between 0.7% and 3.4% of cumulative-excess-return variance. This is what one expects from firm-level returns: most of the variance is idiosyncratic. The contribution of any innovation proxy is necessarily a small slice of an already-small explainable share.

(2) Patents and innovator ΔKL are very nearly additive. Patents alone add +0.06 to +0.23 pp of R^2 above controls (growing with horizon); innovator ΔKL alone adds +0.10 to +0.20 pp (peaking at three years). Their joint contribution P + I is +0.16 to +0.33 pp, essentially the sum of the individual contributions. The two measures share almost no explanatory variance for stock returns; they pick out different firms, and each firm-feature carries independent return information.

(3) Novelty (prospective KL) alone adds essentially nothing. The S-only specification is statistically indistinguishable from controls at every horizon (+0.00 to +0.01 pp), and the marginal R^2 of S in the P+I+S spec relative to P+I is ≤ 0.01 pp at every horizon. The result we documented earlier — that a filing that is merely surprising at the time, without later imitation, does not predict returns — is reproduced cleanly in the R^2 decomposition. The forward-looking component ($-KL_{\text{retr}}$) is what makes ΔKL a return predictor, not the backward-looking surprise itself.

We therefore replace the previous sentence “the two measures appear to be complements” with

the more precise statement: **patents and trademark ΔKL are quantitatively additive on stock returns; their joint R^2 contribution is the sum of the marginal contributions of each in isolation, and novelty alone adds nothing.** A firm’s patent count and a firm’s trademark ΔKL are two substantively different innovation channels and the predictive content of one is not contained in the other.

Four further public comparators. We supplement the patent comparison with four other public innovation proxies. The data and the matching code are released under `data_publish/comparators/`.

The strongest signal among the four is **R&D-to-revenue** ($r = +0.13$ on 1,583 firms), consistent with the intuition that firms that spend more on research file more novel goods/services descriptions. **MIT Technology Review’s 50 Smartest Companies (2015 list)**, of which we matched 17 of 50 to a SEC CIK in our trademark panel, had a mean ΔKL of $+0.052$ versus -0.018 for the rest of the matched universe — a $+0.07$ gap that is borderline-significant on the small sample ($z = +1.63$). **The BCG Most Innovative Companies list (2021–23 union, 80 unique firms, 31 matched)** shows a smaller $+0.02$ on-list-vs-off-list gap that is not distinguishable from zero ($z = +0.64$). **Crunchbase total funding** (using the `notpeter/crunchbase-data` historical mirror, 1,646 matched firms) is essentially uncorrelated with ΔKL ($r = 0.00$ for log funding).

The pattern is the same one we saw with patents: each of these public proxies captures a substantively different innovation construct, and pairwise correlations between proxies sit in the range $\rho \approx 0.0$ – 0.13 . Trademark ΔKL is on the correlated end of that range with R&D and MIT TR’s expert-curated list, on the uncorrelated end with Crunchbase funding totals, and near-zero with patent counts. None of the four can substitute for ΔKL , and ΔKL does not substitute for them.

The four comparator data files are published in `data_publish/comparators/`, alongside the `external_crosswalk.csv` file that records, for every list-based comparator entry, whether and how it matched a SEC CIK in our matched universe. Researchers wishing to replace any of our comparators with a more authoritative source can drop a CSV with the same schema into the same directory and re-run `scripts/external_compare.py`.

What is a trademark really protecting? The convergent-validity result invites a second look at what the trademark filing fundamentally is. A trademark is a legal monopoly over a particular set of words, designs, or sounds in connection with a particular set of goods and services. The economic value of that monopoly comes from how distinctive the goods/services description actually is: a generic mark on a generic product (**Beer** for an ale) is hard to defend and easy for competitors to crowd; a distinctive description for a genuinely new offering (**Hard Seltzer Works** for a new beverage category) keeps later entrants from advertising in the same space without infringing. Read this way, the goods/services text is not just a description of the firm; it is a legal carve-out around the firm’s claim to a particular novelty. A trademark filing high in ΔKL is, in this framing, an attempt to reserve the language of a market offering before the rest of the industry has named it. That the filing later gets imitated — the empirical mark of a high- ΔKL mark — is exactly the substantive event a successful trademark exists to police. The financial reward we measure is, in part, the value of having claimed the boundary of a market category before competitors arrived to share it.

Companion data release. Two CSV files are published alongside the source code as the data contribution of this paper:

- `data_publish/firm_year_dkl.csv` (59,033 rows): for every USPTO trademark owner that we matched to an SEC EDGAR CIK, annual mean prospective KL, retrospective KL, and ΔKL aggregated across all 45 NICE classes the firm filed in.
- `data_publish/firm_year_patents_and_dkl.csv` (174,569 rows): the joint patent-and-trademark panel underlying Table 16, suitable for any follow-up convergent-validity analysis.

16 Templating and survival: do shared descriptions predict failure?

The diagonal-clustering geometry of the topomap suggests that firms filing in the lower-left “common-place” corner — low prospective *and* low retrospective surprise — have the worst post-registration survival. A natural mechanism would be templated copy-paste descriptions: filers using boilerplate that examiners and the marketplace alike learn to discount. We test this directly. For every clean filing, we compute the count of *other* filings sharing the exact same lowercased, punctuation-stripped goods/services text, then bin filings by that multiplicity and report registration and post-registration-survival rates.

The result is the opposite of the templated-failure hypothesis.

Three substantive findings:

1. **More-templated filings survive better, not worse.** Unique-text filings post-register at 48.6% and survive at 38.1%. Filings that share a description with 5,000+ others register at 58.2% and survive at 49.2% — a +11 percentage point survival premium for the most heavily templated tier. The pattern is monotone through the 6–10 multiplicity bin, then plateaus.
2. **The post-registration cancellation rate is mildly flatter for templated filings.** The right-most column of Table 20 shows that, conditional on registering, $\sim 22\%$ of unique-text marks lose their registration (mostly to Section 8 failure at year five), versus $\sim 15\%$ for the 5000+ bin. Both the higher registration rate *and* the higher conditional-survival rate point in the same direction.
3. **Boilerplate signals serious filers.** Inspecting the most-templated normalised G/S strings, the pattern is dominated by commodity professional-services categories: “restaurant services” (52,973 identical filings), “dietary and nutritional supplements” (20,755), “legal services” (16,400), “dietary supplements” (15,982), “alcoholic beverages except beers” (13,286), “banking services” (11,422). These are the descriptions that experienced trademark counsel re-use across clients precisely because they are the safe, examiner-tested phrasings.

Robustness: dropping the $n_{\text{terms}} \geq 3$ filter. The headline analysis applies the full CLEAN filter (Table 1), which excludes 1- and 2-token G/S descriptions because their per-filing KL is degenerate. Many of the most-templated strings are short — “restaurant services”, “wines”, “footwear”, “t-shirts”, “candy”, “cosmetics”, “jewelry” — and so are dropped from the headline panel even when the analysis at hand does not depend on KL at all. To check that the templating-survival pattern is not an artefact of this filter, we re-run the analysis on CLEAN-without- n_{terms} (8,499,455 filings, +591k vs. headline). The pattern is nearly identical for the 1- through 1001–5000 bins (≤ 1 percentage point of drift on either rate). The only meaningful change is the 5000+ bin: still-live rate falls from 49.2% to 44.2%, because the short commodity templates that the headline filter excludes survive at corpus-average rates rather than the elevated rates of the longer structured

templates that dominate the n_{terms} -filtered 5000+ bin (e.g. “athletic apparel namely shirts pants jackets footwear hats and caps athletic uniforms,” 13,047 copies, 61.6% still-live).

So why is the lower-left of the topomap so bad? The diagonal-clustering reading from the topomap survives this audit: the worst-survival cell is firms whose filings cluster on the diagonal at low KL . But the mechanism is not “templated text gets penalised.” Quite the reverse — explicitly templated text helps. What appears to drive the lower-left corner is one-off, generic G/S text that nobody else copies verbatim: descriptions written in the class’s modal vocabulary but unique enough to count as zero-template, filed by firms without the procedural sophistication to use the known-good phrasings their lawyers would otherwise re-use. Boilerplate, when it exists, signals professional repetition; the bottom-left cluster is filled with filers who lack the boilerplate.

17 Both extremes outperform: a U-shape across industries

The decile-rate plot referenced in Section 11 shows that the relationship between ΔKL and survival is not linear; it is U-shaped, with both the most-conservative-debut filings (lowest decile of ΔKL) and the most-innovative filings (highest decile) outperforming the middle deciles. The intuition is that both directions of distance from the modal filing are rewarded: very-novel filings benefit from category creation and the highest decile of ΔKL is dominated by firms that the future imitated; very-conservative filings benefit from being unambiguously defensible and easy for the examiner to parse. Filings in the middle, neither novel enough to claim a new category nor templated enough to coast on legal-clarity grounds, fare the worst.

Quadratic regression. We test the curvature directly. For every industry’s clean filings between 1990 and 2020, we fit a logistic model with both $z(\Delta KL)$ and $z(\Delta KL)^2$ as predictors, with year and year-squared controls. Pooled across all 45 NICE classes and 9.3 million filings, the aggregate quadratic coefficient is positive but small ($\beta_{(\Delta KL)^2}^z = +0.0002$ on 5y survival; $+0.0013$ on 10y survival). The aggregate pooled quadratic is not the primary evidence for the U-shape; the per-industry evidence is. At the per-industry level the quadratic is positive in 27 of the 44 industries we can fit.

The U-shape is present in most — but not all — industries. Fitting the quadratic per industry on the 5y-survival outcome ($n = 44$ industries with at least 5,000 clean filings), the quadratic coefficient on $z(\Delta KL)^2$ is positive in 27 of 44 industries (61%) and is individually significant at $p < 0.05$ in 21 of those 44 (48%). The U-shape dominates in consumer-goods classes (Household Utensils, Furniture, Leather, Textiles, Common Metals, etc.), where heavy templated filing by large incumbents and distinct novelty waves coexist. Several large information-economy classes show the opposite, an *inverted-U*: **Software & Electronics** ($t = -18.2$ on $\beta_{(\Delta KL)^2}$), **Insurance & Financial** ($t = -13.8$), **Scientific & Tech Services** ($t = -12.9$), **Advertising & Retail** ($t = -10.7$), and **Telecommunications** ($t = -10.6$). In these industries ΔKL acts like a linear positive predictor of survival without a harvest tail. Table 22 reports the 15 industries with the strongest U-curvature and the 5 with the strongest inverted-U.

Figure 18 overlays each industry’s per-decile 5y-renewal rate (thin grey lines) on the cross-industry mean (red), making the consistent U-shape visible.

Industries ranked by U-shape strength. Industries differ in how cleanly the U appears. Table 23 ranks all 44 industries with at least 5,000 clean filings by the t -statistic on the quadratic

ΔKL^2 coefficient. Pharmaceuticals (Class 005) tops the list, followed by Materials Treatment (040), Advertising & Retail (035), Chemicals & Industrial (001), and Machines & Power Tools (007). At the bottom of the ranking are Telecommunications (038, $t = -9.8$, statistically inverted-U), Carpets & Wall Hangings (027), Musical Instruments (015), Trim, Lace & Notions (026), Wine & Spirits (033), and Tobacco & Smokers' Articles (034) — in each of these the curvature is absent or slightly inverted.

Industries that lack a U have a stronger linear effect. Among the 29 industries with $t_{\text{quad}} \geq 1.96$, the mean $|\beta_{\text{linear}}|$ is 0.048. Among the 15 industries without a significant U, the mean $|\beta_{\text{linear}}|$ is 0.063 — noticeably higher. These are industries in which ΔKL acts more like a linear (rather than U-shaped) predictor. Telecommunications, the strongest inverted-U, has a positive linear coefficient ($\beta_{\text{lin}} = +0.020$) on top of the negative quadratic; Trim & Notions, Carpets, and Musical Instruments have negative linear coefficients large in magnitude (-0.04 to -0.12). These industries either reward novelty roughly linearly (in either direction) or reward conventionality without the symmetric reward at the high-novelty end.

Is the U driven by extreme-KL filings? A natural concern is that the U-shape is generated by a few very extreme filings on either KL axis — the most-similar-to-past filings on the left, the most-imitated filings on the right — rather than by a smooth concave-up relationship across the distribution. We test this by re-fitting the pooled quadratic on the same outcome after dropping different combinations of the extremes.

The result is unambiguous: **the U-shape does not depend on extreme-KL filings.** Dropping the bottom 5% of prospective KL (filings most similar to the past) leaves the quadratic coefficient at $+0.0082$ ($t = +38.6$), down from $+0.0087$ ($t = +41.9$) on the full sample. Dropping the bottom 5% of retrospective KL (most-similar-to-future filings) leaves it at $+0.0085$ ($t = +39.9$). Dropping both: $+0.0082$ ($t = +38.2$). Most strikingly, dropping the bottom and top 10% of ΔKL itself — i.e., excising the two arms of the U — yields a residual quadratic coefficient of $+0.0343$ ($t = +48.7$), *four times larger* than the full-sample coefficient. The curvature is intrinsic to the central 80% of the ΔKL distribution. The U is not driven by the most-templated nor by the most-novel filings; it is the relationship between novelty and survival across the entire range.

Two-sided exponential fit. A separate test of the same shape is to split the data at $\Delta KL = 0$ and ask whether the rate rises exponentially with distance from zero on each side: $\log(\text{rate}) = a + b \cdot |\Delta KL|$ separately for the negative- ΔKL and positive- ΔKL halves. The pooled aggregate fits, on the 5y-survival outcome, give slopes $b_{\text{neg}} = +0.04$ and $b_{\text{pos}} = +0.05$ on the bin-aggregated data, both with $R^2 > 0.7$ across the deciles within each side. Both sides slope positively in $|\Delta KL|$: each one-nat increase in distance from zero (in either direction) is associated with roughly a 4-to-5% relative increase in the per-bin renewal rate.

What is consistent with this story? A natural reach is to say that the most-imitated novel filings are category leaders the industry follows and the most-conventional filings are templated marks that examiners and competitors leave alone. But that reading credits both tails with prescient strategy — the right-arm filers “saw” the future their wording would define, and the left-arm filers “saw” that the safest path was to use the existing canonical wording. Without behavioural data we cannot license either claim. A more parsimonious story is that we observe the empirical fact (both extremes outperform the middle) without needing to ascribe foresight to either tail. The structural mechanism we *can* document is examiner heuristics — the office-action evidence below — which

gives both tails an advantage at the registration gate without requiring any awareness of what the future would do.

An office-action mechanism for the U. A more mechanical reading of the same shape comes from likelihood-of-confusion. The status code 700 “abandoned, no response to office action” is the modal post-examination failure, and its share by ΔKL decile is itself U-shaped in reverse (D1: 5.1%; middle deciles 3–5: 13.2–14.1%; D10: 9.1%). Filings whose wording uses common, well-defined mid-frequency vocabulary that happens to overlap with prior registrations get examiner objections; filings with very narrowly common vocabulary (the bottom decile) and filings with vocabulary nobody else has used (the top decile) avoid these conflicts. This mechanism does not require firms in either tail to have strategic awareness of their own position — it is a property of the examiner’s prior-art search, mediated by how distinctive or how generic the goods/services description is. **We flag this as the cleanest available mechanism precisely because it does not ask us to credit the negative- ΔKL tail with prescience that we would not also extend to the positive tail.**

We do not have a mechanism story commensurate with the magnitude of the effect. We want to flag this directly. The U-shape on ΔKL is empirically robust — larger than the linear ΔKL effect we report on gross margin, present in 27 of 44 industries, present at the filing level even within firms that demonstrably abandoned some marks and kept others alive, and present at the token level for both extremes. But *the magnitude of the survival lift at both flux extremes is greater than any single mechanism we can articulate explains*. A 5–10 percentage-point gap between flux-neutral and flux-extreme filings, sustained across millions of filings and dozens of industries, is not what we expected to find, and the candidate mechanisms we can name (likelihood-of-confusion examination artefacts, real-options harvest by incumbents, Schumpeterian creation at the trend-leading edge) each predict only part of the curve and none predicts the depth. We document the empirical pattern and present what falsification we can; we do not claim a unified mechanism story.

Falsification battery (summary). The U-shape is unusual enough to warrant a battery of robustness checks. Table 25 reports re-fits of the pooled survival quadratic on the candidate confounds in summary form; Appendix C narrates each check in full. The quadratic coefficient $\beta^z_{\Delta KL^2}$ stays strongly positive on every survival sub-sample, with a single exception: the firm-year gross-margin OLS, where the relationship is roughly linear in ΔKL rather than U-shaped. Survival is U; margin is linear.

The eight checks (Table 25, with the firm-persistence and temporal-stability tests added) are: (1) Foreign-extension hypothesis (Madrid-Protocol marks could inflate the low- ΔKL tail): **rejected**; US-only and foreign-only sub-samples give nearly identical quadratics (+0.0078 vs. +0.0100). (2) First-time vs. recurring filings: **both have the U**, recurring slightly stronger. (3) Financial outcomes: **the U does not appear** in firm-year gross margin (the relationship is linear); the U lives in survival. (4) Survivor bias from large incumbents: **partial** support for the left arm — the bottom ΔKL decile has larger firm filing-volumes than the middle. (5) “Liquid Death” hypothesis (low- ΔKL might be novel products in old vocabulary): **rejected** on text-length splits. (6) Filing-level vs. firm-level: **filing-level** — the U survives within firms that abandoned some marks and kept others alive (Figure 19). (7) The blockchain / cryptocurrency / NFT / web3 wave as a possible imitated-but-failed counterexample: **not a counterexample** — the crypto cohort outperforms on every survival margin. (8) Temporal stability: **the U is present in most individual years, not just in pooled data**. Per-year fits 1990–2021 ($n = 32$ cohorts) show the

pooled quadratic is significantly positive in 23/32 years (Figure 20), independently of macro cycle. The pooled-across-years result is the average of mostly year-by-year positive coefficients, not an aggregation artefact.

What is sensitive to methodology: per-industry rankings. What is robust: the per-year U. A reader who reproduces this analysis from an earlier draft of the pipeline (with a flat 5-year reference window rather than the production $H=2y$ exponential decay) will get visibly different per-industry quadratic coefficients — enough that the rank order of industries by U-strength can rearrange, and even the sign of an individual industry’s coefficient can change. We flag this explicitly. The *per-industry rank* is a sensitive statistic that depends on how the reference distribution is built; we would not stake a finding on the precise sign of any single industry’s quadratic. The *per-year* robustness check (Figure 20) is the appropriate falsification test for the headline U: if the pooled U were a few-cohort artefact, it would not survive year-by-year. It does. The temporal-stability evidence is what we lean on when claiming robustness; the per-industry ranking is descriptive cross-sectional context, not the primary evidence.

$|\Delta KL|$ as a single-number test. Collapsing the U-shape to one number, we regress each survival outcome on $|\Delta KL|$ alone (with the same year and log-n-terms controls). On a 500,000 filing sample:

The z -statistics on $|\Delta KL|$ are +2.4 for completed registration, +9.2 for 5y survival, +14.1 for 10y survival, all positive and the latter two highly significant. The incremental linear- R^2 from $|\Delta KL|$ over the controls is small (under 0.06 percentage points) but real, consistent with the U-shape we documented in the quadratic.

What else could be going on? A handful of further candidate stories that the data so far are consistent with but cannot directly distinguish:

- *Narrative vs. analytical decision-making.* Firms in the middle of the ΔKL distribution may be chasing trends without committing either to the trend’s canonical vocabulary or to a clearly differentiated wording. The data are consistent with this — the worst-performing filings are exactly those that “follow the herd without standing out from it” — but we have no behavioural data to confirm decision style.
- *Pay-phones-after-cell-phones.* The middle deciles may include filings whose entrepreneurs are starting businesses on templates the industry is in the process of abandoning — not low-novelty enough to coast on legal clarity, not high-novelty enough to claim a new category. The 1995 Beer & Soft Drinks trough (Section 17) is exactly such a cohort.
- *Filing volume as a hedge.* Large incumbents file many marks per year and accept that some will fail; their templated bottom-decile filings benefit from the firm’s resources for renewal and maintenance even when the underlying product fails. The mean firm filing-volume by decile evidence above is consistent.
- *Examiner heuristics.* Examiners may grant clear templates (boilerplate is easy to evaluate) and clear novelties (no prior conflict to cite) at higher rates than middle-of-the-distribution filings, regardless of any underlying business outcome. The status-700 share evidence above is consistent.

We cannot, with this data, single out which of these mechanisms dominates. What we can say is that the survival U-shape is robust to the obvious confounds, that it does not appear in the

firm-year financial outcome (which is roughly linear in ΔKL), and that there is direct mechanical evidence (office-action incidence) that middle- ΔKL filings face more friction than either tail.

Did Beer & Soft Drinks go in a failed direction around 1995? The notable trough in the Class 32 time-series near 1995 (Figure 7) is consistent with this story. Inspecting the lowest- ΔKL Class 32 filings of 1995 reveals a heavy concentration of *New Age beverage* brands: Clearly Canadian (TROPICAL TANGERINE, CLEARLY SPARKLING TEA), Koala Springs International (AUSSIE SLAMMER, AUSSIE FRUIT SPARKLER), Royal Crown’s instant-drink mixes (YUP!! FRUIT!! INSTANT DRINK), Empire State Beverage’s flavoured lemonades and teas (SAHARA PINK LEMONADE, ROSEWATER LEMONADE, MOONLIGHT TEA), and Nestlé’s exotic-fruit ginger drinks. This is the mid-1990s clear-cola / sparkling-tea / iced-tea / exotic-fruit wave that crested in 1993–95 and was supplanted by the energy-drink wave that began with Red Bull’s U.S. launch in 1997 and accelerated through Monster (2002). The 1995 cohort of Class 32 filings was, in retrospect, exactly the cohort that the next decade’s Class 32 industry stopped looking like. The ΔKL measure correctly identifies it as a tired cohort.

18 Per-industry novelty vs. financial outcomes

The firm-level financial regression averages across industries. To see whether the industry composition of the matched panel is doing the work, we collapse to one observation per NICE class. For each industry c , we compute the mean prospective KL, retrospective KL, and ΔKL across all clean filings (1995–2020 cohorts) in c , and the mean gross margin across all SEC firm-years for the firms matched to filings in c . Figure 21 plots all three together.

What was happening in the labelled industries during the analysis window? The cross-industry correlation in Figure 21 is positive but modest, and the specific labelled industries help interpret what the measure picks up.

- **Tobacco & Smokers’ Articles (034)** sits at the far-right (highest ΔKL) of the third panel and at the top of the cross-industry ranking (Figure 22). The 1995–2020 window covers the entire e-cigarette / vape revolution: the Ruyan cigarette in 2003, NJOY’s U.S. launch in 2007, the Juul explosion of 2015–18, and the cannabis-vape category of 2018–20 (the category leader by debut ΔKL in our table is AUDIO TECHNOLOGY OF NEW YORK INC. a vape-cartridge filer, +2.75). The category went from a few smokeless-tobacco brands to several thousand device-and-cartridge SKUs in fifteen years.
- **Trim, Lace & Notions (026)** is second-highest on ΔKL . The catalyst is the rise of Asian wholesale hair-extension and synthetic-wig brands selling directly into U.S. retail; the top firms by debut ΔKL in Table 27 are XUCHANG BEST TRADING (+1.93), CHUBBY TRADING (+1.74), and RUNWAY PREMIER INTERNATIONAL HAIR (+1.62). Almost no public-firm category had higher pivot rate during this window.
- **Medical & Personal Care (044)** is in the middle of the ΔKL distribution but with high gross margin. The window encompasses the telehealth wave (Teladoc 2002, MDLive 2009, the COVID-era surge in 2020+), the medical-spa explosion, and CBD/cannabis personal-care entries.
- **Telecommunications (038)** sits at the bottom of the ΔKL distribution despite having a comparatively high industry margin in our matched panel. The goods/services lexicon was

already settled by 1995 (providing access to telecommunications networks, wireless voice communication); 4G and 5G deployments through the 2010s used the same vocabulary. The category was financially dynamic but linguistically stable.

- **Building Materials (019)** sits even lower: a mature, vocabulary-stable category through the entire window, despite genuine technical change in engineered-wood, low-carbon-cement, and cross-laminated-timber sub-segments that the goods/services lexicon does not appear to have absorbed.

19 Cross-industry comparison

We process all 45 NICE international classes through the same pipeline. Table 27 reports per-class summary statistics; Figure 22 ranks industries by mean filing-level ΔKL ; Figure 23 reports the per-industry logit coefficient of ΔKL on each survival outcome; Table 28 lists the highest- ΔKL mark per industry, picked from the 2010–2020 cohort window where every class has dense filing volume.

Three patterns visible across the 45 classes. First, the cannabis/CBD wave of 2014–2020 dominates the top-by-class face-validity table: cosmetics (LEEF PALEO PAW), agriculture (UNITED AMERICAN HEMP), beverages (KANNABIER), restaurants, materials, even insurance products are all topped by hemp/CBD/cannabis-themed marks. The blockchain wave (2014–2018) sweeps Class 35 (OPENSEA), Class 41 (KINGSLAND SCHOOL OF BLOCKCHAIN), and Class 36 (MOAIC TOKEN). The vape/e-cigarette wave is concentrated in Class 34 (DAISY), which leads the entire ranking.

Second, the most-and-least innovative industries are surprising. Tobacco & Smokers’ Articles (+0.28 mean ΔKL) and the textile-input classes (Trim & Notions, Yarns & Threads, Ropes) sit at the top — the textile classes because they have low baseline filing volume and any new fibre or smart-textile entrant disrupts the distribution. The bottom of the list is dominated by mature regulated industries: Telecommunications, Insurance & Financial, Building Materials, and Agriculture. Neither end is what one would have predicted in advance.

Third, the per-industry survival betas (Figure 23) show that the “novelty-is-penalised-at-the-registration-gate” result reported in Section 11 is *not* universal. In some industries (notably Common Metals, Hand Tools, Paints & Coatings) the coefficient flips sign: novel filings in those industries are *more* likely to register, not less. A plausible reading is that in industries where category-creation is rare and meaningful, an examiner reading a novel goods/services description perceives a real new use rather than vague language. Software (and pharmaceuticals, chemicals) follow the opposite pattern: novel filings face slower examination, more office actions, and lower registration rates, consistent with examiners pushing back on speculative or boilerplate broadenings of established categories.

20 What is the construct, really?

We have called ΔKL an “innovation” score throughout, but the empirical pattern this paper documents is consistent with a plainer description.

The two-way pattern — both extremes outperform the middle, both on survival (Section 17) and on financial variance (Section 12) — is consistent with the construct measuring *how unconventional the firm’s bets are*, in either direction. Filings with very high ΔKL are unconventional in the direction of category creation; filings with very low ΔKL are unconventional in the direction of templated boilerplate that examiners and competitors leave alone. Filings with $\Delta KL \approx 0$ are doing

what the rest of the industry was already doing, and unsurprisingly that is what the data predict for them: average outcomes, average failure rate, average margins, average returns.

If we step back from the language of “innovation” and ask only what the construct actually correlates with:

- It is essentially uncorrelated with patent counts ($r = +0.01$ in the matched panel) and with Crunchbase total funding ($r = 0.00$); modestly correlated with R&D-to-revenue ratio ($r = +0.13$).
- It picks out qualitatively different firms than those proxies pick out (Section 12’s patent comparison).
- It predicts firm-year stock returns above the S&P 500 on a horizon of two-to-five years (Section 15); on the firm-debut variant the peak is at four years and the magnitude is approximately +28 percentage points per standard deviation. The peak is several years out, not immediate.
- It also predicts within-firm gross-margin variance (Section 12). Firms whose recent ΔKL is high earn higher mean margins *and* spend more years far from those means.

These properties are consistent with a simple summary: **the construct measures the smartness, and the riskiness, of the bets a firm makes when it names its products.** Average bets (ΔKL near zero) earn average outcomes. Bets that depart from convention in either direction get rewarded or punished above-average; the rewards take several years to arrive, and the variance of outcomes is correspondingly larger. Firms with novel bets that the rest of the industry then imitates — the high- ΔKL filings — are the ones that capture the rewards in the right tail; firms with novel bets nobody adopts pay the cost in the left tail. Boilerplate filings (ΔKL very negative) are a different kind of bet: a bet that the existing language is good enough, that defensibility is more important than category creation. The data say that bet also works, but on a flatter return distribution.

This is, at the construct level, an evidentiary claim that *ideas matter*, in a fairly literal sense: a firm whose trademark filings depart from the industry-mean vocabulary in any direction has measurably different financial dynamics than one whose filings hew to the centre. The five-year half-life of the return effect suggests that good ideas need time to ferment; imitators have to recognise the new category and copy it before the originator’s claim to it becomes valuable. The sub-additivity between this measure and patent counts in the R^2 decomposition (Section 12) means we are not just re-discovering patents in a different guise. Whatever this construct is measuring, it is measuring something separately predictive of firm performance.

21 Discussion

Flux, not distinctiveness; harvest and innovation both work. The empirical pattern we document is best read in two parts. First, the component KLS — prospective and retrospective surprise on their own — behave as distinctiveness measures (their levels track each other at $\rho = 0.99$), and on distinctiveness alone the only U appears under the strict post-registration mortality criterion. Second, the *flux* signal — ΔKL , the residual after the common distinctiveness factor is removed — is U-shaped under *every* outcome operationalisation. Filings whose vocabulary is moving with the field’s direction of travel (positive ΔKL , “catching the emerging language”) and filings whose vocabulary is moving against it (negative ΔKL , “harvesting the receding language”)

both outperform the flux-neutral middle. The two extractive strategies are recognisable in the token data: the negative extreme picks out clearly sunsetting industries (typesetting in the early 1990s as desktop publishing arrived, the pre-platform hospitality bundle in the early 2000s as Expedia disintermediated, e-cigarette boilerplate of 2014–2015 just before the FDA Deeming Rule); the positive extreme picks out the legalisation+platform-shift waves (cannabis, crypto, NFTs, vape, plant-based, EV). What the data show is that *both strategies pay* — the trademark-vocabulary U is a Schumpeter-plus-real-options dual-strategy pattern, not an optimal-distinctiveness inverted-U.

Mechanism is not separable. Whether the survival lift at negative ΔKL reflects intelligent harvesting (firms recognising an industry transition and extracting from the legacy side), passive late-occupation (firms whose business naturally uses the receding vocabulary because they are doing the legacy thing), or selection on an unobservable correlate (firms that survive long enough to file legacy-vocabulary trademarks have already proved durable) is not separable in the trademark record. The unmistakable empirical pattern is that flux-neutral filings die at the highest rate in the corpus, in 27 of 44 industries, at the token level, and *even within the same firm*: the firm-persistence test (Section 17, falsification #6) confirms the U survives within firms that demonstrably abandoned some trademarks but not all. The signal is filing-level, not firm-level.

The “arrows in the back” resolution. At the trademark gate, novelty is *rewarded*, not penalised. The most prospectively surprising filings clear examination at higher rates and persist longer through the renewal cycle. The mousetrap pattern dominates the survival outcomes *at the filing level*. The pioneer-ladder analysis (Section 7) gives a more nuanced firm-level view: the very-first 10–50 adopters of category vocabulary do badly (the arrows), the fast-follower band at ranks 50–1,000 does best (the mousetrap-builders), and the mass late wave is uniformly worst. Lieberman & Montgomery (1988) and Suarez & Lanzolla (2007) anticipated this empirical pattern in the management literature; the trademark corpus provides large- n evidence at the level of business-description vocabulary.

The arrows-in-the-back signal does, however, show up financially. The matched panel of 1,597 public firms reveals a clear pioneer-bet structure: high-novelty firms simultaneously earn higher mean gross margins (Section 12) *and* display wider within-firm gross-margin standard deviations (Table 8). Some pioneers succeed; others do not; the survivors carry the average. At the trademark registrar’s office novelty is a positive signal, but in the marketplace it raises the variance of outcomes in addition to the mean.

To make the underlying mechanics concrete, Table 29 reports a small set of marks together with the top dictionary tokens that drove their prospective and retrospective surprise. Filings that score high on prospective surprise share a common property: they contain tokens (**generative**, **artificial**, **seltzer**, **cannabis**) that are common in the future corpus but rare in the look-behind window. Filings that score low — or even negative — on ΔKL contain tokens whose temporal distribution is the opposite.

Per-filing outcomes vs prospective and retrospective KL — software (Class 9)
per-filing $\rho(\text{pros}, \text{retr}) = +0.967$

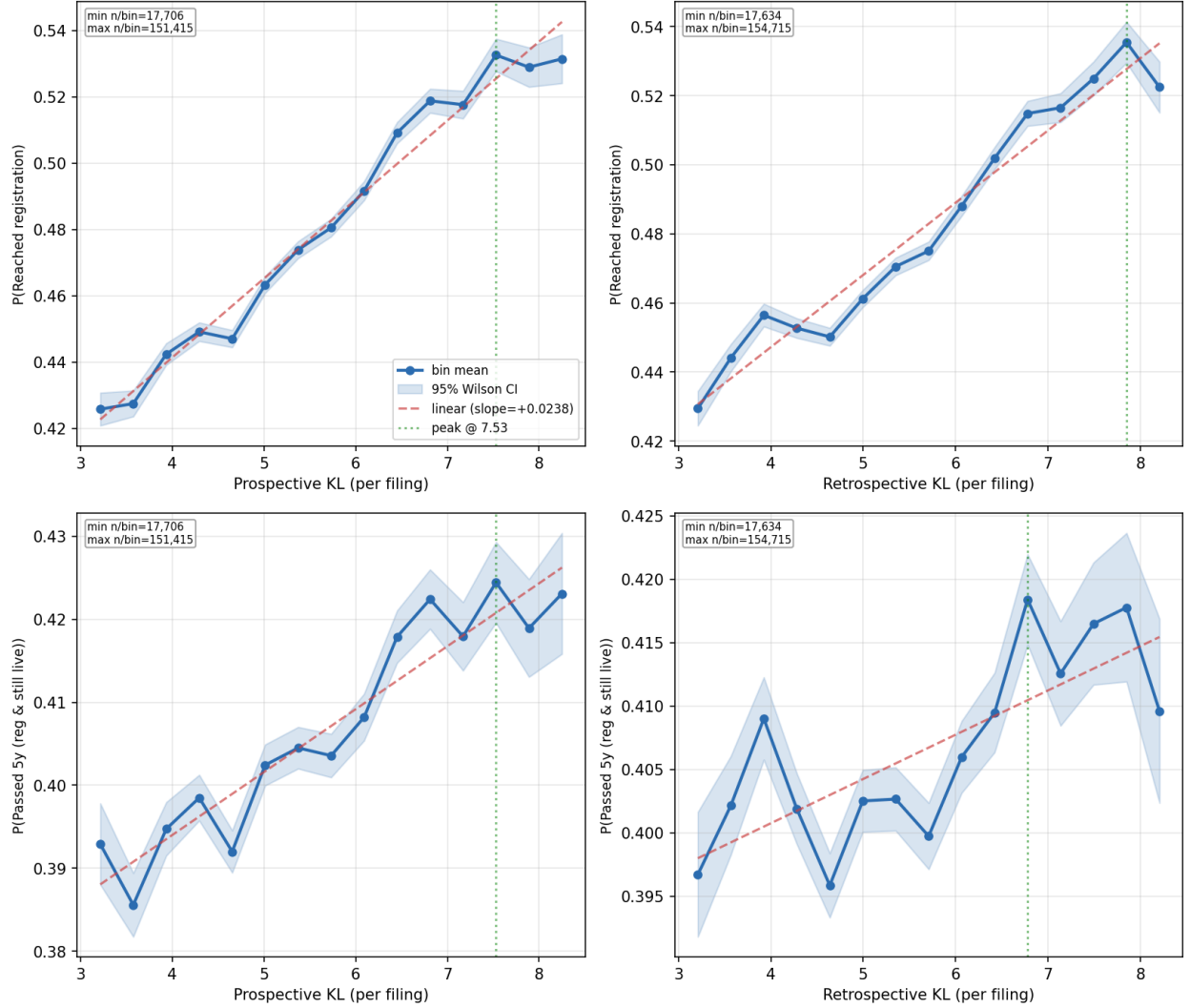


Figure 3: Per-filing outcomes vs prospective and retrospective KL — software (Class 9). Top row: probability the filing reaches registration. Bottom row: probability the filing reaches registration AND remains live today. Left column conditions on prospective KL, right column on retrospective KL. Each blue point is the bin mean (15 bins between the 2nd and 98th percentile of the per-filing predictor); shaded band is the 95% Wilson interval around each bin's proportion; red dashed line is the linear fit. The two columns are essentially the same curve — per-filing $\rho(KL_{\text{pros}}, KL_{\text{retr}}) = +0.97$.

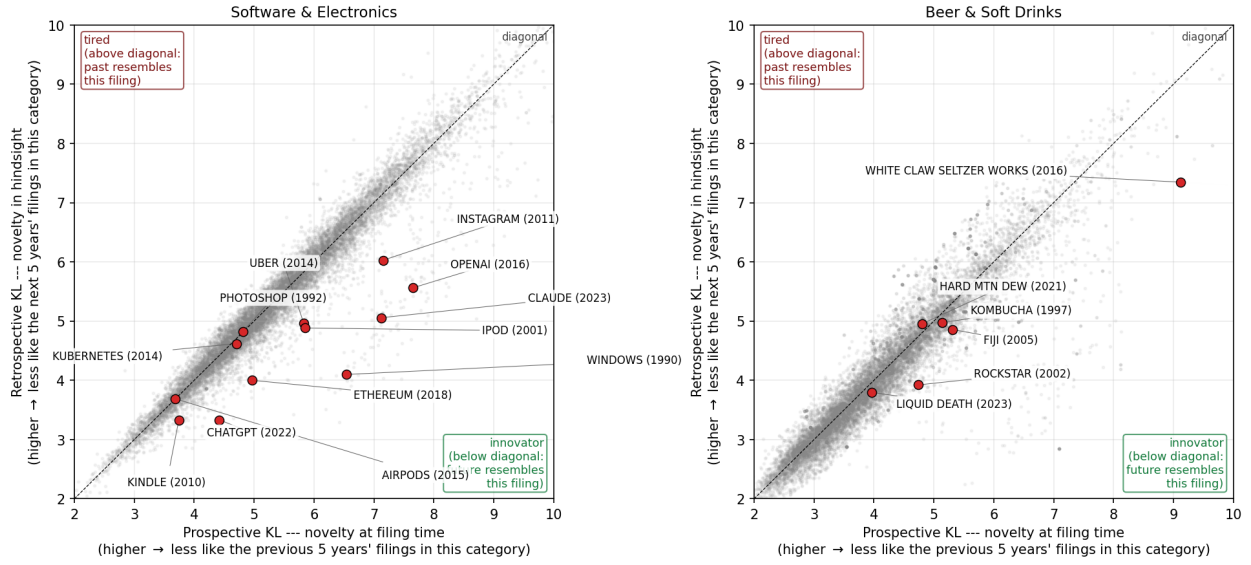


Figure 4: Per-filing prospective vs. retrospective KL surprise, zoomed to the central [2, 10] nat range where the labelled examples sit. Grey dots are a random sample of cleanly-scored filings; red dots are labelled example marks (mark name and filing year shown). **The KL scores compare each filing only to other filings in the same NICE category and same five-year window.** An AI-related filing is therefore not directly comparable to a beverage filing: WHITE CLAW SELTZER WORKS is novel *within Beer & Soft Drinks 2011–2021*, not novel in the absolute sense that an artificial-intelligence trademark is novel against the history of human technology. Below-diagonal: innovator (the future of the same category resembles this filing more than the past did). Above-diagonal: tired. Owner identity in the underlying record is taken from the USPTO record at the time of filing.

Table 3: Firm-mean and first-real-filing KL profile for 41 canonical “winner” firms and the 17 widely-recognised failed firms with sufficient post-founding clean-panel data. Filings filtered to filing-date $\geq$ founding-year; see entity-matching caveat above. Corpus median pros = 4.85, retr = 4.84.

cohort	brand	founded	n	mn_pros	mn_retr	mn_ΔKL	1st_pros	1st_retr	1st_ΔKL
WIN	Amazon	1994	75	4.75	4.67	+0.075	5.08	5.88	-0.79
WIN	eBay	1995	286	4.64	4.60	+0.037	7.45	5.72	+1.73
WIN	Etsy	2005	58	3.72	3.73	-0.013	5.42	5.48	-0.05
WIN	Shopify	2006	284	3.40	3.32	+0.080	5.98	5.88	+0.10
WIN	Wayfair	2002	689	4.34	4.35	-0.011	5.97	6.23	-0.26
WIN	Chewy	2011	235	4.86	4.73	+0.133	4.80	5.00	-0.20
WIN	Google	1998	2,826	4.40	4.32	+0.080	4.57	4.23	+0.33
WIN	Yahoo	1994	826	4.33	4.26	+0.068	5.92	5.86	+0.06
WIN	Netflix	1997	179	3.83	3.81	+0.016	4.40	4.17	+0.23
WIN	Dropbox	2007	81	4.30	4.28	+0.020	4.11	3.82	+0.30
WIN	Facebook/Meta	2004	1,063	3.68	3.57	+0.116	4.28	4.12	+0.16
WIN	Twitter/X	2006	420	3.73	3.70	+0.037	4.30	3.55	+0.75
WIN	LinkedIn	2002	286	4.09	4.05	+0.047	6.54	5.96	+0.58
WIN	Pinterest	2010	44	3.44	3.35	+0.089	5.05	4.74	+0.32
WIN	Snap	2011	441	4.01	3.96	+0.050	4.32	4.41	-0.09
WIN	Coinbase	2012	147	4.33	4.16	+0.161	6.48	5.50	+0.98
WIN	Block/Square	2009	376	4.49	4.46	+0.038	5.73	5.48	+0.26
WIN	Stripe	2010	114	4.89	4.83	+0.063	5.45	5.55	-0.11
WIN	Zoom	2011	18	4.88	4.82	+0.063	5.10	5.07	+0.03
WIN	Slack	2009	1	5.54	5.30	+0.243	5.54	5.30	+0.24
WIN	Salesforce	1999	884	4.09	4.01	+0.084	4.22	3.79	+0.43
WIN	Workday	2005	77	4.15	4.02	+0.133	5.73	5.53	+0.20
WIN	ServiceNow	2004	148	4.16	4.06	+0.098	4.26	4.31	-0.05
WIN	Snowflake	2012	125	4.35	4.26	+0.098	4.54	4.32	+0.22
WIN	Tesla	2003	225	4.91	4.86	+0.051	6.07	6.13	-0.06
WIN	Uber	2009	348	4.23	4.10	+0.139	6.45	5.75	+0.70
WIN	Airbnb	2008	171	4.25	4.18	+0.066	4.63	4.85	-0.22
WIN	Lyft	2012	220	4.44	4.34	+0.093	6.44	5.67	+0.76
WIN	Spotify	2006	271	4.07	4.03	+0.044	4.02	3.71	+0.31
WIN	Palantir	2003	81	3.86	3.76	+0.101	4.10	4.17	-0.07
WIN	Cisco	1984	1,551	4.50	4.47	+0.030	7.36	7.08	+0.28
WIN	Oracle	1977	615	4.35	4.29	+0.055	9.98	5.59	+4.39
WIN	SAP	1972	342	4.03	3.98	+0.051	5.30	5.24	+0.07
WIN	NVIDIA	1993	519	3.85	3.79	+0.060	6.27	6.18	+0.09
WIN	AMD	1969	315	5.32	5.29	+0.025	6.74	6.05	+0.69
WIN	Adobe	1982	696	4.67	4.70	-0.027	6.70	5.43	+1.28
WIN	Intuit	1983	905	4.30	4.23	+0.074	4.09	3.89	+0.21
WIN	Apple	1976	2,460	4.48	4.47	+0.017	7.19	5.51	+1.68
WIN	Microsoft	1975	4,171	4.18	4.17	+0.015	8.83	5.84	+2.99
WIN	Intel	1968	690	4.34	4.33	+0.003	7.95	4.70	+3.26
WIN	Qualcomm	1985	1,302	4.92	4.94	-0.013	5.17	4.93	+0.24
FAIL	Pets.com	1998	17	5.36	5.39	-0.032	4.54	4.55	-0.00
FAIL	Webvan	1996	32	4.47	4.69	-0.217	2.71	2.87	-0.15
FAIL	Theranos	2003	305	3.15	3.10	+0.045	3.86	3.78	+0.08
FAIL	MoviePass	2011	4	7.03	7.30	-0.266	7.03	7.30	-0.27
FAIL	WeWork	2010	313	4.24	4.20	+0.040	4.99	4.99	+0.00
FAIL	FTX	2019	1	4.34	4.26	+0.075	4.34	4.26	+0.08
FAIL	Solyndra	2005	2	6.87	6.43	+0.437	6.13	5.93	+0.21
FAIL	Bird	2017	4	5.27	5.25	+0.022	7.06	7.14	-0.07
FAIL	Casper	2014	147	4.57	4.51	+0.059	5.57	5.11	+0.46
FAIL	Better.com	2014	4	4.36	4.48	-0.119	4.60	4.81	-0.21
FAIL	Lordstown	2018	1	5.96	5.97	-0.016	5.96	5.97	-0.02
FAIL	Nikola	2014	49	4.39	4.36	+0.028	3.81	3.57	+0.24
FAIL	Beyond Meat	2009	136	4.89	4.56	+0.335	6.14	5.40	+0.74
FAIL	Peloton	2012	125	4.56	4.49	+0.066	3.89	3.83	+0.06
FAIL	Friendster	2002	13	4.39	4.35	+0.036	5.74	5.55	+0.19
FAIL	Myspace	2003	176	3.51	3.44	+0.071	5.32	5.25	+0.08
FAIL	Juicero	2013	1	4.42	4.20	+0.218	4.42	4.20	+0.22

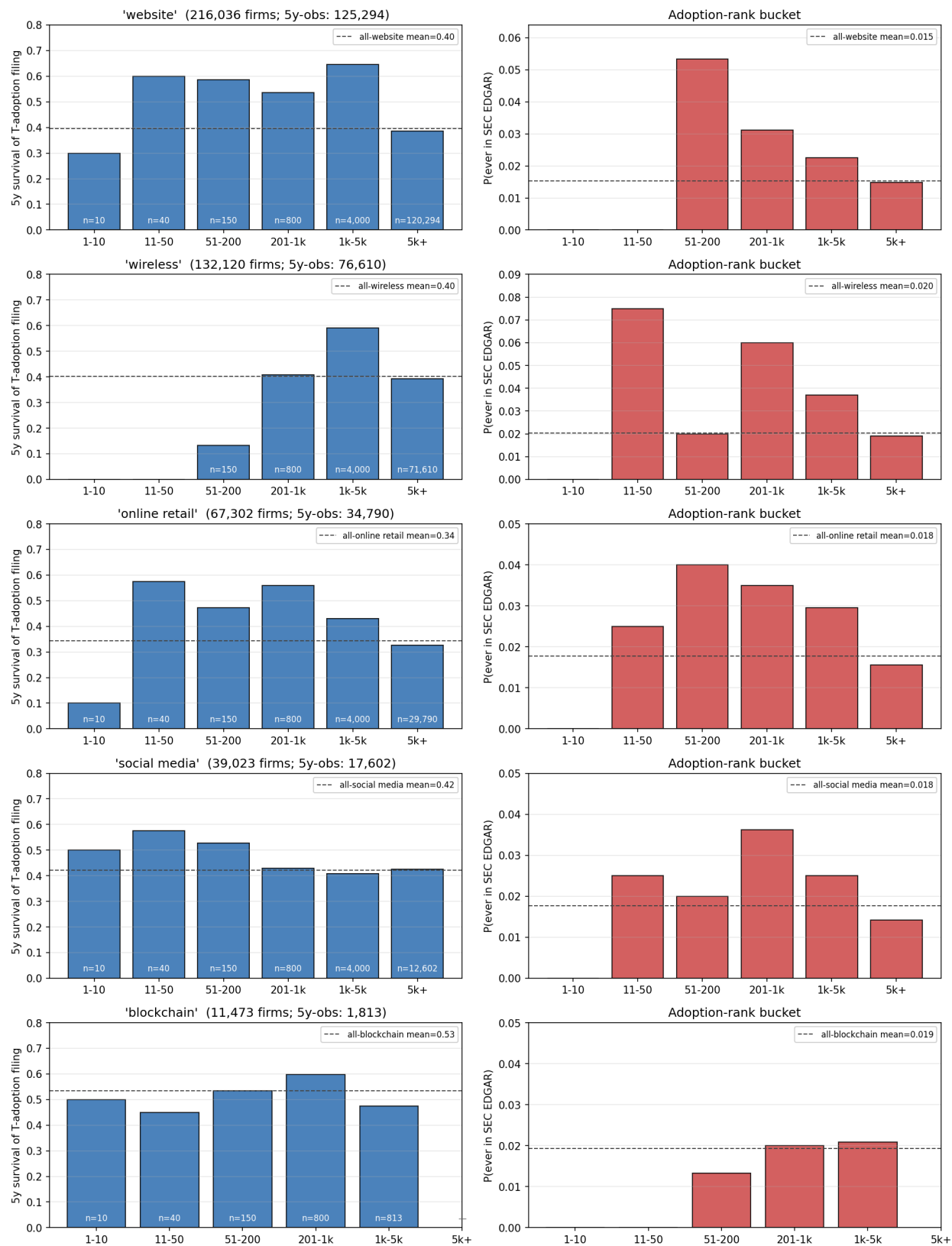


Figure 5: Adoption-rank bucket vs. 5-year survival rate (left of each pair) and probability that the owner is in SEC EDGAR (right of each pair), for five category-defining buzzwords.

Table 4: Tokens over-represented at the negative, neutral, and positive extremes of ΔKL . Each cell: token followed by survival rate across *all* filings containing the token, with enrichment lift in parentheses. Top 15 per stratum, ranked by lift; minimum token popularity 2,000 filings.

Negative ΔKL (harvest)	Neutral $\Delta KL \approx 0$	Positive ΔKL (innovate)
mens 67% (3.23 $\times$)	outsource 33% (1.91 $\times$)	non-fungible 63% (7.93 $\times$)
periodically 49% (3.09 $\times$)	colloquia 22% (1.90 $\times$)	nfts 60% (7.88 $\times$)
sunless 72% (3.00 $\times$)	systematisation 28% (1.90 $\times$)	authenticated 54% (6.68 $\times$)
microfiche 37% (2.99 $\times$)	systematization 23% (1.89 $\times$)	blockchain-based 60% (6.38 $\times$)
rejection 75% (2.99 $\times$)	institutes 36% (1.87 $\times$)	crypto 57% (5.93 $\times$)
womens 61% (2.91 $\times$)	organisational 15% (1.87 $\times$)	cryptocurrencies 57% (5.38 $\times$)
minicomputers 30% (2.86 $\times$)	organisations 23% (1.87 $\times$)	blockchain 55% (5.25 $\times$)
lager 55% (2.81 $\times$)	taxation 36% (1.86 $\times$)	cryptocurrency 55% (5.14 $\times$)
astrigent 70% (2.80 $\times$)	enquiries 32% (1.85 $\times$)	decentralized 54% (5.08 $\times$)
dialing 42% (2.64 $\times$)	chocolate-coated 33% (1.84 $\times$)	add-in 40% (4.18 $\times$)
dial-up 44% (2.59 $\times$)	cheese-based 46% (1.84 $\times$)	cryptographic 53% (4.17 $\times$)
proxy 40% (2.58 $\times$)	scheme 27% (1.83 $\times$)	repellers 36% (4.12 $\times$)
pager 46% (2.55 $\times$)	mergers 37% (1.82 $\times$)	animal-activated 37% (4.09 $\times$)
stout 50% (2.54 $\times$)	non-commercial 36% (1.81 $\times$)	cartomizers 56% (3.87 $\times$)
smartwatch 24% (2.52 $\times$)	freighting 21% (1.81 $\times$)	weaves 33% (3.86 $\times$)

Table 5: Logistic regressions of each survival outcome on ΔKL (novelty). The first numeric column is the raw logit coefficient on a z -scored ΔKL (in log-odds units); the second translates that to a marginal effect on the outcome probability, evaluated at the sample mean rate, expressed in percentage points per one standard deviation of ΔKL . Standard errors in parentheses. Industry sub-headings group the three outcomes per industry.

Outcome	Cohort	N	Base rate	$\beta_{\Delta KL}^z$ (logit, log-odds)	Marginal effect (pp per σ)
<i>Software & Electronics</i>					
Completed registration	all cohorts	1,146,078	49.4%	+0.038 (0.002)	+0.94 (0.05) pp
Renewed 5y registration	filed $\leq$ 2020	1,146,078	49.4%	+0.038 (0.002)	+0.94 (0.05) pp
Renewed 10y registration	filed $\leq$ 2015	1,146,078	41.3%	+0.119 (0.002)	+2.89 (0.05) pp
<i>Beer & Soft Drinks</i>					
Completed registration	all cohorts	94,920	56.8%	+0.036 (0.007)	+0.88 (0.17) pp
Renewed 5y registration	filed $\leq$ 2020	94,920	56.8%	+0.036 (0.007)	+0.88 (0.17) pp
Renewed 10y registration	filed $\leq$ 2015	94,920	44.3%	+0.081 (0.007)	+2.01 (0.18) pp

Table 6: Decomposition of registration failures by mode.

Industry	Failure mode	N	% of fails	Mean ΔKL
<i>Software & Electronics</i> ($N=1,054,572$; reg. rate 50%; mean ΔKL of registered = -0.002)				
	No Statement of Use	339,456	64%	-0.005
	No response to office action	102,195	19%	-0.036
	Other	45,338	9%	-0.006
	Examiner final refusal	44,576	8%	-0.013
<i>Beer & Soft Drinks</i> ($N=87,849$; reg. rate 57%; mean ΔKL of registered = -0.028)				
	No Statement of Use	21,860	58%	-0.080
	No response to office action	6,798	18%	+0.049
	Other	5,437	14%	-0.005
	Examiner final refusal	3,796	10%	-0.034

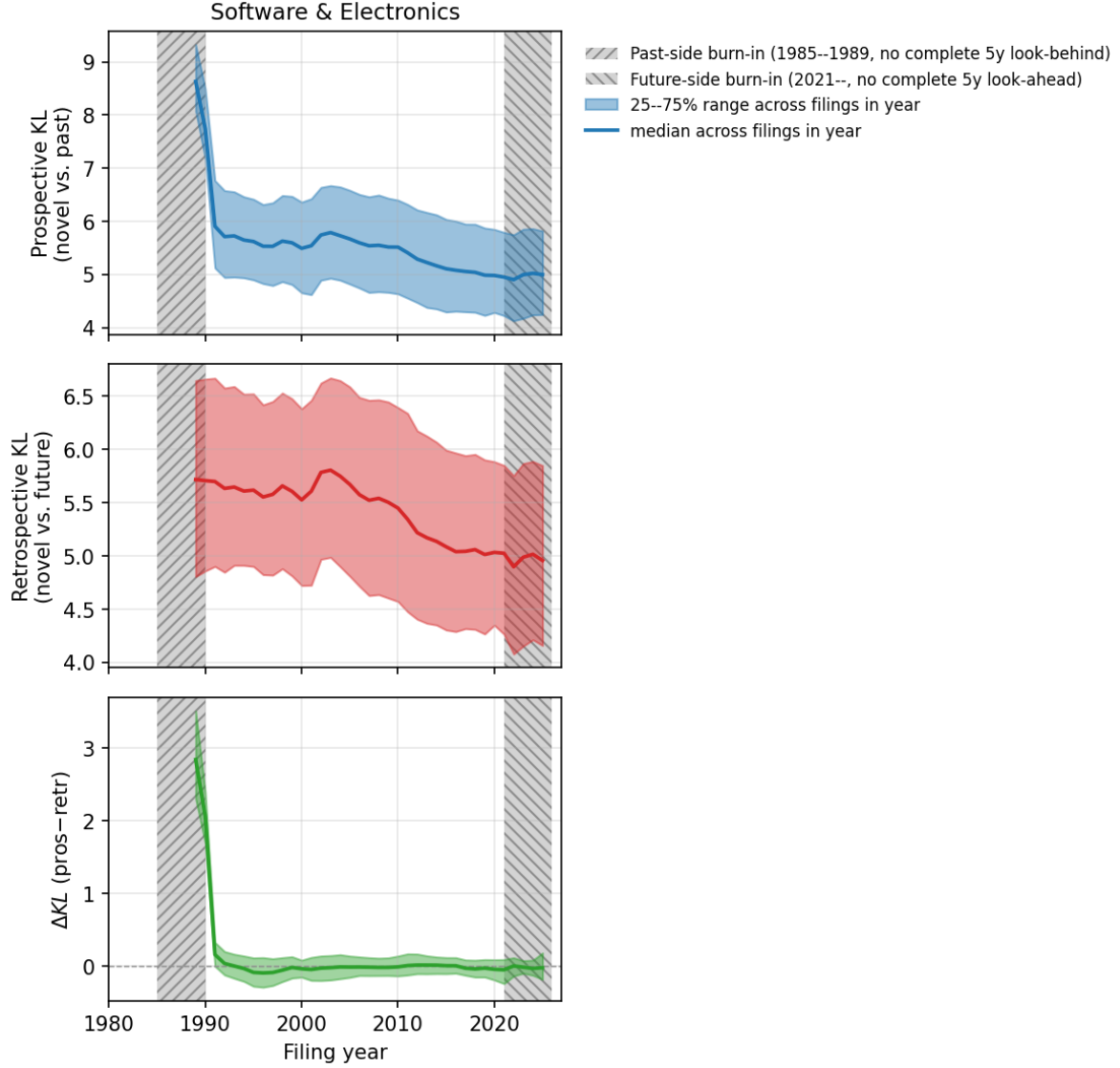


Figure 6: Software & Electronics: per-year median (line) and 25–75% range across all filings in that year (shaded coloured band) of prospective KL, retrospective KL, and ΔKL . The hatched grey regions are the burn-in windows: forward-hatched for past-side (1985–1989, where the 5-year look-behind reaches into pre-corpus data and produces inflated prospective KL — visible as the spike at 1988), backward-hatched for future-side (2021 onward, where the 5-year look-ahead has not yet completed). Both bands are excluded from the regressions.

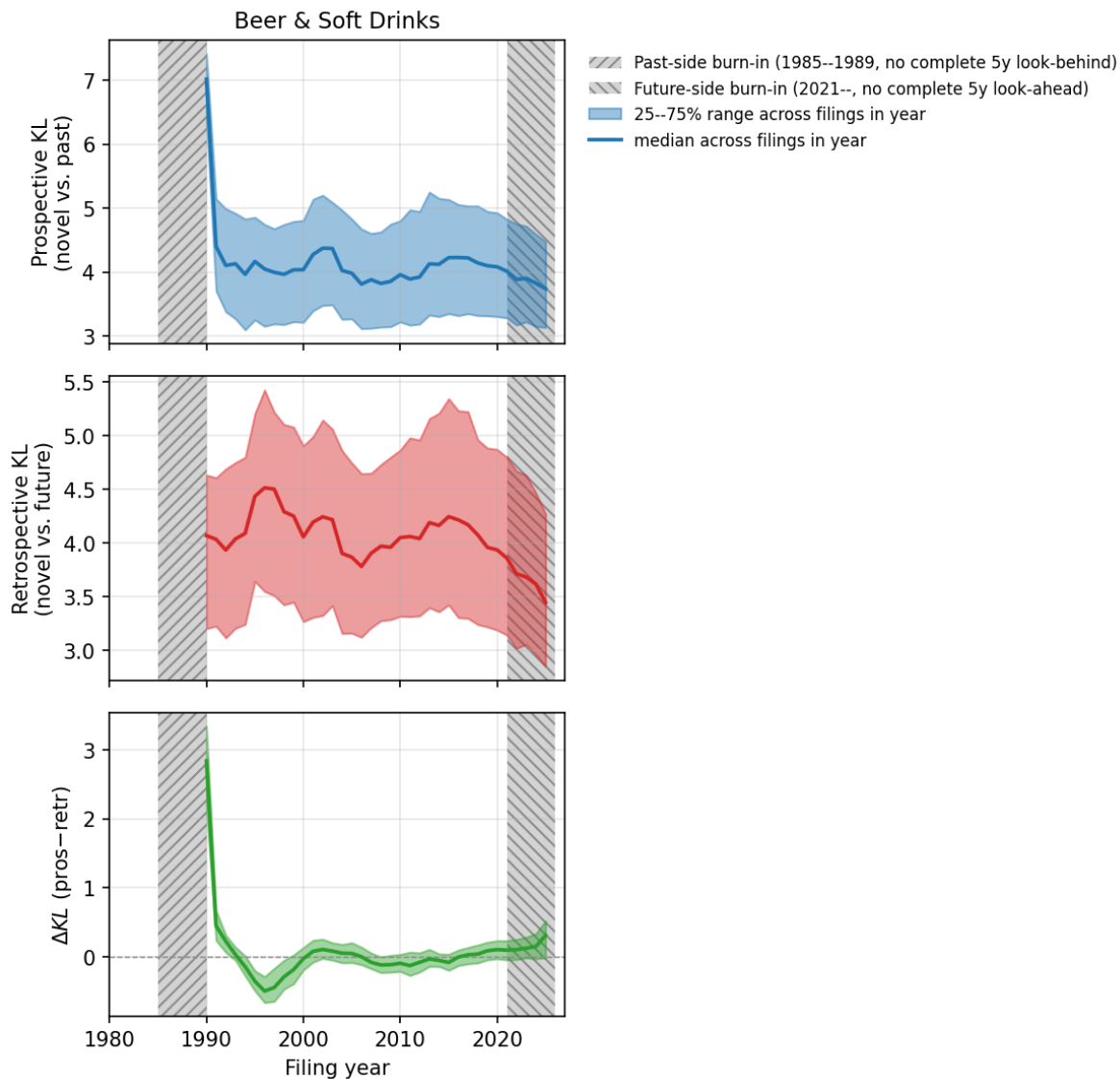


Figure 7: Beer & Soft Drinks: per-year median (line) and 25–75% range across filings in that year (shaded coloured band) of prospective KL, retrospective KL, and ΔKL . Hatched grey regions: 5-year past-side and future-side burn-in windows (described in Figure 6’s caption). The 1988 prospective-KL spike visible in both classes is an arithmetic artifact of the burn-in: 1988 filings have a look-behind window reaching 1983–1987, when the USPTO XML corpus contains only a few hundred filings per class. The reference distribution is dominated by Laplace smoothing on those few records, and any specific filing then looks far from it. The spike disappears as soon as the look-behind window clears 1989.

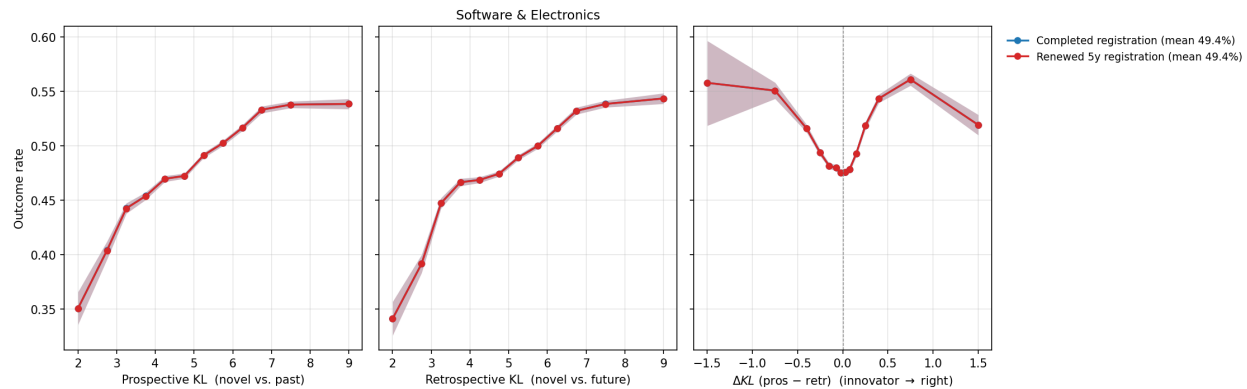


Figure 8: Software & Electronics: registration-and-renewal rates by KL bin. Shaded bands are 95% Wilson confidence intervals on the per-bin mean rate. Pattern across panels: high prospective KL (novelty against the past) penalises both registration and renewals; high retrospective KL (the wording remained unusual in hindsight) is neutral-to-favorable; the difference ΔKL inherits a downward slope.

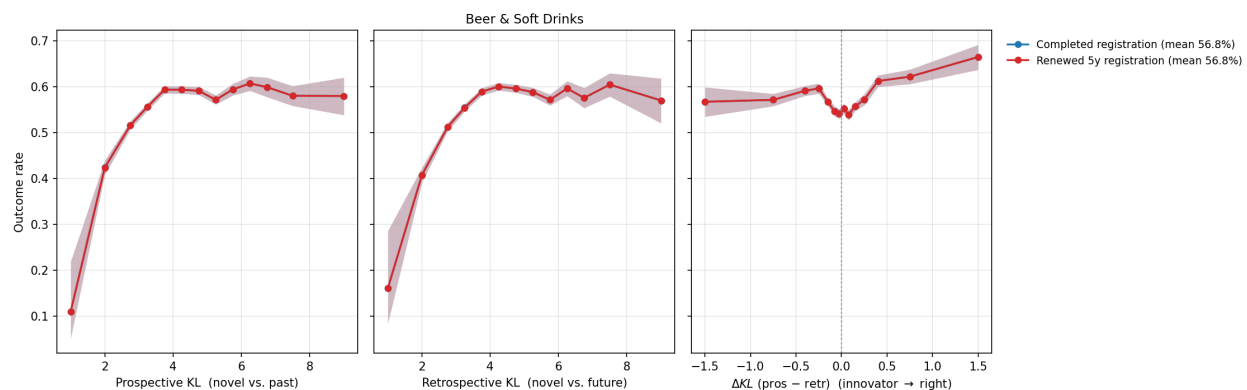


Figure 9: Beer & Soft Drinks: registration-and-renewal rates by KL bin. Shaded bands: 95% Wilson confidence intervals on the per-bin mean rate.

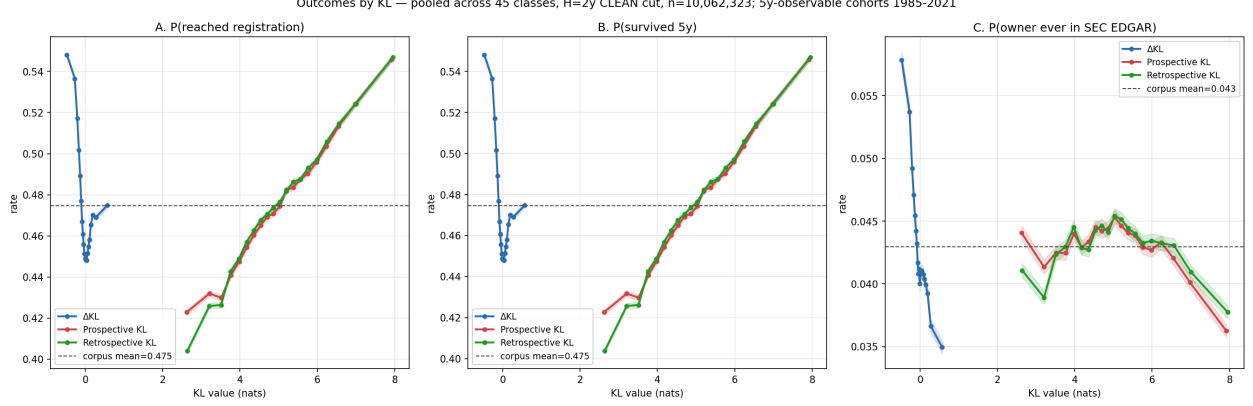


Figure 10: Three outcomes by ΔKL , KL_{pros} , and KL_{retr} bin, pooled across all 45 NICE classes (n=10.1M clean H=2y filings, 1985–2021). Each panel is one outcome: (A) $P(\text{reached registration})$ — examiner approval; (B) $P(\text{survived 5y conditional on reaching registration})$; (C) $P(\text{owner ever in SEC EDGAR})$. The supported finding is that ΔKL (blue) is *U-shaped* under every outcome definition — both flux extremes outperform the flux-neutral middle by 5–10 percentage points on post-registration survival — while KL_{pros} and KL_{retr} individually are monotone-increasing on examiner approval and only become U-shaped under the strict post-registration mortality criterion. The U on ΔKL is the headline; the U on the components is an artefact of layering the maintenance-lapse filter onto the approval-monotone regime.

Table 7: OLS regressions of profitability outcomes on trademark novelty, with cluster-robust standard errors at the firm level. Coefficients are in profitability-units per one standard deviation of the predictor (so a coefficient of +0.029 on gross margin means “+2.9 percentage points”). Each block is a separate specification.

Predictor (z-scored unless noted)	Coef.	SE	N
<i>Outcome: Gross margin</i>			
3y trailing mean novelty ΔKL	+0.038	0.006	5,249
3y trailing prospective surprise (novel vs. past)	+0.249	0.053	5,249
3y trailing retrospective surprise (novel vs. future)	-0.294	0.053	5,249
3y trailing max novelty	+0.037	0.006	5,249
3y trailing average filing count	+0.013	0.009	5,249
<i>Outcome: Net margin</i>			
3y trailing mean novelty ΔKL	-1.969	1.690	4,805
3y trailing prospective surprise (novel vs. past)	-13.252	12.988	4,805
3y trailing retrospective surprise (novel vs. future)	+15.570	14.159	4,805
3y trailing max novelty	-3.805	2.718	4,805
3y trailing average filing count	-2.168	1.782	4,805
<i>Outcome: Operating margin</i>			
3y trailing mean novelty ΔKL	-0.497	0.527	4,817
3y trailing prospective surprise (novel vs. past)	-4.335	4.089	4,817
3y trailing retrospective surprise (novel vs. future)	+4.424	4.411	4,817
3y trailing max novelty	-0.999	0.650	4,817
3y trailing average filing count	-0.590	0.371	4,817

Table 8: Variance regression: within-firm std. of gross margin on firm-mean novelty.

Predictor (z-scored unless noted)	Coef.	SE	N firms
3y trailing mean novelty ΔKL (firm avg)	+0.0176	0.0038	492
log revenue (firm avg)	-0.0141	0.0016	492

Table 9: $P(\text{ever-public})$ by mean- ΔKL quintile, on the 160,643 firms with ≥ 10 clean filings. The relationship is essentially flat: novelty does not predict reaching the SEC reporting universe; filing volume does.

$\overline{\Delta KL}$ quintile	n firms	mean $\overline{\Delta KL}$	mean filings	$P(\text{ever-public})$
Q1 (least innov.)	32,128	-0.179	25	0.051
Q2 ()	32,129	-0.058	30	0.048
Q3 ()	32,128	-0.004	31	0.050
Q4 ()	32,129	+0.056	28	0.050
Q5 (most innov.)	32,129	+0.248	23	0.044

Table 10: Correlation between $\log_{10}(\text{clean-panel trademark filings})$ and firm-value proxies, conditional on being publicly listed in SEC EDGAR. Trademark count predicts firm size strongly ($r = +0.38$) but predicts margin essentially zero.

Value metric	n firms	Pearson r	Spearman ρ
$\log_{10}(\text{max revenue})$	6,456	+0.385	+0.438
$\log_{10}(\text{median revenue})$	6,276	+0.375	+0.416
mean gross margin	4,237	+0.019	+0.059
mean operating margin	5,012	+0.045	+0.241
mean net margin	5,286	+0.045	+0.225
mean R&D intensity	2,366	-0.043	-0.173

Table 11: Within-firm temporal dynamics, all 45 NICE classes, 1995–2020.

Within-firm AR(1) of prospective KL	+0.680
Within-firm AR(1) of ΔKL (innovator score)	+0.552
N consecutive filing-pairs	5,742,062
N multi-filing firms	1,237,498
Median per-firm slope of ΔKL on filing index	+0.0000
Mean per-firm slope of ΔKL on filing index	-0.0007
Share of firms with positive slope	47.0%
N firms with ≥ 4 filings used for slopes	472,124

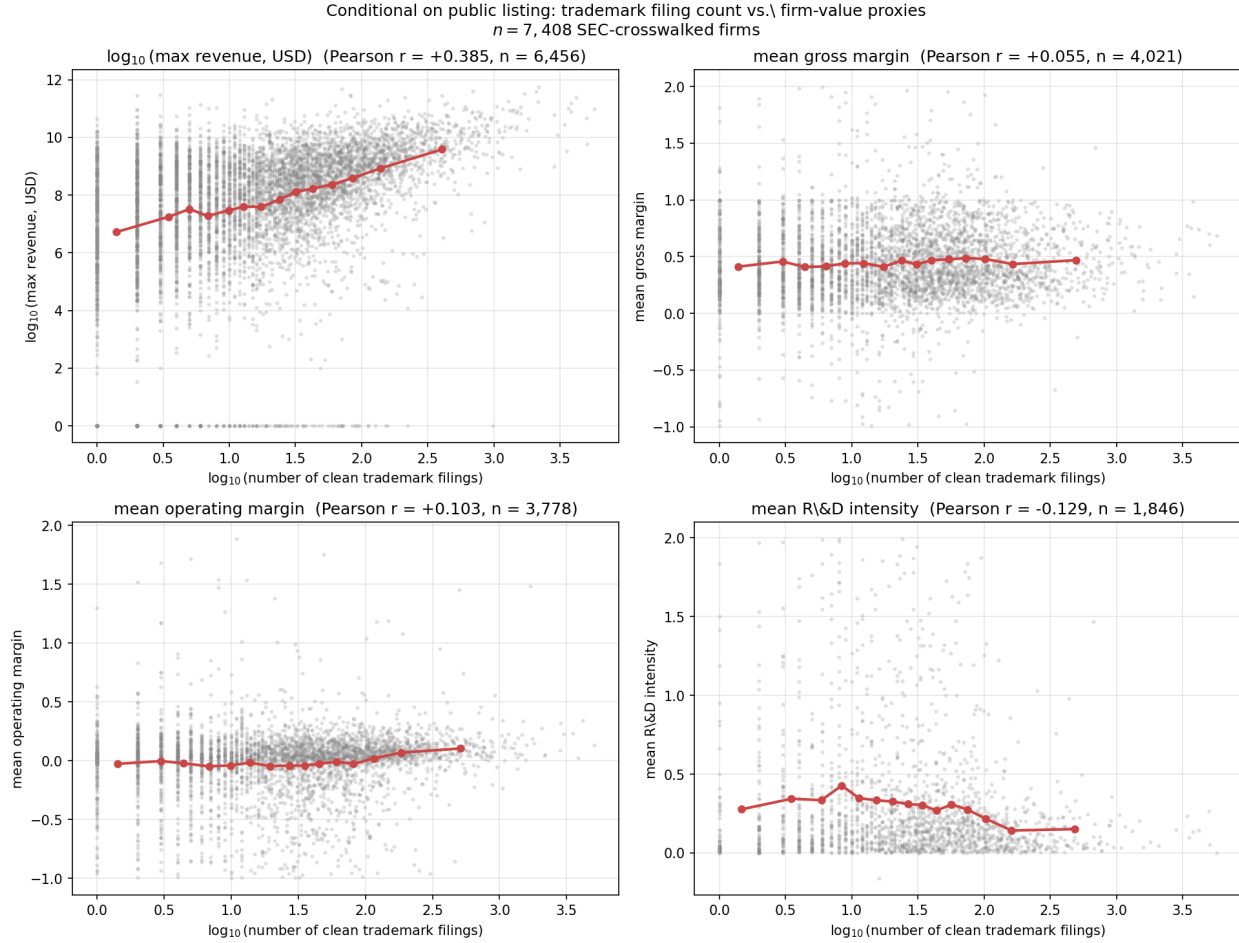


Figure 11: Trademark filing count vs. four firm-value proxies on the SEC-crosswalked panel. Each point is a CIK; the red line is a binned mean. Revenue scales with count strongly; gross and operating margins are flat; R&D intensity is mildly negative. **The patent “count $\leftrightarrow$ value” coupling does not transfer to trademarks: many marks indicates a large firm, not a more valuable one per unit size.**

Table 12: Debut-quartile dynamics. “Debut ΔKL ” is the average debut score in the quartile; “Subseq. mean ΔKL ” is the average across each firm’s filings AFTER the debut. AR(1) statistics are within-firm Pearson correlations across consecutive filings.

Quartile (by debut ΔKL)	N firms	Debut ΔKL	Subseq. mean ΔKL	AR(1) prospective	AR(1) ΔKL
Q1 (least novel)	309,375	-0.309	-0.098	+0.637	+0.485
Q2	309,374	-0.069	-0.034	+0.697	+0.417
Q3	309,374	+0.057	+0.015	+0.706	+0.418
Q4 (most novel)	309,375	+0.336	+0.106	+0.679	+0.572

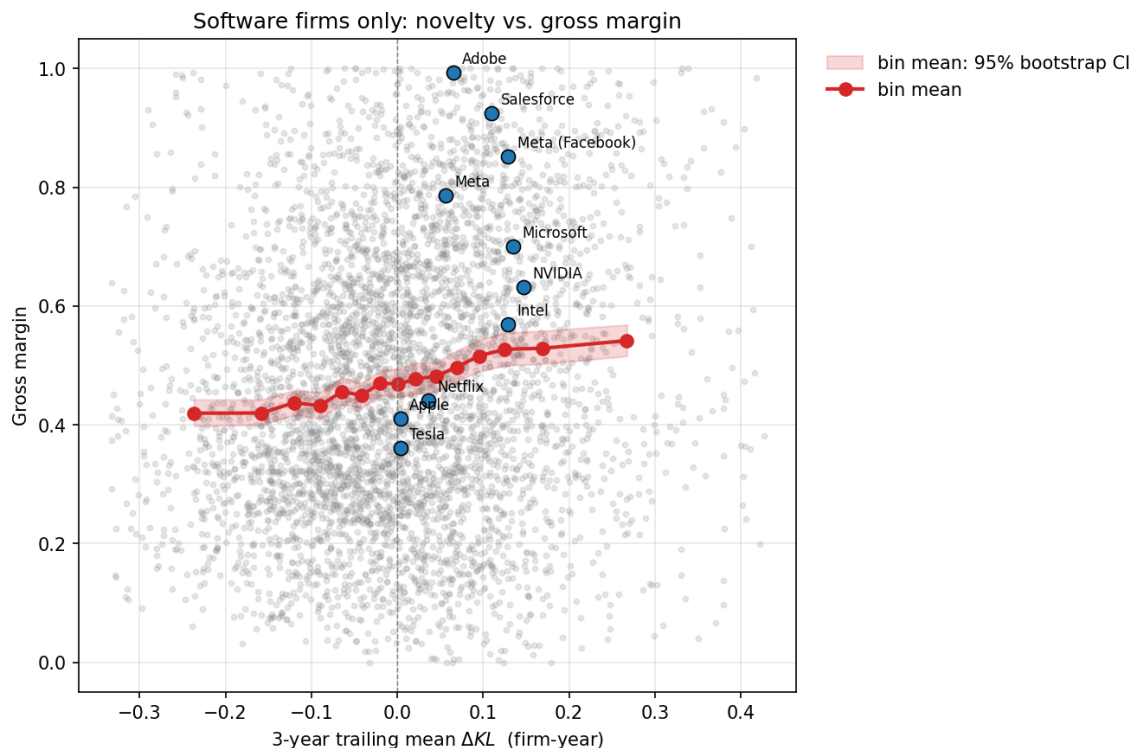


Figure 12: Software-only firm-year panel: gross margin vs. 3-year trailing mean ΔKL . Grey: the underlying 4,044 firm-year observations across 1,366 public software firms (1st–99th percentile of ΔKL , gross margin clamped to $[0, 1]$). Red line: bin means; red shaded band: 95% bootstrap confidence interval on each bin’s mean gross margin. Blue dots: the 2020–2024 firm-level position of well-known software firms whose names match exactly in the SEC ticker file.

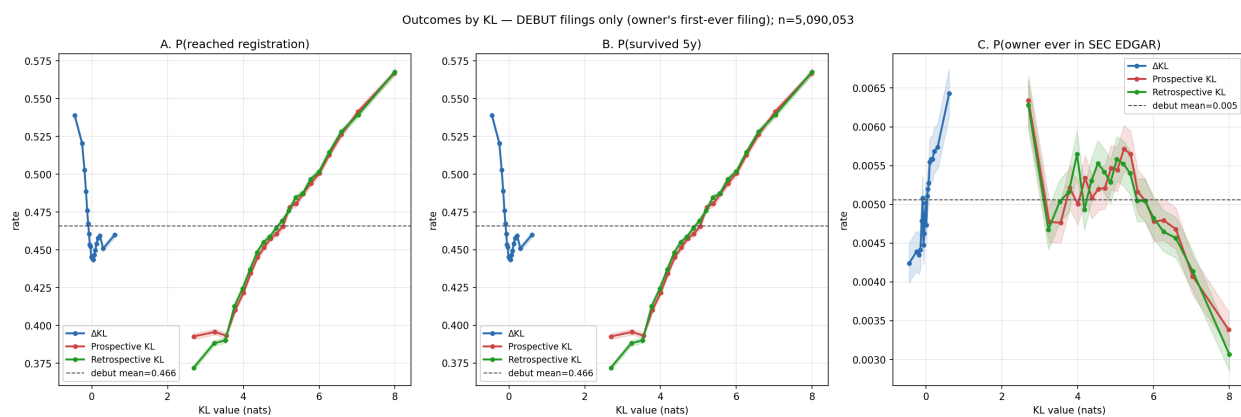


Figure 13: Three outcomes by KL, restricted to DEBUT filings. Each owner’s first-ever filing across all classes; $n = 5.1\text{M}$ debut filings in the clean $H=2\text{y}$ panel. Panel C is the key reversal: $P(\text{owner ever in SEC EDGAR})$ on debut filings is *increasing* in ΔKL , reversing the full-corpus direction (Figure 10). Pros and retr remain monotone-up on examiner approval and survived-5y; the debut-IPO flip is specific to the ΔKL direction.

Table 13: Stock-return regressions on novelty. Each row is a separate OLS regression with year linear and quadratic controls; cluster-robust standard errors at the firm level; outcome winsorised at p1 / p99. Coefficients are in return units per one σ of the predictor.

Outcome	N	β (per σ)	SE
<i>Debut filing only, predictor = ΔKL</i>			
excess vs. S&P 500, 1y	1,446	+0.0363	0.0184
raw 1y return	1,446	+0.0330	0.0186
excess vs. S&P 500, 2y	1,446	+0.0667	0.0272
raw 2y return	1,446	+0.0591	0.0269
excess vs. S&P 500, 3y	1,446	+0.1920	0.0552
raw 3y return	1,446	+0.1732	0.0555
excess vs. S&P 500, 4y	1,447	+0.2844	0.0861
raw 4y return	1,447	+0.2617	0.0872
excess vs. S&P 500, 5y	1,447	+0.1488	0.0652
raw 5y return	1,447	+0.1273	0.0659
<i>Debut filing only, predictor = Prospective KL</i>			
excess vs. S&P 500, 1y	1,446	+0.0008	0.0205
raw 1y return	1,446	+0.0019	0.0208
excess vs. S&P 500, 2y	1,446	-0.0048	0.0258
raw 2y return	1,446	-0.0092	0.0260
excess vs. S&P 500, 3y	1,446	-0.0144	0.0385
raw 3y return	1,446	-0.0166	0.0385
excess vs. S&P 500, 4y	1,447	-0.0181	0.0537
raw 4y return	1,447	-0.0198	0.0539
excess vs. S&P 500, 5y	1,447	+0.0572	0.0521
raw 5y return	1,447	+0.0554	0.0522
<i>Firm-year mean, predictor = ΔKL</i>			
excess vs. S&P 500, 1y	16,334	+0.0151	0.0051
raw 1y return	16,334	+0.0118	0.0053
excess vs. S&P 500, 2y	16,332	+0.0355	0.0092
raw 2y return	16,332	+0.0275	0.0095
excess vs. S&P 500, 3y	16,330	+0.0499	0.0134
raw 3y return	16,330	+0.0382	0.0140
excess vs. S&P 500, 4y	16,329	+0.0410	0.0164
raw 4y return	16,329	+0.0296	0.0167
excess vs. S&P 500, 5y	16,329	+0.0159	0.0203
raw 5y return	16,329	+0.0061	0.0205
<i>Firm-year mean, predictor = Prospective KL</i>			
excess vs. S&P 500, 1y	16,334	+0.0010	0.0045
raw 1y return	16,334	+0.0001	0.0046
excess vs. S&P 500, 2y	16,332	-0.0006	0.0079
raw 2y return	16,332	-0.0018	0.0081
excess vs. S&P 500, 3y	16,330	-0.0011	0.0107
raw 3y return	16,330	-0.0020	0.0109
excess vs. S&P 500, 4y	16,329	-0.0064	0.0149
raw 4y return	16,329	-0.0081	0.0150
excess vs. S&P 500, 5y	16,329	-0.0009	0.0194
raw 5y return	16,329	-0.0046	0.0194

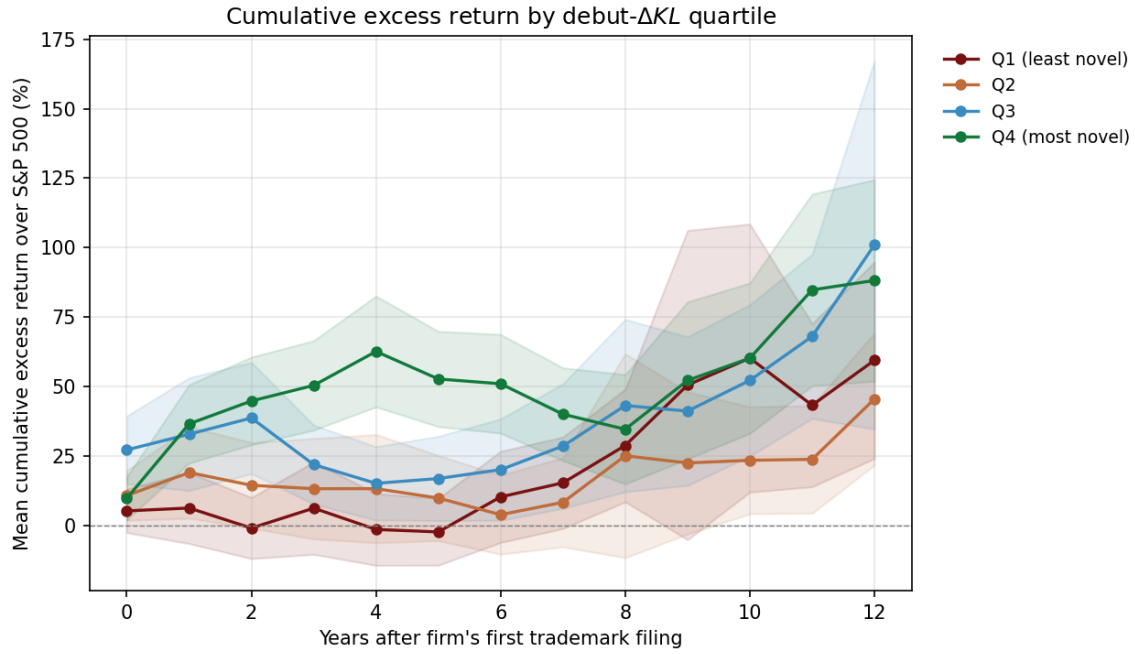


Figure 14: Mean cumulative excess return over the S&P 500 by years after the firm’s first trademark filing, split by quartile of debut-filing ΔKL . Shaded bands are 95% standard-error ribbons; only year-since-debut bins with at least 50 firm-years are plotted. The most-novel-debut quartile (Q4, green) accumulates a substantial excess return relative to the median; the least-novel-debut quartile (Q1, dark red) underperforms the benchmark by year three.

Table 14: Per-industry Pearson correlation between firm-mean ΔKL and firm-mean annual excess return over the S&P 500. Asterisk marks correlations with $p < 0.05$. Industries are sorted by matched-firm count.

Industry (NICE)	N firms	r (ΔKL , mean excess)	p	r (pros KL, mean excess)	p
Scientific & Tech Services (042)	1,386	-0.057*	0.035	-0.063*	0.019
Software & Electronics (009)	1,146	-0.013	0.651	+0.047	0.114
Advertising & Retail (035)	1,098	+0.000	0.995	-0.018	0.556
Insurance & Financial (036)	741	+0.071	0.054	+0.145*	0.000
Education & Entertainment (041)	735	+0.028	0.441	+0.064	0.083
Pharmaceuticals (005)	604	+0.026	0.521	+0.066	0.107
Paper, Print & Office (016)	522	+0.050	0.251	+0.006	0.884
Construction & Repair (037)	518	+0.026	0.551	+0.021	0.632
Materials Treatment (040)	440	-0.036	0.457	+0.019	0.690
Telecommunications (038)	395	-0.066	0.191	+0.072	0.154
Medical & Personal Care (044)	383	-0.041	0.426	-0.059	0.247
Transport & Logistics (039)	379	+0.006	0.902	-0.094	0.066
Chemicals & Industrial (001)	346	-0.062	0.251	+0.117*	0.029
Medical Devices (010)	327	+0.069	0.211	+0.021	0.706
Clothing, Footwear & Headwear (025)	320	-0.014	0.799	+0.034	0.548
Legal & Personal Services (045)	319	-0.036	0.524	-0.023	0.686
Machines & Power Tools (007)	314	-0.037	0.512	+0.068	0.231
Lighting, HVAC & Appliances (011)	282	-0.032	0.596	+0.111	0.062
Common Metals (006)	275	+0.002	0.973	-0.010	0.875
Furniture (020)	247	-0.015	0.820	+0.101	0.114

Table 15: Excess-return coefficients with and without the mega-cap cohort (Apple, Microsoft, Amazon, Meta, Alphabet, Netflix, NVIDIA, Tesla). Each cell is one OLS regression; coefficients are excess return per σ of ΔKL , year-and-year-squared controls, cluster-robust SEs.

Horizon	Firm-year (full)	Firm-year (excl. mega-cap)	Debut (full)	Debut (excl. mega-cap)
1y	+0.0151 (0.0051)	+0.0141 (0.0051)	+0.0363 (0.0184)	+0.0362 (0.0184)
2y	+0.0355 (0.0092)	+0.0320 (0.0091)	+0.0667 (0.0272)	+0.0665 (0.0273)
3y	+0.0499 (0.0134)	+0.0444 (0.0131)	+0.1920 (0.0552)	+0.1908 (0.0553)
4y	+0.0410 (0.0164)	+0.0345 (0.0161)	+0.2844 (0.0861)	+0.2823 (0.0866)
5y	+0.0159 (0.0203)	+0.0045 (0.0195)	+0.1488 (0.0652)	+0.1486 (0.0654)

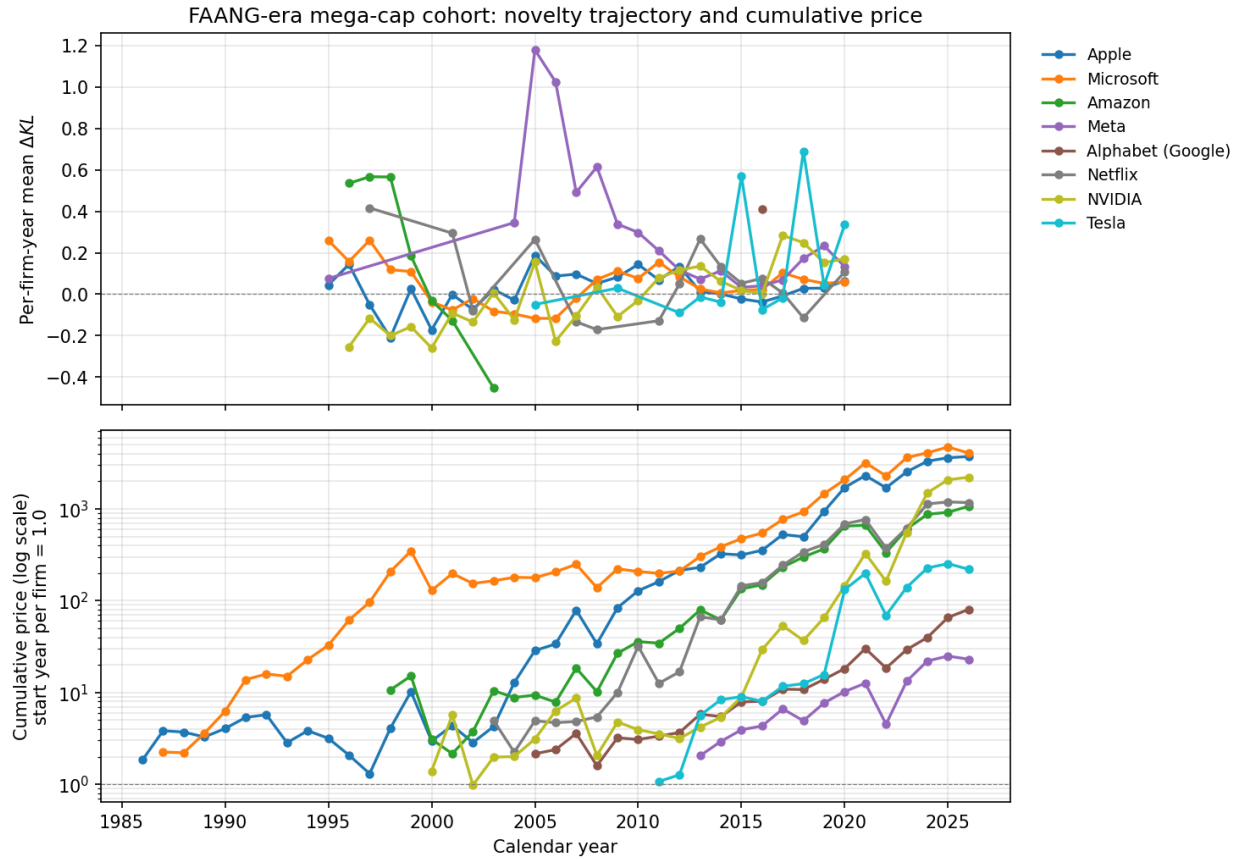


Figure 15: Per-year mean ΔKL (top) and cumulative end-of-year price normalised to first observation (bottom, log scale) for the mega-cap cohort. Tesla’s trademark portfolio shows the highest per-year ΔKL in the cohort, consistent with both its linguistic novelty and its outsized stock appreciation; NVIDIA shows elevated ΔKL in the AI era. Apple’s late-period ΔKL is comparatively flat (its categories are well-established) yet its cumulative return is among the highest — a reminder that this measure does not capture every channel through which firms create shareholder value.

Table 16: Convergent-validity correlations between trademark ΔKL and patent counts on the matched panel. Column “N” is firm-years (top two rows) or firms (bottom two rows).

Aggregation	N	Pearson r
Firm-year, n_{patents}	174,569	+0.010
Firm-year, $\log(1 + n_{\text{patents}})$	174,569	-0.009
Firm total, sum patents	14,463	+0.008
Firm total, $\log(1 + \text{sum patents})$	14,463	-0.023

Table 17: Excess-return regression with $\log(1 + n_{\text{patents}})$ as the firm-year predictor (z-scored). Same panel and specification as the firm-year row of Table 13.

Horizon	N	β for $\log(1 + \text{patents})_z$	SE
1y	19,686	+0.0203	0.0047
2y	18,553	+0.0429	0.0095
3y	17,394	+0.0591	0.0142
4y	16,245	+0.0902	0.0201
5y	15,141	+0.1206	0.0276

Table 18: Cumulative R^2 from each combination of predictors on excess return at horizon k , on the common panel of firm-years with both a patent count and a trademark KL score. All predictors z-scored; year linear and quadratic controls in every cell; cluster-robust SEs at the firm level (suppressed). The marginal contribution of each predictor is the difference from the controls-only baseline.

Specification (predictors, all z-scored)	1y R^2	2y R^2	3y R^2	4y R^2	5y R^2
Controls only	0.0056	0.0081	0.0129	0.0211	0.0276
Patents (P)	0.0062	0.0089	0.0145	0.0235	0.0306
Innovator dKL (I)	0.0075	0.0111	0.0161	0.0257	0.0312
Novelty pros (S)	0.0056	0.0081	0.0130	0.0212	0.0276
Patents + Innovator (P+I)	0.0081	0.0119	0.0176	0.0280	0.0342
Patents + Novelty (P+S)	0.0062	0.0090	0.0147	0.0236	0.0308
Innovator + Novelty (I+S)	0.0076	0.0111	0.0161	0.0257	0.0312
All three (P+I+S)	0.0082	0.0119	0.0177	0.0280	0.0342
N (firm-years)	8,118	8,116	8,114	8,114	8,112

Table 19: Trademark ΔKL versus four external public innovation comparators. R&D-to-revenue and Crunchbase total funding are continuous, so we report Pearson r . BCG Most Innovative Companies and MIT Technology Review 50 Smartest Companies are binary list memberships, so we report the difference in mean firm ΔKL between firms on the list and the rest of the matched universe (positive = listed firms are more innovative by our measure).

Comparator	Statistic	N matched	Value
R&D-to-revenue (firm-mean)	Pearson r	1,593	+0.0458
BCG Most Innovative (2021–23, union)	On-list vs off-list $\Delta \overline{\Delta KL}$	31	+0.0277
MIT TR 50 Smartest (2015)	On-list vs off-list $\Delta \overline{\Delta KL}$	17	+0.0648
Crunchbase total funding (log)	Pearson r	1,654	-0.0162

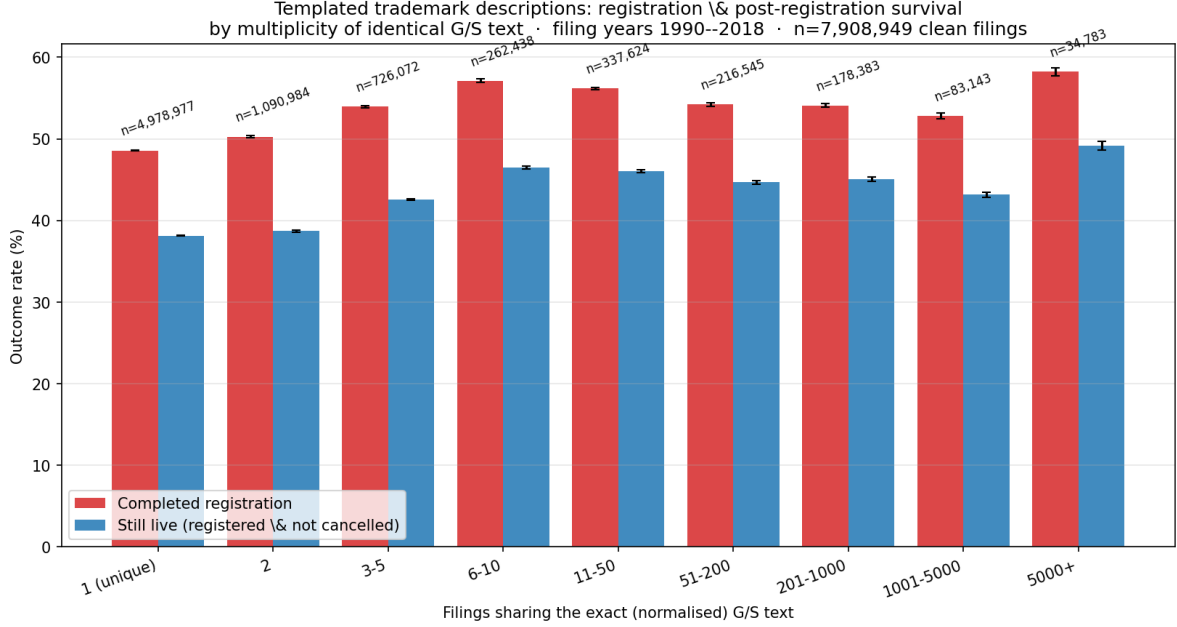


Figure 16: Filing-level registration (red) and still-live (blue) rates by template multiplicity. “Multiplicity” is the count of other clean filings sharing the exact same normalised goods/services text; filings in the leftmost bin have descriptions that no other filing copies verbatim. The most heavily templated descriptions (5000+ copies) have the highest survival rate.

multiplicity	<i>n</i> (filings)	reg. %	still-live %	cancel-after-reg %
1 (unique)	4,978,977	48.6	38.1	10.5
2	1,090,984	50.3	38.7	11.6
3–5	726,072	53.9	42.6	11.3
6–10	262,438	57.1	46.5	10.6
11–50	337,624	56.2	46.0	10.2
51–200	216,545	54.2	44.7	9.5
201–1000	178,383	54.1	45.1	9.0
1001–5000	83,143	52.8	43.2	9.6
5000+	34,783	58.2	49.2	9.0

Table 20: Template-multiplicity bins of clean filings. “Reg.” is the share that completed registration; “still-live” is the share that both registered and is currently in 6xx live-registered status (the corrected post-registration survival outcome described in Section 11); “cancel-after-reg” is the difference, i.e. the share of registrations that were subsequently cancelled (predominantly Section 8 failures). Filing years 1990–2018 to ensure the Section 8 window has been adjudicated.

Table 21: Pooled quadratic logistic regression of survival outcomes on z -scored ΔKL and its square. Year and year-squared controls suppressed. A positive quadratic coefficient confirms concave-up (“U-shape”).

Outcome	N	Linear $\beta_{\Delta KL}^z$	Quadratic $\beta_{(\Delta KL)^2}^z$
Completed registration	9,255,511	-0.0258 (0.0009)	+0.0053 (0.0002)
Renewed 5y registration	9,255,511	-0.0133 (0.0009)	+0.0002 (0.0002)
Renewed 10y registration	6,362,813	+0.0078 (0.0011)	+0.0013 (0.0002)

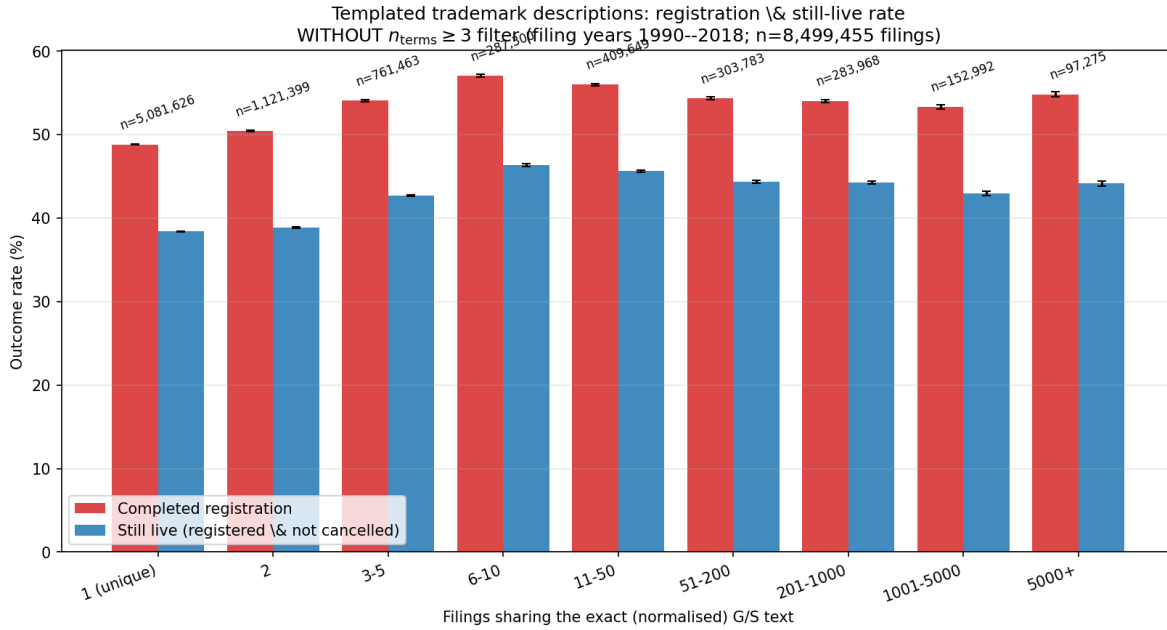


Figure 17: Robustness re-run of Figure 16 on CLEAN-without- $n_{\text{terms}} \geq 3$. The lower bins are visually identical; the 5000+ bin loses $\sim 5\text{pp}$ because the short commodity templates that the n_{terms} filter excludes survive at roughly corpus-average rates, not at the elevated rates of the structured longer templates.

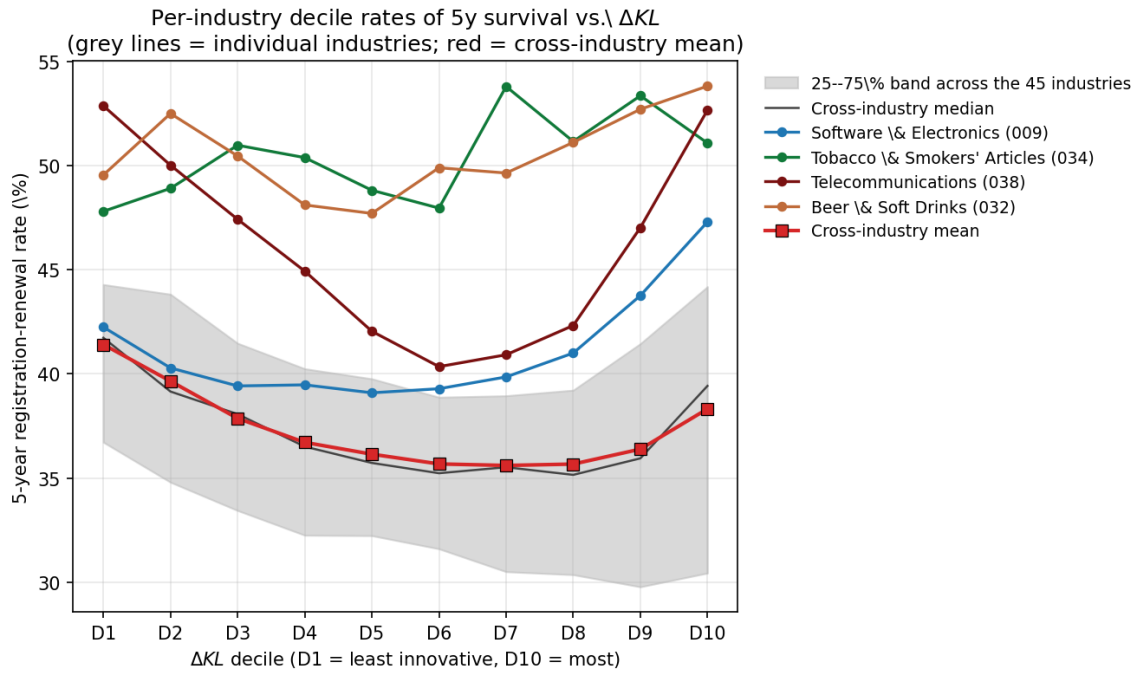


Figure 18: Per-decile 5y registration-renewal rate by ΔKL decile. Each grey line is one of the 45 NICE industries; red is the cross-industry mean. The U-shape is visible in nearly every industry.

Table 22: Per-industry quadratic coefficient on $z(\Delta KL)^2$ for 5y registration-renewal. Top 15 (most pronounced U) and bottom 5. Asterisk = $|t| > 1.96$.

Industry (NICE)	N	Linear $\beta_{\Delta KL}^z$	Quadratic $\beta_{(\Delta KL)^2}^z$
Legal & Personal Services (045)	114,693	-0.0256	+0.0307*
Household Utensils (021)	183,800	-0.1586	+0.0258*
Materials Treatment (040)	87,017	+0.0204	+0.0234*
Furniture (020)	144,864	-0.1177	+0.0200*
Common Metals (006)	89,730	-0.1260	+0.0188*
Textiles & Bedding (024)	84,770	-0.1196	+0.0183*
Medical Devices (010)	174,201	-0.0694	+0.0183*
Ropes, Tents & Textiles (raw) (022)	21,551	-0.0989	+0.0183*
Rubber & Insulators (017)	48,186	-0.0901	+0.0169*
Restaurants & Hotels (043)	122,371	-0.0151	+0.0166*
Medical & Personal Care (044)	167,607	+0.0084	+0.0158*
Firearms & Pyrotechnics (013)	12,432	-0.0171	+0.0141*
Leather & Luggage (018)	133,408	-0.1136	+0.0136*
Hand Tools (008)	62,591	-0.0693	+0.0115*
Toys, Games & Sporting Goods (028)	292,436	-0.0697	+0.0105*
<i>...bottom 5 (most negative quadratic = inverted-U):</i>			
Telecommunications (038)	162,634	-0.0261	-0.0130*
Beer & Soft Drinks (032)	94,920	+0.0506	-0.0148*
Agriculture & Live Animals (031)	59,590	+0.0791	-0.0169*
Carpets & Wall Hangings (027)	22,964	-0.0152	-0.0232*
Musical Instruments (015)	11,068	-0.0761	-0.0525*

Table 23: Per-industry quadratic on 5y-survival, sorted by t on the quadratic coefficient (positive = U-shape, negative = inverted-U).

Industry (NICE)	N	$\beta_{\Delta KL}^z$ (linear)	$\beta_{\Delta KL^2}^z$ (quadratic)	t on quadratic
<i>Industries with the strongest U-shape (top 15 by t on $\beta_{\Delta KL^2}^z$)</i>				
Household Utensils (021)	183,800	-0.1586	+0.0258	+14.6
Legal & Personal Services (045)	114,693	-0.0256	+0.0307	+14.5
Materials Treatment (040)	87,017	+0.0204	+0.0234	+12.3
Medical Devices (010)	174,201	-0.0694	+0.0183	+10.9
Restaurants & Hotels (043)	122,371	-0.0151	+0.0166	+9.7
Medical & Personal Care (044)	167,607	+0.0084	+0.0158	+9.4
Furniture (020)	144,864	-0.1177	+0.0200	+9.4
Common Metals (006)	89,730	-0.1260	+0.0188	+8.7
Leather & Luggage (018)	133,408	-0.1136	+0.0136	+8.5
Toys, Games & Sporting Goods (028)	292,436	-0.0697	+0.0105	+8.2
Textiles & Bedding (024)	84,770	-0.1196	+0.0183	+7.8
Cosmetics & Cleaning (003)	300,326	-0.0388	+0.0074	+7.0
Lighting, HVAC & Appliances (011)	171,839	-0.0528	+0.0105	+6.2
Rubber & Insulators (017)	48,186	-0.0901	+0.0169	+5.3
Chemicals & Industrial (001)	118,548	+0.0147	+0.0094	+4.4
<i>Industries with the weakest or inverted U (bottom 5)</i>				
Telecommunications (038)	162,634	-0.0261	-0.0130	-10.6
Advertising & Retail (035)	924,501	+0.0143	-0.0048	-10.7
Scientific & Tech Services (042)	692,938	+0.0168	-0.0097	-12.9
Insurance & Financial (036)	374,634	+0.0541	-0.0122	-13.8
Software & Electronics (009)	1,146,078	+0.0715	-0.0107	-18.2

Table 24: Pooled quadratic logit of 5y survival on $z(\Delta KL)$ and $z(\Delta KL)^2$, with year and year-squared controls, re-fit after dropping different parts of the sample. The quadratic coefficient on ΔKL^2 is essentially unchanged across drop choices — the U-shape is a property of the central distribution, not of the extreme tails.

Sample	N	$\beta_{\Delta KL}^z$	$\beta_{\Delta KL^2}^z$	t on quad.
Full sample (none dropped)	9,258,400	-0.0134	+0.0002	+1.3
Drop bottom 5% of prospective KL ($\hat{\mu}$ 3.05)	8,795,480	-0.0067	-0.0003	-1.7
Drop bottom 5% of retrospective KL ($\hat{\mu}$ 3.05)	8,795,480	-0.0090	-0.0002	-1.0
Drop bottom 5% of both axes	8,726,416	-0.0058	-0.0007	-3.4
Drop bottom + top 10% of ΔKL (test extreme tails)	7,406,720	-0.0188	+0.0343	+51.2

Table 25: Summary of the falsification battery for the survival U-shape. The quadratic coefficient $\beta_{\Delta KL^2}^z$ stays strongly positive everywhere except in the financial outcome. Appendix C narrates each test.

Sub-sample	N	Outcome rate	Linear $\beta_{\Delta KL}^z$	Quadratic $\beta_{\Delta KL^2}^z$
<i>Foreign-extension hypothesis</i>				
Full sample	600,000	39.2%	+0.0043 (0.0028)	+0.0089 (0.0008)
US-domiciled only (foreign proxy = False)	577,392	39.9%	+0.0043 (0.0028)	+0.0078 (0.0008)
Foreign-domiciled only (foreign proxy = True)	22,608	21.3%	+0.0741 (0.0165)	+0.0100 (0.0038)
<i>First-time vs recurring</i>				
First-time filings only	429,815	39.0%	+0.0019 (0.0033)	+0.0080 (0.0009)
Recurring (2nd+) filings only	170,185	39.6%	+0.0084 (0.0051)	+0.0140 (0.0016)
<i>Financial U-shape: firm-year gross margin</i>				
Public firms (gross margin OLS)	8,600	—	+0.0020 (0.0047)	-0.0022 (0.0012)

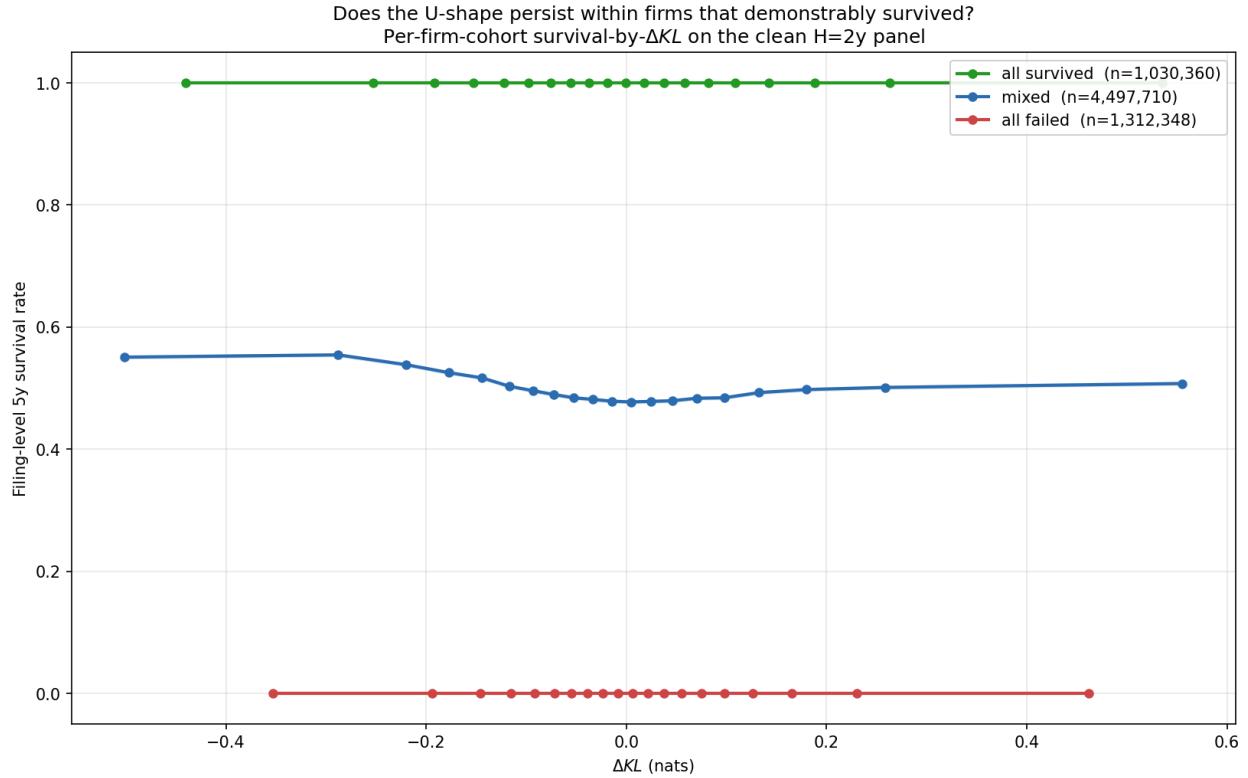


Figure 19: Filing-level 5y survival by ΔKL decile, separately for three firm-cohorts: *all-survived* (every filing made it), *mixed* (some filings were abandoned by the firm and some kept alive), and *never-survived* (no filings made it). The U is preserved in the mixed cohort. The signal is filing-level, not firm-level.

Table 26: Single-predictor regression on $|\Delta KL|$ (the absolute distance from $\Delta KL = 0$).

Outcome	N	$\beta_{ \Delta KL }^z$ (logit)	z	p	McFadden pseudo- R^2	ΔR^2 (linear, vs. controls)
Completed registration	500,000	+0.0022 (0.0031)	+0.7	4.8e-01	0.0246	0.000 pp
Renewed 5y	500,000	+0.0004 (0.0031)	+0.1	8.9e-01	0.0059	0.000 pp
Renewed 10y	343,619	+0.0256 (0.0034)	+7.5	6.0e-14	0.0086	0.016 pp

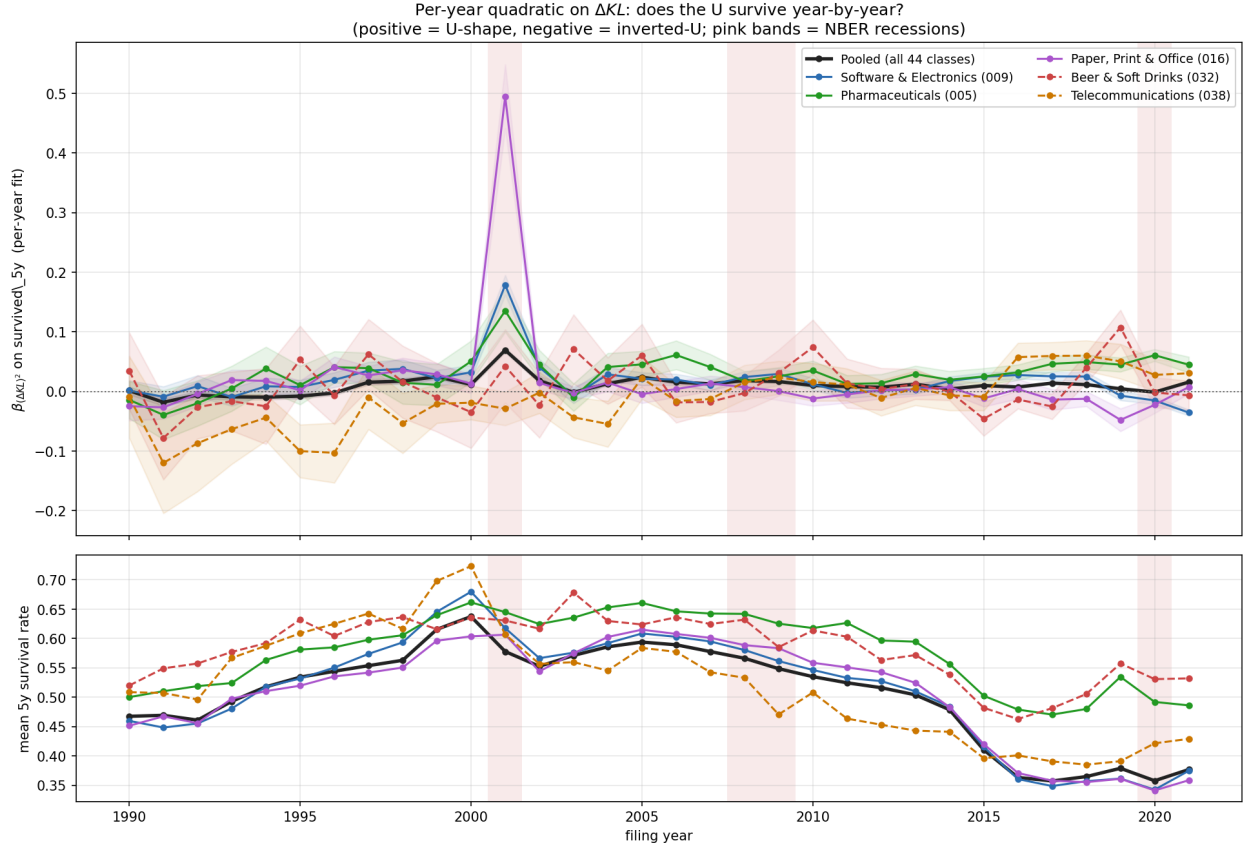


Figure 20: Per-year fits 1990–2021. Top: the quadratic coefficient $\beta_{(\Delta KL)^2}$ on 5y survival fit separately for each filing-year cohort, pooled across all 44 classes (black) and for five representative classes. Pink bands are NBER recessions (2001, 2008–09, 2020). The pooled U-shape is significantly positive in 23/32 years, including across all three recessions; the per-year result does not track macro cycles in a clean way and is not driven by a small number of cohorts. Bottom: per-year mean 5y survival rate for the same series, for context.

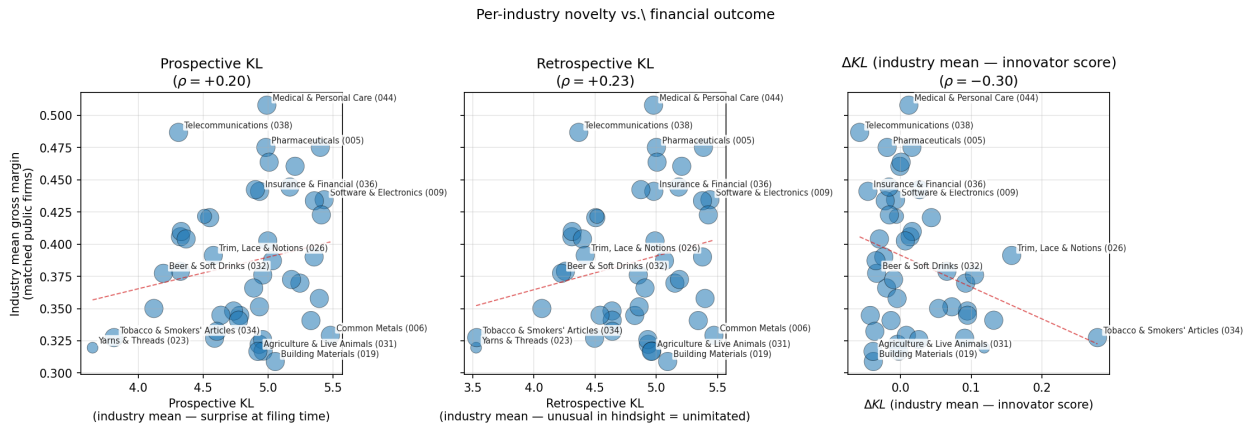


Figure 21: Each dot is one of the 45 NICE international classes; dot size is proportional to the number of matched public firms in the class. Left panel: industry-mean prospective KL (surprise at filing time) on the x -axis. Centre: industry-mean retrospective KL (remained-novel-in-hindsight). Right: industry-mean ΔKL (imitated-innovator score). All three are plotted against industry-mean gross margin on the y -axis. Dashed lines are OLS fits, ρ is the Pearson correlation across the 45 industries.

Table 27: Per-industry summary statistics, restricted to filings from 1995–2020 with both 5-year reference windows populated.

Industry (NICE)	Filings	Clean	Owners	Years	ΔKL med.	ΔKL mean	5y reg. rate	Reg. rate	Top 3 firms by firm-mean ΔKL (≥ 3 filings)
Tobacco & Smokers' Articles (034)	82,520	32,818	16,646	1990–2020	+0.15	+0.29	50%	56%	PHILLIPS & KING INTERNATIONAL, INC. (+3.12); AUDIO TECHNOLOGY OF NEW YORK INC. (+2.75); BAMBÚ SALES, INC. (+2.68)
Trim, Lace & Notions (026)	84,034	35,638	24,532	1990–2020	+0.10	+0.17	36%	43%	Selected Trading Corp. (+2.49); LITTLE ROSEY ENTERPRISES, INC. (+2.19); XUCHANG BEST TRADING CO.,LTD (+1.93)
Ropes, Tents & Textiles (raw) (022)	47,287	21,551	14,335	1990–2020	+0.08	+0.16	31%	42%	International Hairgoods, Inc. (+2.10); TENSILE SHADE PRODUCTS, LLC (+1.96); Hoechst Celanese Corporation (+1.88)
Household Utensils (021)	403,933	183,800	114,891	1990–2020	+0.06	+0.12	36%	43%	BARRON, THOMAS ARCHIBALD (+2.90); Franzus Company, Inc. (+2.82); Villeroy & Boch S.a.r.l (+2.60)
Yarns & Threads (023)	15,210	2,889	1,756	2006–2020	+0.10	+0.12	24%	31%	QINGDAO BANGTE FIBRE CO., LTD. (+0.82); Champion Thread Company (+0.72); Jiangsu Lugang Science & Technology Co., (+0.65)
Hand Tools (008)	128,745	62,591	38,099	1990–2020	+0.06	+0.12	32%	42%	EDDIE BAUER, INC. (+2.36); O.M.S. Inc. (+2.26); WASHINGTON FORGE, INCORPORATED (+2.08)
Carpets & Wall Hangings (027)	56,986	22,964	12,990	1990–2020	+0.07	+0.12	35%	43%	Hallmark Cards, Incorporated (+2.31); Armstrong World Industries, Inc. (+1.85); Kentile Floors Inc. (+1.85)
Furniture (020)	326,621	144,864	87,163	1990–2020	+0.04	+0.11	36%	44%	Express Yourself, Inc. (+3.02); Helikon Furniture Company, Inc. (+2.93); Ohio Mattress Company Licensing And Comp (+2.70)
Lighting, HVAC & Appliances (011)	333,507	171,839	95,942	1990–2020	+0.04	+0.09	31%	41%	Hiros, Inc. (+2.79); Melitta-Werke Bentz & Sohn (+2.66); Water Resources International Incorporat (+2.63)
Textiles & Bedding (024)	173,759	84,770	50,657	1990–2020	+0.05	+0.08	38%	45%	FORBES INC. (+2.55); AD PROPERTIES CORP. (+2.47); Hallmark Cards, Incorporated (+2.46)
Firearms & Pyrotechnics (013)	37,403	12,432	6,331	1992–2020	+0.07	+0.08	33%	47%	Wright Leather Works LLC (+1.52); G.A. Precision LLC (+1.43); MAGPUL INDUSTRIES CORP. (+1.13)
Wine & Spirits (033)	222,138	61,865	29,925	1990–2020	+0.03	+0.07	46%	53%	Two Beers Brewing Company (+2.60); SNOW BRAND MILK PRODUCTS CO., LTD. (+2.50); Bodegas "El Pomar" S. A. de C.V. (+2.48)
Vehicles (012)	242,485	116,643	61,001	1990–2020	+0.01	+0.06	34%	45%	MIYATA INDUSTRY CO., LTD. (+3.53); Marui Co., Ltd. (+3.52); JENSEN CAR COMPANY LIMITED (+3.28)
Jewelry & Watches (014)	234,908	98,618	64,105	1990–2020	+0.02	+0.06	38%	45%	BURRELL, KIRK (+2.63); IRENEW BIO ENERGY SOLUTIONS LLC (+2.51); National Trading Mfg., Inc. (+2.51)
Medical Devices (010)	302,192	174,201	79,685	1990–2020	-0.01	+0.05	38%	48%	Mikrona Technologie AG (+2.86); AudioScience, Inc. (+2.85); Inventions Extraordinaire Inc. (+2.82)
Toys, Games & Sporting Goods (028)	583,108	292,436	144,975	1990–2020	-0.01	+0.04	42%	49%	American Video Entertainment, Inc. (+3.21); VIRUSTATIC INC. (+3.07); Abu Aktiebolag (+2.93)
Cosmetics & Cleaning (003)	564,014	300,326	139,169	1990–2020	+0.02	+0.04	43%	51%	MARKETING SERVICES OF BEND, LLC (+3.64); Image Laboratories, Inc. (+3.06); PALEO PAW CORP. (+2.95)
Common Metals (006)	175,437	89,730	51,240	1990–2020	+0.01	+0.04	29%	45%	Shell Oil Company (+3.03); SOMBUR MACHINE AND TOOL (+2.64); LANAI RESORT PARTNERS (+2.47)
Leather & Luggage (018)	249,052	133,408	84,643	1990–2020	-0.01	+0.04	38%	45%	Tonka Corporation (+2.97); Grifs Western, Inc. (+2.72); Impuls International, Inc. (+2.62)
Rubber & Insulators (017)	88,430	48,186	24,876	1990–2020	+0.00	+0.04	28%	44%	Filmco Industries, Inc. (+2.31); Oregon Research and Development Corporat (+1.97); Stone Container Corporation (+1.86)
Lubricants & Fuels (004)	78,345	32,824	17,794	1990–2020	-0.02	+0.03	38%	49%	Ashland Oil, Inc. (+2.66); Geo. Pfau's Sons Company, Inc. (+1.92); TOPCO ASSOCIATES, INC. (+1.62)
Restaurants & Hotels (043)	285,610	122,371	74,815	2003–2020	+0.02	+0.03	39%	49%	Burger Bench, LLC (+2.82); RIB CITY GROUP, LLC (+2.32); Pizza Joints Inc. (+1.83)
Software & Electronics (009)	1,807,636	1,146,078	558,481	1990–2020	-0.02	+0.02	41%	49%	Til Systems Limited (+3.60); Northgate Computer Systems, Inc. (+3.30); JAMES RIVER II, INC. (+3.24)
Machines & Power Tools (007)	286,930	148,670	76,188	1990–2020	-0.02	+0.02	27%	41%	Republic Machinery Co., Inc. (+2.99); Rietschle, Inc. (+2.64); NLB CORP. (+2.59)
Coffee, Tea, Bakery & Spices (030)	428,854	192,476	97,520	1990–2020	-0.01	+0.02	43%	52%	GMB Enterprises, Inc. (+3.24); Buy The Weigh. Inc. (+3.07); Cafeteleira Zardain Hermanns S.A. de C.V. (+2.96)
Paper, Print & Office (016)	712,163	419,972	229,601	1990–2020	-0.01	+0.02	41%	51%	LawPrep, Inc. (+2.84); Anderson Import-Export Company, Inc. (+2.79); CLASSIC GAMES, INC. (+2.77)
Education & Entertainment (041)	1,344,519	850,779	491,343	1990–2020	-0.01	+0.02	38%	48%	Corporacion Wilfrido Vargas, S.A. (+3.98); CENTURY OCN PROGRAMMING, INC. (+3.86); Video Challenge, Inc. (+3.67)
Materials Treatment (040)	145,590	87,017	55,201	1990–2020	-0.01	+0.02	33%	45%	Re-USA-ble, Inc. (+3.33); Lynn, Robert C (+3.05); SUNMARKS, INC. (+3.00)
Chemicals & Industrial (001)	218,420	118,548	49,757	1990–2020	-0.01	+0.02	33%	48%	Seattle FilmWorks, Inc. (+3.07); Artis, Inc. (+2.86); International Flavors & Fragrances, Inc. (+2.57)
Construction & Repair (037)	284,361	166,789	99,723	1990–2020	-0.00	+0.01	30%	43%	VERITECHNOLOGY ELECTRONICS CORPORATION (+3.12); All State Plumbing Inc. (+2.73); American Lubrication Services, Inc. (+2.69)
Agriculture & Live Animals (031)	136,656	59,590	30,003	1990–2020	-0.03	+0.01	40%	50%	Oxnard Vegetable Exchange, Inc. (+3.37); CENTRAL GARDEN SUPPLY (+3.37); PET-AG, INC. (+3.32)
Medical & Personal Care (044)	312,270	167,607	97,110	2003–2020	-0.00	+0.01	37%	46%	Golden, Joseph (+2.16); Logue, Charles G. (+2.16); My Therapy Session, Inc. (+1.87)
Pharmaceuticals (005)	600,473	336,093	124,260	1990–2020	-0.01	+0.01	49%	57%	Marlyn Company, Inc. (+4.80); WINNING LAB. INC. (+4.46); P. Leiner Nutritional Products, Inc. (+4.34)
Scientific & Tech Services (042)	1,148,416	692,938	387,431	1990–2020	+0.01	+0.01	39%	47%	SOUTHWEST TEXAS METHODIST HOSPITAL (+3.54); Treehouses of America, Inc. (+3.48); Premier Cruise Lines, Ltd. (+3.37)
Beer & Soft Drinks (032)	206,304	94,920	46,682	1990–2020	-0.02	+0.01	51%	57%	BONFARE FOOD STORES, INC. (+4.24); WILLINGER, DOUGLAS A. (+4.24); Kola Colombiana Inc. (+4.20)
Meat, Dairy & Prepared Foods (029)	253,936	127,008	60,556	1990–2020	-0.03	+0.01	43%	52%	Mrs. Kinney's Kitchen, Inc. (+2.98); Seven Seas Foods, Inc. (+2.98); Curtice-Burns Foods, Inc. (+2.94)
Building Materials (019)	122,596	66,282	33,012	1990–2020	-0.02	+0.00	32%	46%	Manville Corporation (+2.86); FLORIDA TILE INDUSTRIES, INC. (+2.65); SENECA TILES, INC. (+2.65)
Paints & Coatings (002)	64,892	31,911	15,703	1990–2020	-0.04	+0.00	30%	46%	Tintas Coral S/A (+3.15); ABC Protective Coating Corporation (+2.56); Graphic Color Corporation (+2.49)
Legal & Personal Services (045)	203,476	114,693	74,406	2003–2020	-0.01	+0.00	36%	46%	Sproose, Inc. (+1.96); Katz, Daniel (+1.81); SkyWard IO, Inc. (+1.74)
Transport & Logistics (039)	187,535	112,578	63,048	1990–2020	-0.01	+0.00	34%	46%	Alamo Rent-A-Car (+3.42); Novair International Airways Ltd. (+3.03); First Family of Travel, Ltd. (+2.88)
Advertising & Retail (035)	1,418,345	924,501	542,287	1990–2020	-0.03	-0.00	38%	47%	TRATEC OF CALIFORNIA, INC. (+3.77); Pioneer TeleTechnologies, Inc. (+3.65); American Family Publishers (+3.65)
Clothing, Footwear & Headwear (025)	1,265,966	604,850	382,965	1990–2020	-0.05	-0.01	47%	53%	Pacific Coast Management Team (+3.38); Ashley Productions, Inc. (+3.35); Carter, Gregory, C (+3.35)
Musical Instruments (015)	31,792	11,068	6,747	1993–2020	-0.00	-0.01	31%	45%	Original Fuzz, LLC (+1.43); Sun Jianjun (+0.85); SUNLAND INFORMATION TECHNOLOGY CO., LTD. (+0.82)
Insurance & Financial (036)	602,901	374,634	183,490	1990–2020	-0.05	-0.02	36%	48%	MEDVITT, MICHAEL J. (+3.66); Visconti, James A. (+3.65); Medical Bill Management, Inc. (+3.63)
Telecommunications (038)	230,267	162,634	82,158	1990–2020	-0.03	-0.03	46%	52%	United Artists Holding, Inc. (+4.37); Ingram, William H. (+4.21); Ohringer, Sheldon S. (+3.91)

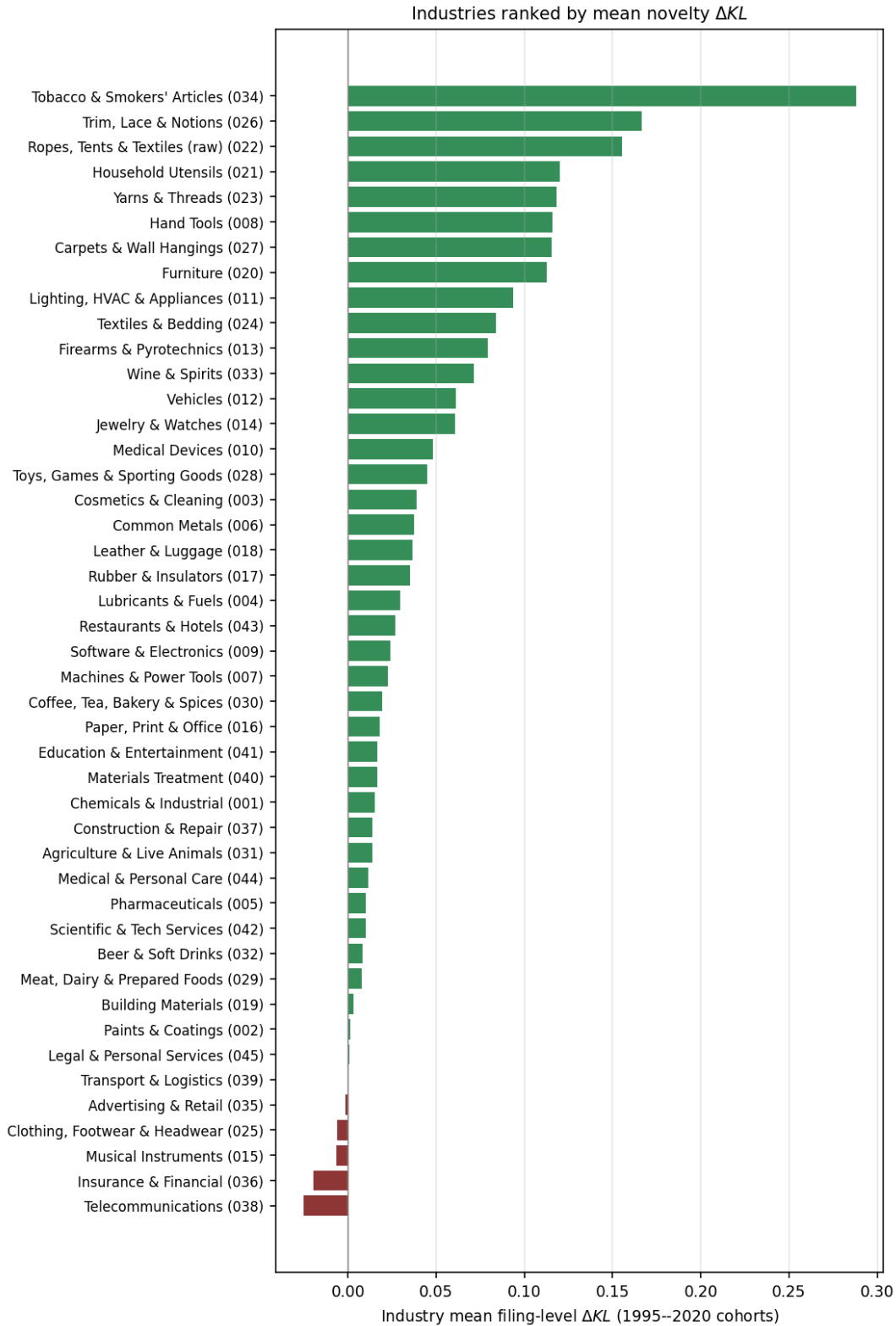


Figure 22: Industries ranked by mean filing-level ΔKL across 1995–2020 cohorts. Green: industries whose typical filing was prospectively novel and retrospectively obvious (the future imitated the past). Red: industries whose typical filing was the opposite. Tobacco & Smokers' Articles tops the list, driven by the 2014–2020 e-cigarette / vape vocabulary explosion. Telecommunications, Insurance & Financial, and Paints & Coatings sit at the bottom: their goods-and-services lexicon was already settled before 1995 and has been stable since.

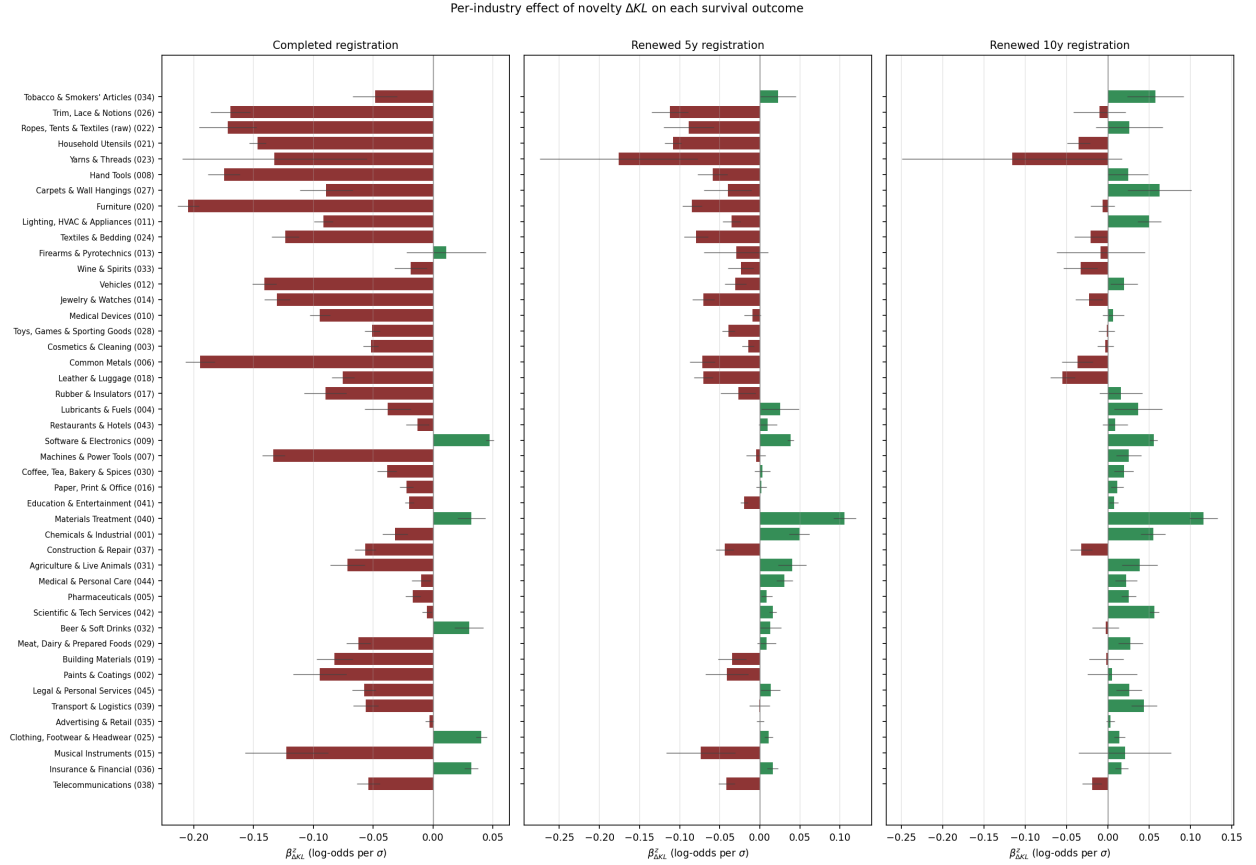


Figure 23: Per-industry effect of ΔKL on survival outcomes (logistic-regression z -score coefficient with 95% CI). Industries are ordered by mean ΔKL (the same order as Figure 22). **The sign of the innovation reward varies by industry.** Some industries (e.g. Common Metals, Hand Tools, Paints & Coatings) actually *reward* novel filings at the registration gate, presumably because in those industries a novel goods/services description can describe a real new use the examiner recognises as patentable activity. Most industries match the software pattern: novelty penalised at registration, neutral or favourable at renewal.

Table 28: Top-1 highest- ΔKL mark per industry, 2010–2020 cohort. The 2014–2020 cannabis/CBD/blockchain wave appears in nearly every consumer-goods class.

Industry (NICE)	Year	ΔKL	Mark	Owner at filing
Tobacco & Smokers' Articles (034)	2016	+4.51	DAISY	Green Tank Technologies Corp.
Agriculture & Live Animals (031)	2018	+4.20	UNITED AMERICAN HEMP	United American Hemp LLC
Cosmetics & Cleaning (003)	2018	+4.05	LEEF	PALEO PAW CORP.
Chemicals & Industrial (001)	2017	+4.03	REEFERTILIZER	Reefertilizer Inc.
Coffee, Tea, Bakery & Spices (030)	2012	+3.93	RED & WHITE	Forefront Enterprises, LLC
Household Utensils (021)	2017	+3.59	IREPELLER	10093939 Canada Corp.
Beer & Soft Drinks (032)	2018	+3.56	KANNABIER	Advanced Creative Intelligent
Wine & Spirits (033)	2014	+3.49	CHATEAU DE CAYX	Boutique du Château de Cayx
Ropes, Tents & Textiles (raw) (022)	2013	+3.33	SUNAMI	TENSILE SHADE PRODUCTS, LLC
Scientific & Tech Services (042)	2014	+3.30	ZENPOOL	Geniuses at Work Corporation
Toys, Games & Sporting Goods (028)	2012	+3.20	BETCLOUD	BALLY GAMING, INC.
Materials Treatment (040)	2018	+3.17	VERITAS — FARMS	271 Lake Davis Holdings LLC
Meat, Dairy & Prepared Foods (029)	2016	+3.15	PHIVIDA	PHIVIDA HOLDINGS INC.
Jewelry & Watches (014)	2013	+3.11	OFF THE CLOSET	Rahhal, Nour
Software & Electronics (009)	2018	+3.09		Unify Earth UEX
Furniture (020)	2016	+3.07	GO PET CLUB	Pel Wholesale, Inc.
Paper, Print & Office (016)	2020	+3.05	KAYLEEMAY LLC	Fall, Cayle M
Clothing, Footwear & Headwear (025)	2010	+3.03	CHARMANT FEMME	2BB UNLIMITED, INC.
Medical Devices (010)	2020	+3.03	ELASTIMASK	Bedford Industries, Inc.
Education & Entertainment (041)	2017	+3.01	ACADEMY	Kingsland School of Blockchain
Trim, Lace & Notions (026)	2015	+2.99	WOWQUEEN	XUE YONG BO
Advertising & Retail (035)	2018	+2.85	OPENSEA	Ozone Networks, Inc.
Lubricants & Fuels (004)	2017	+2.82		Mary Jane's Medicinals, LLC
Pharmaceuticals (005)	2017	+2.74	CANINES BEST DEFENSE CBD	Nobrega, Kathryn
Insurance & Financial (036)	2017	+2.71	MOSAIC TOKEN	MOSAIC RESEARCH LIMITED
Medical & Personal Care (044)	2018	+2.64	FARMING A HEALTHIER FUTURE	Southern Hemp, LLC
Carpets & Wall Hangings (027)	2016	+2.63	LOVEMAT	Wuhan Xuanyuanlong Science and
Transport & Logistics (039)	2017	+2.63	MARIJUANA EXPRESS	Marijuana Express LLC
Lighting, HVAC & Appliances (011)	2010	+2.42	KESSIL	DiCon Fiberoptics, Inc.
Vehicles (012)	2013	+2.34	PRECISION DRONE WWW.PRECISIONDRO	Precision Drone, LLC
Leather & Luggage (018)	2018	+2.31	NOMADZ	Ambit Global
Textiles & Bedding (024)	2017	+2.27	BESTDROP	Zhiyong Xie
Legal & Personal Services (045)	2011	+2.26	DIAMONDS IN THE RUFF	CINDY ROSS
Restaurants & Hotels (043)	2013	+2.20	GREEN FEET BREWING	Green Feet Brewing
Hand Tools (008)	2016	+2.02	THE NAZOR	Nazor, LLC
Firearms & Pyrotechnics (013)	2010	+1.94	ALPHA RAIL	TROY INDUSTRIES, INC.
Machines & Power Tools (007)	2017	+1.86	COUNTERMATE	IMPAK Corporation
Musical Instruments (015)	2012	+1.75	ERGODOCK	Hickey, Bruce
Common Metals (006)	2017	+1.70	UP YOUR CASTER	CASTER CONNECTION, LLC
Yarns & Threads (023)	2011	+1.69	BERROCO NANUK	Berroco, Inc.
Rubber & Insulators (017)	2014	+1.69	B3D	Brandon Bell
Construction & Repair (037)	2010	+1.52	WILKS MASONRY	Wilks Masonry Corporation
Building Materials (019)	2012	+1.51	FREE FORM YOGA FLOOR	Isabelle J. Barger
Telecommunications (038)	2017	+1.44	NCRYPT	NCHAIN HOLDINGS LTD.
Paints & Coatings (002)	2016	+1.32	TRIPLE BEST	Sike Wu

Table 29: Selected filings with per-token KL contributions to prospective and retrospective surprise. **Surprise is an interaction across tokens, not a property of any single word.** A filing’s KL score is $\sum_v P(v) \log[P(v)/Q(v)]$, summed across every token v that appears in the filing’s goods/services description. The token **software** is by itself common in Class 9, but its *combination* with rare neighbors — **generative artificial, conversational agents, cryptographic keys** — can give a filing high prospective surprise relative to the look-behind window’s mix of tokens. Tokens shown below are the top five by per-token contribution $P(v) \log[P(v)/Q(v)]$ to each direction’s KL; common tokens like **software** can appear in the prospective list when their multinomial weight in the filing is much larger than their weight in the look-behind reference.

Mark / Year	Owner at filing	KL_{pros}	KL_{retr}	ΔKL	Top tokens driving prospective surprise	Top tokens driving retrospective surprise
OPENAI (2016)	OpenAI, Inc.	7.65	5.56	+2.09	software artificial , artificial intelligence , intelligence , artificial , software	software artificial , artificial intelligence , artificial , intelligence , software
CLAUDE (2023)	Anthropic, PBC	7.12	5.06	+2.06	model performing , document question-answering , tasks natural , generative text , performing generative appearance enabling , modifying appearance , software modifying , enabling transmission , appearance	tasks , natural , model performing , question-answering simulating , document question-answering appearance enabling , modifying appearance , software modifying , enabling transmission , appearance
INSTAGRAM (2011)	INSTAGRAM, LLC	7.16	6.03	+1.13	appearance , software modifying , enabling transmission , appearance	appearance , software modifying , enabling transmission , appearance
PHOTOSHOP (1992)	ADOBE INC.	4.81	4.83	-0.01	manipulating graphic , creating manipulating , images computer , graphic images , manipulating	manipulating graphic , creating manipulating , images computer , graphic images , programs creating
KUBERNETES (2014)	LF PROJECTS, LLC	4.70	4.62	+0.08	servers secure , secure private , private networks , networks , private	servers secure , secure private , private networks , networks , private
WHITE CLAW SELTZER WORKS (2016)	MARK ANTHONY INTERNATIONAL S	9.12	7.35	+1.78	sugar-based beer , sugar-based , seltzer , beer	sugar-based beer , sugar-based , seltzer , beer
LIQUID DEATH (2023)	Supplying Demand, Inc.	3.97	3.80	+0.17	drink mixes , water nutritional , mixes concentrates , water sparkling , concentrates	drink mixes , mixes concentrates , water nutritional , concentrates , flavor enhanced
HARD MTN DEW (2021)	PEPSICO, INC.	4.81	4.96	-0.15	flavored brewed , brewed , brewed malt-based , malt-based alcoholic , nature beer	flavored brewed , brewed , brewed malt , nature beer , malt-based alcoholic
ROCKSTAR (2002)	ROCKSTAR ENERGY LLC	4.74	3.92	+0.81	drinks , energy drinks , energy	drinks , energy drinks , energy

Limitations

- **Goods/services text is one of several novelty channels.** Brand novelty (LIQUID DEATH), design novelty (AIRPODS casing), and pricing novelty are invisible to this corpus. Estimates here should be read as innovation-in-business-description novelty.
- **The crosswalk is exact-match only.** A fuzzy-match second pass and manual review of the top-by-volume USPTO owners would plausibly double or triple the matched panel size and reduce survivorship bias toward firms that did not change their registered name.
- **Survivorship bias in the financial panel.** Firms appear in SEC FSDS only while public. Pre-IPO innovators (Anthropic, OpenAI, Stripe at most timepoints) and dead-but-once-public firms drop out. The financials reward we estimate is conditional on already being a public concern.
- **The reference distributions are smoothed multinomials, not topic models.** An LDA replication is the obvious artifact-check.

Next steps

The cross-industry universal sweep (all 45 NICE classes pooled) would pin down which industries are innovative by this measure and would allow new-class births (cannabis beverages 2018, generative AI tools 2022) to be scored against a cross-industry background. A fuzzy-match crosswalk would expand the financial panel from $\sim 1,400$ firms toward the $\sim 5,000$ ceiling of the SEC EDGAR universe. With either or both extensions in place, the natural follow-up is a *variance* regression: do high- ΔKL firms exhibit higher margin variance, the empirical fingerprint of pioneers either spectacularly succeeding or spectacularly failing?

A Per-class summary statistics

Industry (NICE)	Filings	Clean	Years	Owners	ΔKL med.	p5	p95
Software & Electronics (009)	1,807,636	1,515,300	1958–2025	754,869	-0.02	-0.36	+0.45
Beer & Soft Drinks (032)	206,304	125,512	1980–2025	62,180	+0.01	-0.53	+0.60

B Selected filings, full text

OPENAI (2016) Owner at filing: *OpenAI, Inc.*. $KL_{\text{pros}} = 7.65$, $KL_{\text{retr}} = 5.56$, $\Delta KL = +2.09$.
 Goods/services text: *Computer software for artificial intelligence*

CLAUDE (2023) Owner at filing: *Anthropic, PBC*. $KL_{\text{pros}} = 7.12$, $KL_{\text{retr}} = 5.06$, $\Delta KL = +2.06$.
 Goods/services text: *Downloadable computer software in the nature of an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a theme, summarizing text, document question-answering and simulating natural conversation; downloadable software in the nature of a downloadable mobile application featuring an artificial intelligence model for performing generative text AI tasks and natural language processing AI tasks and for writing content based on a t...*

INSTAGRAM (2011) Owner at filing: *INSTAGRAM, LLC*. $KL_{\text{pros}} = 7.16$, $KL_{\text{retr}} = 6.03$, $\Delta KL = +1.13$.

Goods/services text: *Downloadable computer software for modifying the appearance and enabling transmission of photographs*

PHOTOSHOP (1992) Owner at filing: *ADOBE INC.*. $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.83$, $\Delta KL = -0.01$.

Goods/services text: *computer programs for creating and manipulating graphic images on a computer [and manuals for use therewith, sold as a unit]*

KUBERNETES (2014) Owner at filing: *LF PROJECTS, LLC*. $KL_{\text{pros}} = 4.70$, $KL_{\text{retr}} = 4.62$, $\Delta KL = +0.08$.

Goods/services text: *Computer software for accessing, browsing, sharing, defining, maintaining, virtualizing and communicating information over computer networks, servers and secure private networks; computer software for use in creating, managing, controlling, changing, replicating, deploying, naming and linking information over computer networks, servers and secure private networks; computer software for assisting users in navigating through computer networks, servers and secure private networks; computer software for running web app. . .*

WHITE CLAW SELTZER WORKS (2016) Owner at filing: *MARK ANTHONY INTERNATIONAL SRL*. $KL_{\text{pros}} = 9.12$, $KL_{\text{retr}} = 7.35$, $\Delta KL = +1.78$.

Goods/services text: *Hard seltzer Brewed sugar-based beer*

LIQUID DEATH (2023) Owner at filing: *Supplying Demand, Inc.*. $KL_{\text{pros}} = 3.97$, $KL_{\text{retr}} = 3.80$, $\Delta KL = +0.17$.

Goods/services text: *Powdered drink mixes for making flavored water, sports drinks, and energy drinks; powders and concentrates for making flavor enhanced water and sparkling water Powdered nutritional supplement drink mixes and concentrates*

HARD MTN DEW (2021) Owner at filing: *PEPSICO, INC.*. $KL_{\text{pros}} = 4.81$, $KL_{\text{retr}} = 4.96$, $\Delta KL = -0.15$.

Goods/services text: *flavored brewed malt-based alcoholic beverages in the nature of beer Alcoholic flavored brewed malt beverages, except beers*

ROCKSTAR (2002) Owner at filing: *ROCKSTAR ENERGY LLC*. $KL_{\text{pros}} = 4.74$, $KL_{\text{retr}} = 3.92$, $\Delta KL = +0.81$.

Goods/services text: *Sports drinks, namely, energy drinks*

C Falsification battery for the U-shape

The compact summary in Section 17 (Table 25) listed seven candidate explanations for the survival U-shape on ΔKL and the outcome of each re-fit. This appendix narrates each check in full.

(1) Foreign-extension hypothesis is rejected. A common worry is that the very-stale (low- ΔKL) end is inflated by Madrid-Protocol or Section-44 international extensions of marks already registered abroad — such marks would use vocabulary established years before they reached the US filing record. Using a foreign-owner proxy (name suffixes GMBH, S.A., AB, KK, Pty, Pte, PLC, N.V., B.V., etc.; 3.8% of the sample) and re-fitting on US-only and foreign-only sub-samples, the quadratic coefficient is essentially identical (+0.0078 and +0.0100 respectively). The U-shape is not a foreign-extension artefact.

(2) The U is present in both first-time and recurring filings ($\beta_{\text{quad}} = +0.0080$ vs. +0.0140), with recurring filings showing a slightly stronger curvature. The fast-followers-do-best pattern from Section 7 is consistent with stronger curvature in the recurring subset (firms that are still filing are firms that are still operating).

(3) The financial U-shape is essentially absent. A quadratic refit of the firm-year gross-margin OLS gives a linear coefficient indistinguishable from zero ($\beta_{\text{lin}} = +0.002$, SE 0.005) and a marginally negative quadratic ($\beta_{\text{quad}} = -0.002$, SE 0.001) on 8,600 firm-years. **The U-shape lives in survival; gross margin scales roughly linearly with ΔKL .** The +3.2 pp / σ linear effect we report in Section 12 is the right summary.

(4) Survivor bias from large incumbents partially supports the left arm. Mean firm filing-volume by ΔKL decile shows the bottom decile is dominated by larger filers (mean firm-size 103.7 filings) than the middle deciles (63–71); the top decile is back to 71. Templated marks disproportionately come from firms with deep portfolios who can afford to file — and maintain — many similar boilerplate marks. This is consistent with part of the left-arm uplift being *firm capability* rather than *filing distinctiveness*.

(5) The “Liquid Death” hypothesis is rejected. A sceptic might worry that the bottom-decile arm is high-novelty firms whose products are described in old vocabulary — like Liquid Death’s canned water marketed as “still water; sparkling water.” Splitting the bottom decile by goods/services text length does not show this: the shortest quartile and the longest quartile both renew at 44–46% at five years. Templated marks are not novel products in disguise; they really are templated, and they really do survive.

(6) The U is filing-level, not firm-level. A further worry: the U-shape might be a property of the firms that file rather than of any given filing — “on-trend” firms might simply fail more often, dragging their filings down, while “harvest” and “innovate” firms might survive and keep their filings alive collectively. To test this we classified each firm with ≥ 3 clean filings by whether *all* of its filings survived 5y (“always-survives,” 197,583 firms), *all* failed (“never-survives,” 253,064 firms), or some survived while others were let lapse (“mixed,” 349,819 firms). In the mixed cohort — firms that demonstrably persisted past at least one Section-8 lapse — the U-shape in filing-level survival by ΔKL is preserved: bottom-decile rate 0.551, middle 0.479, top 0.508 (depth $\approx +0.07$ relative to middle). Within the same firm, filings whose vocabulary places them at either flux extreme are more likely to be the ones the firm keeps alive than the flux-neutral filings the firm lets lapse. The U is therefore a property of the specific filing, not just of the firm behind it; trademarks on the flux-neutral ridge really are less valuable to the firms that file them. Figure 19 shows the three cohorts.

(7) Imitated-but-failed waves: the blockchain check. A natural counterexample to the “imitated novelty pays” claim is to ask whether any recognisable hype wave was both widely imitated (high ΔKL) and disproportionately a failure. The blockchain / cryptocurrency / NFT / web3 cohort is a clean test: 87,463 filings 1995–2025 mention at least one crypto-related keyword in their goods/services text; 9,408 are clean by our usual restrictions. The cohort’s mean ΔKL is +0.412 vs. −0.002 for the corpus average — a genuinely high-novelty wave. Survival-wise the wave is *out-performing*, not failing: registration rate 51.5% vs. 48.3% corpus mean; five-year survival 51.4% vs. 39.1% corpus mean. Restricted to the top decile of ΔKL , the crypto cohort beats non-crypto top-decile filings on both margins (57.3% vs. 47.5% on registration; 57.2% vs. 40.9% on five-year survival). Whatever else can be said about the blockchain wave’s commercial track record, the trademark filings themselves are surviving well, consistent with the same likelihood-of-confusion mechanic — the crypto vocabulary did not yet exist in the trademark register before the wave, so applicants did not collide with prior marks.

(8) Temporal stability: the U is present year-by-year. The cleanest test of whether the pooled U is an aggregation artefact is to refit the quadratic separately for each filing-year cohort and ask whether the U appears in individual years or only when years are pooled. We fit a logistic regression $P(\text{survived}_{.5y}) = \text{logit}^{-1}(\beta_0 + \beta_1 z(\Delta KL) + \beta_2 z(\Delta KL)^2 + \gamma \log(n_{\text{terms}}))$ separately for each year 1990–2021, both pooled across all 44 classes and for five representative classes covering the range of cross-sectional outcomes (Pharmaceuticals 005, Software & Electronics 009, Paper & Print 016, Beer & Soft Drinks 032, Telecommunications 038). Figure 20 plots the per-year β_2 with 95% CIs alongside NBER recession bars; the underlying per-year counts are in `u_shape_temporal_stability.csv`.

The pooled per-year coefficient is positive in 24 of 32 years and *significantly* positive ($t > 1.96$) in 23 of 32. It is significantly inverted in only 4 of 32 years. The U is present in the 1990s, in the 2000s, and in the 2010s; it strengthens slightly in the 2020 cohort but does not depend on a single decade. Pharmaceuticals is the cleanest case at the class level: 22/32 years significantly positive, 0 significantly inverted. Software is 19/32 significantly positive, 2 significantly inverted, with the inverted years concentrated in the late-1990s pre-dot-com cohort. Telecommunications is genuinely inverted in 10/32 years; the inverted-U cross-sectional result for that class is not a pooling artefact but a real industry-level phenomenon.

The U does not track the macro cycle in a clean way. The 2001 dot-com recession is the one year in which the pooled coefficient softens (it remains positive but only marginally significant); the 2008–09 financial crisis leaves the pooled coefficient roughly unchanged; the 2020 recession shows a *strengthening* of the U. Recessions do not produce or destroy the U-shape; the U-shape is more stable than the macro cycle is.

The temporal-stability test answers the question of whether the U is a property of specific vintages of firms that happen to cluster at flux-extreme positions. It is not: the pattern holds in nearly every cohort independently.

D Per-industry deep dives

For each NICE class with at least 5,000 clean H=2y filings in the 1990–2021 window, this appendix gives summary statistics, the top three firms by firm-mean ΔKL (firms with ≥ 10 filings), and one worked example each from the positive-flux extreme, the flux-neutral middle, and the negative-flux extreme. The brief cross-industry table in Section 19 (`industry_summary.tex`) gives an aggregated view; this appendix is the per-class expansion for readers who want to drill into a specific domain.

Class 001: Chemicals & Industrial

Clean filings:	125,272
Unique owners:	52,278
Year range:	1990–2021
Mean KL_{pros} :	5.36 nats
Mean KL_{retr} :	5.37 nats
Mean / median ΔKL :	-0.009 / -0.010
5y survival rate:	47.1%
Share with SEC-listed owner:	6.47%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): HerbalScience Singapore Pte Ltd (+1.20); HERBALSCIENCE, INC. (+1.04); NICCA U.S.A., Inc. (+1.03)

Worked examples:

Positive flux: REEFERTILIZER (2017, $\Delta KL = +3.49$):

Plant nutrients, for use only with non-cannabis plants and cannabis with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis

Flux-neutral: COSMCPREP (2003, $\Delta KL = +0.00$):

reagents and magnetic beads used to purify plasmid, cosmid, fosmid, and BAC DNA for scientific research

Negative flux: CLOUD CANDY (2014, $\Delta KL = -1.83$):

Electronic cigarette liquid (e-liquid) comprised of propylene glycol; Electronic cigarette liquid (e-liquid) comprised of vegetable glycerin

Class 002: Paints & Coatings

Clean filings:	33,653
Unique owners:	16,695
Year range:	1990–2021
Mean KL_{pros} :	4.78 nats
Mean KL_{retr} :	4.79 nats
Mean / median ΔKL :	-0.012 / -0.023
5y survival rate:	44.6%
Share with SEC-listed owner:	4.40%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): J. M. Huber Corporation (+0.68); AS-CONN PRINT TECHNOLOGY CO.,LTD (+0.57); CLOVER TECHNOLOGIES GROUP, LLC (+0.53)

Worked examples:

Positive flux: BLOXX-IT (1990, $\Delta KL = +2.38$):

paint primer, bonder and sealer coating for interior and exterior use

Flux-neutral: OVP (1997, $\Delta KL = -0.00$):

light interference pigments for use in such goods as inks, dyes, paints, plastics, and textiles

Negative flux: OPTICS BY SOLA (1999, $\Delta KL = -1.50$):

Coatings for ophthalmic lenses, anti-reflective coatings for ophthalmic lenses, abrasion resistant coatings for ophthalmic lenses; impact resistant coatings for ophthalmic lenses, anti-fog coatings fo

Class 003: Cosmetics & Cleaning

Clean filings:	326,430
Unique owners:	154,551
Year range:	1990–2021
Mean KL_{pros} :	4.32 nats
Mean KL_{retr} :	4.31 nats
Mean / median ΔKL :	+0.005 / +0.005
5y survival rate:	50.0%
Share with SEC-listed owner:	2.94%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): MNG 2005 INC (+1.46); Revlon-Realistic Professional Products, Inc. (+1.19); Infused MFG, LLC (+0.94)

Worked examples:

Positive flux: ADARA (2008, $\Delta KL = +2.85$):

Hair care preparations consisting of organic coconut virgin oil and coconut virgin oil; Non-medicated cosmetic skin care preparations consisting of organic coconut virgin oil and coconut virgin oil

Flux-neutral: ELEANOR MAE COLLECTION (2002, $\Delta KL = +0.00$):

non-medicated foot cream; skin lotion; milk bath; lip balm; goat milk soap

Negative flux: PICKLED ESSENCE FAMILY (2009, $\Delta KL = -1.55$):

Aromatherapy pillows that may be used to freshen clothing or lay over the eyes

Class 004: Lubricants & Fuels

Clean filings:	36,160
Unique owners:	19,926
Year range:	1990–2021
Mean KL_{pros} :	4.91 nats
Mean KL_{retr} :	4.91 nats
Mean / median ΔKL :	+0.004 / -0.017
5y survival rate:	47.8%
Share with SEC-listed owner:	5.58%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Frostbeard Studio, LLC (+0.81); hep global GmbH (+0.59); Victoria's Secret Stores, Inc. (+0.55)

Worked examples:

Positive flux: HEMPTHOLOGY (2018, $\Delta KL = +2.82$):

Candles; Wax melts; all of the foregoing goods containing CBD derived solely from hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis

Flux-neutral: TE HACE FALTA PRENDER MAS CARBON! (2017, $\Delta KL = +0.00$):

Charcoal briquettes; Coal briquettes; Firewood; Lighter fluid; Lighter fluid for charcoal; Wood chips for use as fuel; Wood chips of mesquite for smoking and grilling; Solid fire starters

Negative flux: HARNESSING THE POWER OF HEMP, TO POWER Y (2021, $\Delta KL = -1.92$):

Candles containing hemp with a delta-9 tetrahydrocannabinol concentration of not more than 0.3 percent on a dry weight basis

Class 005: Pharmaceuticals

Clean filings:	359,560
Unique owners:	134,184
Year range:	1990–2021
Mean KL_{pros} :	4.98 nats
Mean KL_{retr} :	4.99 nats
Mean / median ΔKL :	-0.011 / -0.011
5y survival rate:	56.4%
Share with SEC-listed owner:	10.60%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Silver Spur Corporation (+2.28); SILVER SPUR CORPORATION (+1.54); Earthmade Natural Vitamins, Inc. (+1.44)

Worked examples:

Positive flux: COLON MAGIC (2009, $\Delta KL = +2.72$):

Dietary supplement for eliminating toxins from the intestinal tract; Nutritional supplement for eliminating toxins from the intestinal tract

Flux-neutral: NEXT GENERATION CURCUMIN (2019, $\Delta KL = +0.00$):

Dietary supplements containing curcumin Dietary ingredient, namely, curcumin, for use as an ingredient in the manufacture of dietary supplements

Negative flux: HEART SMART (2011, $\Delta KL = -1.47$):

Foods and snacks which are free of Wheat to accommodate special medical and health conditions

Class 006: Common Metals

Clean filings:	97,003
Unique owners:	55,943
Year range:	1990–2021
Mean KL_{pros} :	5.48 nats
Mean KL_{retr} :	5.46 nats
Mean / median ΔKL :	+0.020 / +0.013
5y survival rate:	43.4%
Share with SEC-listed owner:	4.04%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Quad Performance Products, Inc. (+0.42); Kason Industries, Inc. (+0.36); General Electric Company (+0.29)

Worked examples:

Positive flux: GOLD DIAMOND (1990, $\Delta KL = +2.73$):

key chains, key tags and fobs, not of precious metal

Flux-neutral: GVS-K (2015, $\Delta KL = -0.00$):

Junctions of metal for pipes; valves of metal, other than parts of machines; reinforcing materials of metal for pipes; elbows of metal for pipes; clips of metal for pipes; drain traps valves of metal;

Negative flux: VALUE PRESS (1995, $\Delta KL = -1.10$):

metal closure caps for containers and bottles

Class 007: Machines & Power Tools

Clean filings:	160,666
Unique owners:	83,246
Year range:	1990–2021
Mean KL_{pros} :	5.34 nats
Mean KL_{retr} :	5.34 nats
Mean / median ΔKL :	+0.003 / -0.016
5y survival rate:	40.3%
Share with SEC-listed owner:	4.35%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Goodnature Products, Inc. (+0.50); Arovast Corporation (+0.40); ANC Technology INC (+0.38)

Worked examples:

Positive flux: FAFNIR TDC (1990, $\Delta KL = +2.34$):

bearings for axles, transmissions, drive shafts and controls for land vehicles

Flux-neutral: AXIS AUTOMATION (2015, $\Delta KL = -0.00$):

Design of standard and custom machinery for others Food processing machines, namely, tray denesters for separating food trays, and conveyor lines in the nature of vacuum conveyors Dispenser for dispen

Negative flux: STANNAH STAIRLIFTS (1992, $\Delta KL = -0.94$):

mobility aids for elderly or disabled persons; namely, stairlifts and parts thereof

Class 008: Hand Tools

Clean filings:	71,252
Unique owners:	44,508
Year range:	1990–2021
Mean KL_{pros} :	4.65 nats
Mean KL_{retr} :	4.58 nats
Mean / median ΔKL :	+0.070 / +0.032
5y survival rate:	40.3%
Share with SEC-listed owner:	2.66%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): BORBOLETA BEAUTY INC. (+0.68); STILETTO TOOLS, INC. (+0.66); Marut Enterprises LLC (+0.52)

Worked examples:

Positive flux: GARD'N'GRIP (1990, $\Delta KL = +2.70$):

hand tools for lawn and garden use; namely, rakes, shovels, cutters, hoes, and forks

Flux-neutral: RIPLEY (1996, $\Delta KL = +0.00$):

wire, cable, and optical fibre preparation tools, namely, cutters, strippers, rippers, crimpers, trimmers, and slitters for electrical, electronic, CATV, telecommunication, telephone, and industrial a

Negative flux: COMFORT REACH (1999, $\Delta KL = -0.89$):

hand pruners, weeders, bulb planters and loppers

Class 009: Software & Electronics

Clean filings:	1,226,806
Unique owners:	602,439
Year range:	1990–2021
Mean KL_{pros} :	5.42 nats
Mean KL_{retr} :	5.43 nats
Mean / median ΔKL :	-0.011 / -0.023
5y survival rate:	48.6%
Share with SEC-listed owner:	5.52%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Leading Technology, Inc. (+2.50); Redmond Productions, Inc. (+1.37); Geniuses at Work Corporation (+1.18)

Worked examples:

Positive flux: MEDINET (1990, $\Delta KL = +3.40$):

a computer-based network including computer hardware and software

Flux-neutral: PRIME PROTECTION (2002, $\Delta KL = -0.00$):

Computerized ear tags or chips for the identification of cattle and related computer software for animal health record keeping Plastic ear tags for the identification of cattle

Negative flux: UNDONE (2011, $\Delta KL = -1.44$):

Electric hair curlers; Electric hair curling irons; Electric hair straightener; Electric hair straightening irons; Electric hair-curlers

Class 010: Medical Devices

Clean filings:	190,121
Unique owners:	88,206
Year range:	1990–2021
Mean KL_{pros} :	5.35 nats
Mean KL_{retr} :	5.35 nats
Mean / median ΔKL :	-0.001 / -0.017
5y survival rate:	47.4%
Share with SEC-listed owner:	7.80%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Societe Vygon (+0.67); NuGeneration Technologies, LLC (+0.60); CALIFORNIA EXOTIC NOVELTIES, LLC (+0.58)

Worked examples:

Positive flux: COVAFLU (2020, $\Delta KL = +3.20$):

Face covers being sanitary masks for protection against viral infection; Face shields for medical use

Flux-neutral: TRI-CHEW (2010, $\Delta KL = -0.00$):

Oral therapy apparatus, namely, a tool used by a professional therapist or trained caregiver for jaw movement, tongue movement, lip movement, oral exploration, or for the development of feeding skills

Negative flux: XEO (2019, $\Delta KL = -1.29$):

Catheters; catheters for percutaneous transluminal coronary angioplasty (PTCA catheters) and catheters for percutaneous transluminal angioplasty (PTA catheters), namely, catheters for balloon dilatati

Class 011: Lighting, HVAC & Appliances

Clean filings:	194,994
Unique owners:	112,215
Year range:	1990–2021
Mean KL_{pros} :	4.84 nats
Mean KL_{retr} :	4.79 nats
Mean / median ΔKL :	+0.045 / +0.014
5y survival rate:	38.8%
Share with SEC-listed owner:	2.80%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): G.K.L. Corp. (+0.59); ALPHARERRY INC (+0.59); VANJO LLC (+0.54)

Worked examples:

Positive flux: VENTEX (1990, $\Delta KL = +2.50$):

ventilation (air conditioning) installations and apparatus, fans, air conditioning apparatus

Flux-neutral: MASTER VALET (2012, $\Delta KL = +0.00$):

[Electric laundry irons and parts therefor] Garment steamers [; steam generators for household purposes and parts therefor]

Negative flux: SOOPSPOON (2019, $\Delta KL = -1.74$):

Electrically-powered handheld vaporizers for vaporizing vaporizable constituents of herbal and plant matter for creating an aroma for household use; Gas powered handheld vaporizers for vaporizing vapo

Class 012: Vehicles

Clean filings:	128,694
Unique owners:	68,694
Year range:	1990–2021
Mean KL_{pros} :	4.93 nats
Mean KL_{retr} :	4.90 nats
Mean / median ΔKL :	+0.034 / +0.013
5y survival rate:	43.7%
Share with SEC-listed owner:	5.01%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Legendary Auto Interiors, Ltd. (+0.99); Drop Stop, LLC (+0.70); Hooey, LLC (+0.62)

Worked examples:

Positive flux: PF 815 (1990, $\Delta KL = +2.36$):

vehicle tires for automobiles, trucks and vans

Flux-neutral: I WHEEL (2021, $\Delta KL = +0.00$):

Wheelchair spoke covers; Wheelchair spoke guards; Wheelchair tire covers; Handrims and pushrims for wheelchairs

Negative flux: ZF-POWERLINE (2011, $\Delta KL = -1.05$):

Transmissions, manual transmissions, automated and automatic transmissions, dual clutch transmissions, hybrid transmissions, synchronized transmissions, powershift transmissions, continuously variable

Class 013: Firearms & Pyrotechnics

Clean filings:	13,777
Unique owners:	7,027
Year range:	1991–2021
Mean KL_{pros} :	4.53 nats
Mean KL_{retr} :	4.47 nats
Mean / median ΔKL :	+0.063 / +0.059
5y survival rate:	45.9%
Share with SEC-listed owner:	3.94%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): BATTLEARMS IP, LLC (+0.63); Ghost Inc. (+0.56); Battle Arms Development, Inc. (+0.50)

Worked examples:

Positive flux: STS (2009, $\Delta KL = +2.14$):

(Based on Use in Commerce) Firearms (Based on Intent to Use) Ammunition for firearms; Firearm attachments, namely, modular external rail systems for attaching accessories to firearms

Flux-neutral: SHAMAN (2021, $\Delta KL = +0.00$):

Hand gun accessories, namely, belt clips for securing a gun without the use of a holster

Negative flux: (1994, $\Delta KL = -1.60$):

non-explosive defensive weapon in the nature of an organic irritant packaged in a pressurized spray canister

Class 014: Jewelry & Watches

Clean filings:	109,375
Unique owners:	72,078
Year range:	1990–2021
Mean KL_{pros} :	4.52 nats
Mean KL_{retr} :	4.50 nats
Mean / median ΔKL :	+0.019 / +0.004
5y survival rate:	43.2%
Share with SEC-listed owner:	1.79%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Miki Corporation (+0.61); MIKI CORPORATION (+0.60); Spence Diamonds Ltd. (+0.49)

Worked examples:

Positive flux: OFF THE CLOSET (2013, $\Delta KL = +3.11$):

Women's jewelry; Earrings; Necklace and earring combinations that can be worn separately or as one piece; Necklaces as jewelry; Rings as jewelry; Bracelets

Flux-neutral: ANCHOR2ANCHOR (2021, $\Delta KL = +0.00$):

Hats; Scarfs; Shoes; Shorts; Sneakers; Socks; Sweaters; Sweatshirts; Underwear; Bomber jackets; Denim jackets; Denim jeans; Denim shorts; Denim skirts; Knit shirts; Knit tops; Knitted caps; Knitted un

Negative flux: SJ (1992, $\Delta KL = -1.29$):

gold or platinum jewelry with or without diamonds precious and/or semi-precious stones

Class 015: Musical Instruments

Clean filings:	12,496
Unique owners:	7,733
Year range:	1991–2021
Mean KL_{pros} :	4.47 nats
Mean KL_{retr} :	4.48 nats
Mean / median ΔKL :	-0.008 / -0.015
5y survival rate:	43.2%
Share with SEC-listed owner:	0.74%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Lone Wolf Blues Company (+0.38); Zdenka Infeld Asset Management GmbH (+0.32); Godlyke, Inc. (+0.29)

Worked examples:

Positive flux: EZ CHORD (2006, $\Delta KL = +1.27$):

Guitar playing assistance devices, namely, a sliding clamp-like structure positioned onto the guitar neck to form and sound guitar chords and notes

Flux-neutral: O-PORT (2009, $\Delta KL = -0.00$):

Guitar playing assistance devices, namely, a sliding clamp-like structure positioned onto the guitar neck to form and sound guitar chords and notes

Negative flux: RHYTHM TOOLS (2003, $\Delta KL = -1.04$):

Musical instruments, namely drums; drumheads; a click clack, namely a set of jingles, usually metal, mounted within a generally u-shaped frame; a timpano, namely a tensioned membrane mounted in a rect

Class 016: Paper, Print & Office

Clean filings:	444,018
Unique owners:	245,687
Year range:	1990–2021
Mean KL_{pros} :	5.39 nats
Mean KL_{retr} :	5.40 nats
Mean / median ΔKL :	-0.008 / -0.012
5y survival rate:	50.2%
Share with SEC-listed owner:	3.87%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Llummar Star Kites, Inc. (+1.08); Brown & Bigelow, Inc. (+0.90); Atkins Daniel D (+0.82)

Worked examples:

Positive flux: IMPRINT JIM DIESO (2010, $\Delta KL = +2.53$):

Art prints comprised of digital illustrations originating from photographs

Flux-neutral: STAND & WITHSTAND INTEGRITY GROUP (2021, $\Delta KL = +0.00$):

Educational services, namely, developing curriculum for others in the field of how to get an education; Training services in the field of how to get an education; Arranging and conducting of seminars

Negative flux: ITC MENDOZA ROMAN (1992, $\Delta KL = -2.08$):

typefaces, typefonts, and type designs of alphanumeric characters and/or typographical symbols type-faces, typefonts and type designs of alphanumeric characters and/or typographical symbols recorded on

Class 017: Rubber & Insulators

Clean filings:	51,254
Unique owners:	26,584
Year range:	1990–2021
Mean KL_{pros} :	4.97 nats
Mean KL_{retr} :	4.96 nats
Mean / median ΔKL :	+0.018 / +0.010
5y survival rate:	42.6%
Share with SEC-listed owner:	4.28%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Manco, Inc. (+0.44); MadeSolid, Inc. (+0.42); Insta-Foam Products, Inc. (+0.33)

Worked examples:

Positive flux: ADHESIVE PLUS (1990, $\Delta KL = +2.22$):

chemical adhesive compound for general, construction and household use in sealing, caulking and insulating

Flux-neutral: RENEWLENE (2008, $\Delta KL = +0.00$):

Extruded plastic in the form of bars, blocks, pellets, rods, sheets and tubes for use in manufacturing; Plastic in bars, blocks, pellets, rods, sheets and tubes for general industrial use; Plastic in

Negative flux: DURRA BUILDING STRONG SOLUTIONS (2004, $\Delta KL = -0.90$):

acoustical wall and roof panels for buildings

Class 018: Leather & Luggage

Clean filings:	147,807
Unique owners:	95,097
Year range:	1990–2021
Mean KL_{pros} :	4.32 nats
Mean KL_{retr} :	4.31 nats
Mean / median ΔKL :	+0.008 / -0.015
5y survival rate:	43.4%
Share with SEC-listed owner:	1.98%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Piper, Diane E. (+0.51); Romar International Corp. (+0.50); Bresnick, Tiffani (+0.38)

Worked examples:

Positive flux: STEPHANO SABETTI (1990, $\Delta KL = +2.63$):

men's, women's, and children's clothing - namely, pants, shirts, t-shirts, sweatshirts, jackets, coats, sweaters, scarves, gloves, socks, shoes, ties, and hats leather articles, namely, carry on bags,

Flux-neutral: ROTUNDA (1996, $\Delta KL = +0.00$):

business card cases, shoe bags for travel, tote bags

Negative flux: AXIOM EQUINE SPORT BOOT (2012, $\Delta KL = -0.98$):

Horse flanks and shanks, namely, horse protective padding, braces, and leg boots for the flanks and legs

Class 019: Building Materials

Clean filings:	70,015
Unique owners:	34,729
Year range:	1990–2021
Mean KL_{pros} :	5.08 nats
Mean KL_{retr} :	5.10 nats
Mean / median ΔKL :	-0.015 / -0.014
5y survival rate:	45.6%
Share with SEC-listed owner:	3.45%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Saroyan Lumber Company (+0.82); SCHLUTER-SYSTEMS KG (+0.38); Holyfield Management Inc. (+0.35)

Worked examples:

Positive flux: ACRAJOINT (1990, $\Delta KL = +2.94$):

building materials; namely, a system comprised of glass blocks and adhesive used to form a wall, sold as a unit

Flux-neutral: MANTA (1999, $\Delta KL = +0.00$):

material in the form of bricks, tiles, sheets or mortar for screening electromagnetic interference; material comprising a blend of carbon and iron oxides (ferrite) in the form of bricks, tiles, sheets

Negative flux: HOMESEAL (1998, $\Delta KL = -1.03$):

Asphalt, modified asphalt and rubberized asphalt based peel and stick membranes for use on roofing

Class 020: Furniture

Clean filings:	165,306
Unique owners:	102,530
Year range:	1990–2021
Mean KL_{pros} :	5.16 nats
Mean KL_{retr} :	5.09 nats
Mean / median ΔKL :	+0.075 / +0.042
5y survival rate:	42.3%
Share with SEC-listed owner:	2.76%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Xiamen Youjing E-commerce Co., Ltd. (+0.66); Everglades College, Inc. (+0.63); Shanxi Huacai Zhongxing E-commerce Co., Ltd. (+0.62)

Worked examples:

Positive flux: THE SIMPSONS (1990, $\Delta KL = +2.05$):

tables, chairs, desks, beds, children's furniture, mattresses, pillows, mirrors, picture frames, plastic key ring tags

Flux-neutral: TILANG (2020, $\Delta KL = +0.00$):

Pet toys Planters for flowers and plants Shoe bags for travel Curtain rods; Mirrors

Negative flux: ULTRA VIVID (2016, $\Delta KL = -0.83$):

Non-metal and non-leather key chains Treatment of materials by laser beam

Class 021: Household Utensils

Clean filings:	219,996
Unique owners:	143,153
Year range:	1990–2021
Mean KL_{pros} :	4.85 nats
Mean KL_{retr} :	4.78 nats
Mean / median ΔKL :	+0.067 / +0.037
5y survival rate:	40.8%
Share with SEC-listed owner:	2.23%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): ALPHARERRY INC (+0.81); InnovatioNxt, LLC (+0.72); FLYCAMP TRADING LLC (+0.67)

Worked examples:

Positive flux: IREPELLER (2017, $\Delta KL = +1.96$):

Plug-in diffusers for mosquito repellents; Ultrasonic pest repellents

Flux-neutral: HALO (2017, $\Delta KL = +0.00$):

Wine preservation devices, namely, bottle stoppers specifically adapted for use with wine bottles

Negative flux: BEER RINGS (2016, $\Delta KL = -1.03$):

Charms for attachment to beverage glassware for identification purposes

Class 022: Ropes, Tents & Textiles (raw)

Clean filings:	24,515
Unique owners:	16,565
Year range:	1990–2021
Mean KL_{pros} :	4.70 nats
Mean KL_{retr} :	4.59 nats
Mean / median ΔKL :	+0.111 / +0.072
5y survival rate:	40.1%
Share with SEC-listed owner:	2.31%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Caravan Canopy Int'l, Inc. (+0.69); Xiamen Youjing E-commerce Co., Ltd. (+0.58); High Caliper Growing, Inc. (+0.54)

Worked examples:

Positive flux: (2013, $\Delta KL = +2.41$):

Awnings; Awnings comprised primarily of tensile fabric membranes; Awnings for vehicles

Flux-neutral: FAYONLINE (2018, $\Delta KL = -0.00$):

Sleeping bag pads; Futons Hammocks Blanket throws; Baby blankets; Children's blankets; Liners adapted to sleeping bags for camping

Negative flux: STONE CONTAINER CORPORATION (1990, $\Delta KL = -0.95$):

[logs, trees, wood chips and bark] agricultural and forestry products; namely, market pulp sawdust

Class 024: Textiles & Bedding

Clean filings:	97,945
Unique owners:	59,850
Year range:	1990–2021
Mean KL_{pros} :	4.25 nats
Mean KL_{retr} :	4.21 nats
Mean / median ΔKL :	+0.042 / +0.016
5y survival rate:	42.9%
Share with SEC-listed owner:	2.23%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): FISHER TEXTILES, LLC (+0.61); An Le (+0.51); Zhu Tianding (+0.42)

Worked examples:

Positive flux: RECORD BREAKERS WORLD OF SPEED (1990, $\Delta KL = +2.91$):

bed sheets, pillow cases, bedspreads, comforters, draperies, curtains, blankets

Flux-neutral: ARM & HAMMER (2010, $\Delta KL = +0.00$):

box spring covers, bed blankets

Negative flux: APPLETON LANE (1996, $\Delta KL = -0.94$):

vinyl wallcoverings and wallcovering borders textile fabrics for use in draperies and upholstery sample books of wallcoverings

Class 025: Clothing, Footwear & Headwear

Clean filings:	668,593
Unique owners:	426,635
Year range:	1990–2021
Mean KL_{pros} :	4.36 nats
Mean KL_{retr} :	4.40 nats
Mean / median ΔKL :	-0.041 / -0.051
5y survival rate:	51.6%
Share with SEC-listed owner:	1.39%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): CEO Lux, LLC (+0.72); CHILD SAFETY PRODUCTS, INC. (+0.66); Warner Bros. Inc. (+0.57)

Worked examples:

Positive flux: ÜBERION (2007, $\Delta KL = +3.14$):

Triathlon clothing, namely, triathlon tights, triathlon shorts, triathlon singlets, triathlon shirts and triathlon suits

Flux-neutral: COBALT (2019, $\Delta KL = -0.00$):

wetsuits; wetsuit gloves; water socks; rash guards; aqua shoes; flip-flops for use as footwear

Negative flux: GEFFEN BABY (2011, $\Delta KL = -1.53$):

Cloth diapers; Infant cloth diapers; Inserts specially adapted for cloth diapers made of hemp; Textile diapers

Class 026: Trim, Lace & Notions

Clean filings:	42,323
Unique owners:	29,873
Year range:	1990–2021
Mean KL_{pros} :	4.47 nats
Mean KL_{retr} :	4.34 nats
Mean / median ΔKL :	+0.124 / +0.075
5y survival rate:	40.4%
Share with SEC-listed owner:	1.76%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Celebrity Signatures International, Inc. (+1.07); RED SLATE BRANDS LLC (+0.82); KIM, HYUN JOON (+0.77)

Worked examples:

Positive flux: BEAUTY BUN (2011, $\Delta KL = +2.08$):

Hair accessories, namely, plastic styling inserts that create height and volume on the crown of the head

Flux-neutral: STOGEDEST (2021, $\Delta KL = +0.00$):

Wigs; Clown wigs; False hair; Hair extensions; Hair pieces; Hair pieces and wigs; Hair weaves; Human braiding hair; Wig caps

Negative flux: TEXTURED LOCS (2020, $\Delta KL = -2.66$):

Wigs, hairpieces, and add-in and add-on hair accessories constructed primarily of synthetic and/or human hair

Class 027: Carpets & Wall Hangings

Clean filings:	26,727
Unique owners:	15,621
Year range:	1990–2021
Mean KL_{pros} :	4.55 nats
Mean KL_{retr} :	4.49 nats
Mean / median ΔKL :	+0.053 / +0.019
5y survival rate:	40.9%
Share with SEC-listed owner:	3.14%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Marketing and Sales Management Corp. (+0.75); J. Josephson, Inc. (+0.59); Watershed Geosynthetics LLC (+0.54)

Worked examples:

Positive flux: THE SIMPSONS (1990, $\Delta KL = +2.63$):

carpets, rugs, vinyl wall coverings, wall hangings, wall paper

Flux-neutral: MILLICARE (1996, $\Delta KL = +0.00$):

cleaning preparations consisting of porous, absorbent particles in powder form and a liquid concentrate for use in the cleaning of commercial carpets

Negative flux: GENCORP (1997, $\Delta KL = -2.06$):

wallcovering having an outer surface of plastic

Class 028: Toys, Games & Sporting Goods

Clean filings:	325,521
Unique owners:	167,749
Year range:	1990–2021
Mean KL_{pros} :	4.94 nats
Mean KL_{retr} :	4.94 nats
Mean / median ΔKL :	-0.006 / -0.030
5y survival rate:	47.1%
Share with SEC-listed owner:	3.83%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Aruze Gaming America, Inc. (+1.17); KING SHOW GAMES, INC. (+0.73); Nabisco Brands Company (+0.73)

Worked examples:

Positive flux: WASHINGTON SENATORS (1990, $\Delta KL = +3.02$):

toys and sporting goods; namely, Christmas tree ornaments

Flux-neutral: CORTEZ (2001, $\Delta KL = -0.00$):

fishing products, namely, rods, reels, lures, fishing line, terminal tackle, tackle bags and tackle boxes

Negative flux: SWAY BUSTER (2013, $\Delta KL = -2.26$):

Golf Training Apparatus, namely, devices used to help groove a repetitive swing and stance

Class 029: Meat, Dairy & Prepared Foods

Clean filings:	134,683
Unique owners:	64,279
Year range:	1990–2021
Mean KL_{pros} :	4.63 nats
Mean KL_{retr} :	4.65 nats
Mean / median ΔKL :	-0.020 / -0.023
5y survival rate:	51.8%
Share with SEC-listed owner:	3.42%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): BEYOND MEAT, INC. (+0.94); Aphria Inc. (+0.78); Crunchy Munchies LLC (+0.62)

Worked examples:

Positive flux: PHIVIDA (2016, $\Delta KL = +2.68$):

Healthy base beverages, namely, drinking water with vitamins containing hemp derived from industrial hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry

Flux-neutral: MUSCLE BROTH (2016, $\Delta KL = +0.00$):

liquid or powdered meat extracts containing amino acids, vitamins and minerals

Negative flux: EXCITE MY BITE (2021, $\Delta KL = -1.31$):

bouillon not containing or derived from cannabis with a delta-9 THC concentration of more than 0.3 percent on a dry weight basis or containing CBD

Class 030: Coffee, Tea, Bakery & Spices

Clean filings:	205,442
Unique owners:	104,841
Year range:	1990–2021
Mean KL_{pros} :	4.90 nats
Mean KL_{retr} :	4.91 nats
Mean / median ΔKL :	-0.011 / -0.010
5y survival rate:	51.3%
Share with SEC-listed owner:	2.72%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): ECBlend LLC (+1.15); VMR Products, LLC (+0.97); New Generation Foods, Inc. (+0.85)

Worked examples:

Positive flux: SWITCH (2012, $\Delta KL = +2.67$):

Electronic cigarette batteries; Electronic cigarette battery chargers Electronic cigarette liquid (e-liquid) comprised of flavorings in liquid form used to refill electronic cigarettes Electronic ciga

Flux-neutral: FRUITABLES (1995, $\Delta KL = -0.00$):

fruit rolls, fruit leathers, dried fruit, fruit flavored confections, namely candied fruit, fruit flavored snacks

Negative flux: EMERALD CITY VAPE PREMIUM VAPOR LIQUID S (2015, $\Delta KL = -2.01$):

Electronic cigarette liquid (e-liquid) comprised of flavorings in liquid form used to refill electronic cigarette cartridges

Class 031: Agriculture & Live Animals

Clean filings:	64,135
Unique owners:	32,411
Year range:	1990–2021
Mean KL_{pros} :	4.92 nats
Mean KL_{retr} :	4.92 nats
Mean / median ΔKL :	-0.004 / -0.013
5y survival rate:	49.8%
Share with SEC-listed owner:	2.21%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): TRIM READY INC (+1.16); DAVID AUSTIN ROSES LIMITED (+1.09); CMG PARTNERS, INC (+0.76)

Worked examples:

Positive flux: BARK AVENUE (2017, $\Delta KL = +3.20$):

Edible dog treats not containing CBD and not containing any ingredients with a delta-9 tetrahydrocannabinol concentration of more than 0.3 percent on a dry weight basis

Flux-neutral: DOC'CHOPS (2004, $\Delta KL = -0.00$):

Animal snack food; Snack food for animals designed to contain medications

Negative flux: 12 TWENTY (2019, $\Delta KL = -1.24$):

Pet food Transdermal patches for therapeutic use in the treatment of addictions, for relief of pain, nausea, and discomfort, for relaxation, for reducing stress, nervousness and fatigue, for maintaini

Class 032: Beer & Soft Drinks

Clean filings:	100,951
Unique owners:	49,681
Year range:	1990–2021
Mean KL_{pros} :	4.18 nats
Mean KL_{retr} :	4.20 nats
Mean / median ΔKL :	-0.014 / -0.017
5y survival rate:	56.6%
Share with SEC-listed owner:	3.11%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): HONEST INC. (+1.98); Duclaw Brewery, LLC (+0.96); All Market Inc. (+0.89)

Worked examples:

Positive flux: PHIVIDA (2016, $\Delta KL = +2.77$):

Healthy base beverages, namely, drinking water with vitamins containing hemp derived from industrial hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry

Flux-neutral: TRUE CAL-ORGANIC FARMS (2013, $\Delta KL = -0.00$):

Organic fresh fruits; organic fresh vegetables Organic fruit juices and beverages; organic vegetable juices and beverages

Negative flux: NORTHERN EXPOSURE (1997, $\Delta KL = -1.15$):

fermented alcoholic beverages made with malt, namely, assorted beers

Class 033: Wine & Spirits

Clean filings:	68,232
Unique owners:	32,766
Year range:	1990–2021
Mean KL_{pros} :	4.09 nats
Mean KL_{retr} :	4.06 nats
Mean / median ΔKL :	+0.038 / +0.020
5y survival rate:	53.1%
Share with SEC-listed owner:	0.90%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Stranger & Stranger Ltd (+0.76); MAE-STRO TEQUILERO, S.A. DE C.V. (+0.72); Destiladora Del Valle de Tequila, S.A. de C.V. (+0.70)

Worked examples:

Positive flux: MARYGIN (2018, $\Delta KL = +2.60$):

Liquor; all of the foregoing containing ingredients solely derived from hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis and not containi

Flux-neutral: UNITED STARS (2018, $\Delta KL = +0.00$):

Alcoholic beverages, namely, vodka, prepared alcoholic cocktails, peppermint liqueurs, distilled spirits, alcoholic beverages containing fruit

Negative flux: CHATEAU SOCIANDO-MALLET (2012, $\Delta KL = -1.04$):

Wines of guaranteed label of origin Haut Medoc from a particular wine-making estate

Class 034: Tobacco & Smokers' Articles

Clean filings:	36,678
Unique owners:	18,795
Year range:	1990–2021
Mean KL_{pros} :	3.66 nats
Mean KL_{retr} :	3.50 nats
Mean / median ΔKL :	+0.163 / +0.115
5y survival rate:	55.4%
Share with SEC-listed owner:	1.42%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Freedom Smokeless LLC (+1.65); FRESH BAKED CO. (+1.37); Vaporfuel, LLC (+1.26)

Worked examples:

Positive flux: DAISY (2016, $\Delta KL = +3.90$):

Oral vaporizers for smoking purposes, namely, for smoking cannabis solely derived from hemp with a delta-9 tetrahydrocannabinol (THC) concentration of not more than 0.3 percent on a dry weight basis

Flux-neutral: SATIVAH (2010, $\Delta KL = +0.00$):

Herbal smoking blend for cigarettes, cigars and pipes [and smoking articles used therewith, namely, cigarette papers, cigar or cigarette lighters not made of precious metals and matches]

Negative flux: SITRA (2021, $\Delta KL = -1.39$):

Hookahs; hookah parts, namely, containers for housing charcoal separate from bowls while permitting heat transfer to tobacco placed in the bowls; smokers articles, namely, containers for housing hooka

Class 035: Advertising & Retail

Clean filings:	996,559
Unique owners:	584,964
Year range:	1990–2021
Mean KL_{pros} :	5.38 nats
Mean KL_{retr} :	5.40 nats
Mean / median ΔKL :	-0.019 / -0.027
5y survival rate:	47.0%
Share with SEC-listed owner:	3.65%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Steven S. Lazar (+1.17); Beverly Hills Internet (+1.08); FINANZIARIA D'INVESTIMENTO - FININVEST SPA (+0.75)

Worked examples:

Positive flux: ACI AUDIO COMMUNICATIONS INCORPORATED (1990, $\Delta KL = +3.44$):

promoting the sale of the goods and/or services of others by recording telephone commercials and providing access thereto by the members of the purchasing public

Flux-neutral: SAVE IT MAKES CENTS (2014, $\Delta KL = -0.00$):

Retail store services featuring convenience store items and gasoline

Negative flux: RECLAIM YOUR SPACE (2011, $\Delta KL = -1.51$):

Providing organizing services for belongings and spaces in a residence or office such as papers and time

Class 036: Insurance & Financial

Clean filings:	399,896
Unique owners:	196,121
Year range:	1990–2021
Mean KL_{pros} :	4.92 nats
Mean KL_{retr} :	4.96 nats
Mean / median ΔKL :	-0.042 / -0.055
5y survival rate:	47.3%
Share with SEC-listed owner:	6.69%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): The Upper Deck Company (+0.70); Philip Lilien (+0.69); AMERICAN TELEPHONE AND TELEGRAPH COMPANY (+0.67)

Worked examples:

Positive flux: WE SPEAK YOUR LANGUAGE (1990, $\Delta KL = +3.23$):
insurance administration services in the field of health care

Flux-neutral: SPACE WE'VE GOT SPACE TO FILL (1999, $\Delta KL = +0.00$):
Commercial real estate services in the nature of property leasing Commercial real estate services in the nature of commercial property development

Negative flux: MOST-WANTED (1992, $\Delta KL = -1.31$):
fund-raising services for basic and applied research into neuromuscular disorders, and for funding of hospital services for persons afflicted by such disorders

Class 037: Construction & Repair

Clean filings:	178,756
Unique owners:	106,312
Year range:	1990–2021
Mean KL_{pros} :	5.18 nats
Mean KL_{retr} :	5.18 nats
Mean / median ΔKL :	+0.001 / +0.002
5y survival rate:	42.5%
Share with SEC-listed owner:	4.84%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Aerus Concepts, L.P. (+0.39); Irvine Company, The (+0.38); Coastal Corporation, The (+0.32)

Worked examples:

Positive flux: MODEM EXPRESS (1990, $\Delta KL = +2.53$):
REPAIR SERVICES IN THE FIELD OF [COMPUTERS] AND COMPUTER PERIPHERALS
[retail store services in the field of computers and computer peripherals]

Flux-neutral: SONERI (2015, $\Delta KL = -0.00$):
Hotel services; Restaurant services Retail store services featuring groceries Financial advisory and consultancy services; Financial investment brokerage; Real estate agencies; Real estate brokerage B

Negative flux: THE MAJORS (2007, $\Delta KL = -1.21$):
Homebuilding services; development of real property for single-family and multi-family residential and master-planned community purposes

Class 038: Telecommunications

Clean filings:	170,826
Unique owners:	86,334
Year range:	1990–2021
Mean KL_{pros} :	4.32 nats
Mean KL_{retr} :	4.37 nats
Mean / median ΔKL :	-0.050 / -0.040
5y survival rate:	51.5%
Share with SEC-listed owner:	7.36%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Los Angeles Cellular Telephone Company (+0.90); Iowa Network Services, Inc. (+0.81); Graphnet, Inc. (+0.76)

Worked examples:

Positive flux: CORPORATE TELEMAGEMENT (1990, $\Delta KL = +3.43$):
telecommunication services, namely, providing long distance telephone services

Flux-neutral: VZCREW (2002, $\Delta KL = -0.00$):

Telecommunication services, namely wireless telephone communication services and wireless messaging services providing online information in the field of entertainment

Negative flux: SIMPLIFY THE CONNECTION (1998, $\Delta KL = -1.07$):

Long distance telecommunications services, namely, long-distance telephone calling services

Class 039: Transport & Logistics

Clean filings:	120,417
Unique owners:	67,340
Year range:	1990–2021
Mean KL_{pros} :	5.03 nats
Mean KL_{retr} :	5.05 nats
Mean / median ΔKL :	-0.020 / -0.017
5y survival rate:	45.4%
Share with SEC-listed owner:	6.90%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Go America Tours, Inc. (+0.67); Coastal Corporation, The (+0.55); Sun Run Generation LLC (+0.45)

Worked examples:

Positive flux: JAL JAPAN AIR LINES A WORLD OF DIFFERENC (1990, $\Delta KL = +2.68$):

transportation of passengers and cargo by air; and storage services, namely the storage of cargo and luggage which has been or is to be transported by air

Flux-neutral: VICTORIA CLIPPER (2017, $\Delta KL = +0.00$):

transportation of passengers by ship without on-board overnight accommodations

Negative flux: YOUR ENERGY EDGE (1999, $\Delta KL = -1.24$):

public utility services, namely supplying natural gas to others (

Class 040: Materials Treatment

Clean filings:	94,015
Unique owners:	59,227
Year range:	1990–2021
Mean KL_{pros} :	5.37 nats
Mean KL_{retr} :	5.37 nats
Mean / median ΔKL :	-0.004 / -0.011
5y survival rate:	44.6%
Share with SEC-listed owner:	5.75%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Intervet Inc. (+0.32); United States Department of Energy (+0.31); Cherry Street Energy, LLC (+0.30)

Worked examples:

Positive flux: ETERNITY HANES HEAVYWEIGHT (2007, $\Delta KL = +3.02$):

Imprinting messages on wearing apparel, accessories and mugs

Flux-neutral: MAGSHIRE BEL (2020, $\Delta KL = -0.00$):

Cryogenic preservation; Cryopreservation of human tissue; Cryopreservation services; Vitrification drying and dehydration preservation services for biologically active materials

Negative flux: SUPERME! (2010, $\Delta KL = -1.47$):

Creation of custom portraits and paintings based on photographs, snapshots or images provided by the customer

Class 041: Education & Entertainment

Clean filings:	920,449
Unique owners:	531,807
Year range:	1990–2021
Mean KL_{pros} :	5.18 nats
Mean KL_{retr} :	5.19 nats
Mean / median ΔKL :	-0.006 / -0.016
5y survival rate:	47.0%
Share with SEC-listed owner:	3.06%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): CPI Church Street, Inc. (+1.11); Digital Gaming Corporation Limited (+0.93); Dapper Labs Inc. (+0.86)

Worked examples:

Positive flux: THE COASTERS (1990, $\Delta KL = +3.88$):

entertainment services in the nature of a musical group

Flux-neutral: BEN MIYARES' PACKAGING MANAGEMENT UPDATE (2005, $\Delta KL = +0.00$):

Providing online electronic newsletters and trade journals in the field of packaging machinery, the packaging machinery industry and association membership affairs; providing electronic newsletters an

Negative flux: MLE MAJOR LEAGUE EMPIRE (2019, $\Delta KL = -1.29$):

Entertainment services, namely, an on-line activity where you create your own music videos

Class 042: Scientific & Tech Services

Clean filings:	743,414
Unique owners:	411,644
Year range:	1990–2021
Mean KL_{pros} :	5.17 nats
Mean KL_{retr} :	5.20 nats
Mean / median ΔKL :	-0.022 / -0.006
5y survival rate:	46.1%
Share with SEC-listed owner:	5.61%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): Geniuses at Work Corporation (+1.28); Advance Publications, Inc. (+0.98); Albin, James R. (+0.92)

Worked examples:

Positive flux: PIPELINE (1993, $\Delta KL = +3.47$):

computer software for use in providing multiple-user access to a global computer information network for the transfer and dissemination of a wide range of information providing multiple-user access to

Flux-neutral: IEEE: ADVANCING TECHNOLOGY FOR HUMANITY (2009, $\Delta KL = -0.00$):

Providing news and information in the field of science, engineering and engineering technology via the Internet; Accreditation services, namely, developing, evaluating, and testing standards with rega

Negative flux: TAILGATE TEAM (1996, $\Delta KL = -2.13$):

retail grocery store and supermarket services

Class 043: Restaurants & Hotels

Clean filings:	136,699
Unique owners:	83,211
Year range:	2002–2021
Mean KL_{pros} :	4.87 nats
Mean KL_{retr} :	4.85 nats
Mean / median ΔKL :	+0.014 / +0.012
5y survival rate:	48.4%
Share with SEC-listed owner:	2.80%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): EX3 (+0.42); BRINKER INTERNATIONAL PAYROLL COMPANY, L.P. (+0.37); Dunhill 1530 Main, L.P. (+0.34)

Worked examples:

Positive flux: MARYLAND FRIED CHICKEN COMPLETE DINNERS (2008, $\Delta KL = +2.39$):

Restaurant services, including sit-down service of food and take-out restaurant services

Flux-neutral: PANNA NEW LATINO FOOD (2015, $\Delta KL = +0.00$):

Café and restaurant services; Restaurant services featuring tequenos, cachitos, breads, sandwiches, hamburger, hotdogs, arepas, pastry, cachapas, empanadas, cheese sticks, salads, omelets, natural jui

Negative flux: KIVO (2007, $\Delta KL = -0.90$):

Contract food services; preparing and serving foods in institutional dining facilities

Class 044: Medical & Personal Care

Clean filings:	188,794
Unique owners:	109,333
Year range:	2002–2021
Mean KL_{pros} :	4.96 nats
Mean KL_{retr} :	4.95 nats
Mean / median ΔKL :	+0.005 / -0.000
5y survival rate:	45.8%
Share with SEC-listed owner:	5.56%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): ClearChoice Holdings LLC (+0.78); Aspiration Partners, Inc. (+0.53); Bittner, Craig A. (+0.43)

Worked examples:

Positive flux: AMERICAN COLLECTIVE (2009, $\Delta KL = +2.30$):

Medical cannabis collective and dispensary services, namely, providing medical cannabis to medical cannabis patients

Flux-neutral: NO BARRIERS. NO LIMITS. INDEPENDENCE FOR (2019, $\Delta KL = +0.00$):
Health care consulting in the field of occupational therapy; Occupational therapy services

Negative flux: ROMAN HEMP CO. (2021, $\Delta KL = -1.05$):

Farming services in the field of hemp and hemp flower derived from hemp with a delta-9 tetrahydrocannabinol or a delta-8 tetrahydrocannabinol concentration of not more than 0.3 percent on a dry-weight

Class 045: Legal & Personal Services

Clean filings:	126,420
Unique owners:	81,811
Year range:	2002–2021
Mean KL_{pros} :	4.96 nats
Mean KL_{retr} :	4.99 nats
Mean / median ΔKL :	-0.022 / -0.022
5y survival rate:	45.2%
Share with SEC-listed owner:	3.47%

Top 3 firms by firm-mean ΔKL (≥ 10 filings): YUGA LABS, INC. (+0.56); FamFriend, LLC (+0.50); Blend Images, LLC (+0.39)

Worked examples:

Positive flux: A COURSE IN MIRACLES MINISTER (2014, $\Delta KL = +2.06$):

Ecclesiastical services, namely, the licensing of ministers, namely, ordaining ministers to perform religious ceremonies

Flux-neutral: ARTHUR MILLER JR ADNF FOUNDATION A DAUGH (2017, $\Delta KL = +0.00$):

Charitable outreach services, namely, providing counseling services in the field of interpersonal communication between law enforcement officers and community members who have faced tragedies or other

Negative flux: NEFA FOUNDATION (2008, $\Delta KL = -1.50$):

Providing information regarding understanding and preventing terrorism